Estimating Investor Demand Elasticity with Endogenous Firm Responses*

Yicheng Wang†

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Abstract

I estimate investor demand elasticity when firms respond to flow-driven prices they also move, and trace what it implies for capital misallocation. Using a shift-share design, I show that a one-standard-deviation mutual fund flow shock raises stock prices by 1.9% and equity issuance by 0.2% of assets, funding investment and deleveraging. In a dynamic model where firms and investors jointly determine the price each acts on, market clearing splits market value into fundamental value and price pressure. Because issuance raises fundamental value, observed market value blends price pressure with the improvement it triggers, which accounts for 17% of the initial price response; separating the two raises the estimated elasticity from 1.3 to 1.6. That elasticity is not a single number but source-, size-, and state-dependent, and that structure determines its real consequences. The correction closes little of the distance to perfectly elastic demand, so price pressure should give firms ample incentive to over- or under-invest. Yet demand-side counterfactuals move aggregate TFP far less than a modest reduction in real capital adjustment costs.

Keywords: Demand Elasticity, Structural Estimation, Equity Issuance, Market Timing, Mutual Fund Flows, Capital Misallocation
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†Carlson School of Management, University of Minnesota, wang9543@umn.edu.

Introduction

Two well-documented features of equity markets, one about the demand for equity and one about its supply, bear directly on how capital gets allocated. Investor demand for individual stocks is inelastic, so a shift in demand moves a firm's stock price well beyond what its fundamentals warrant (Koijen and Yogo, 2019). And firms act on the price they see, issuing equity when their shares are highly priced, buying it back when they are not, and financing investment out of the proceeds (Bolton et al., 2013). The price guiding those real decisions is then not the price a firm's prospects justify, and capital can follow non-fundamental demand instead of productivity. How far prices detach, and how much capital follows, are both governed by how elastic investor demand is.

Demand and supply are not independent: the market value a firm responds to is a price its own response then changes, along with the fundamental value underneath it. Standard approaches leave that response out by treating equity supply as fixed, which creates one problem of measurement and one of real consequence. The measurement question is what a standard estimate of demand elasticity actually recovers. When demand lifts a firm's *market value* above its *fundamental value*, the gap, *price pressure*, makes issuance attractive, and the proceeds firms invest raise fundamental value itself. A price rise therefore has two parts, and investors treat them differently, selling into one they expect to partly unwind but not into one they do not, so an elasticity estimated on observed prices blends the two responses. The real question is how much capital inelastic demand actually misallocates. Price pressure is what real decisions get made on, so the less elastic demand is, the further investment drifts from what productivity warrants, while the same responses absorb part of it first.

Answering either question requires separating price pressure from fundamental value, and since neither can be observed directly, I combine reduced-form evidence with a dynamic structural model to recover the elasticity that governs price pressure and to trace what it implies for capital allocation. Empirically, I use a shift-share design based on mutual fund flow shocks to establish that firms respond to demand-driven price movements, and on which margins: equity issuance, investment, and debt paydown. Theoretically, I build a model in which firms and investors internalize how their own actions move the price, so that market clearing splits market value into a fundamental-value component and price pressure. I estimate the model by indirect inference, matching the impulse responses and other moments from the flow-shock data.

On measurement, I find that ignoring firm responses understates demand elasticity and misses its source-, size-, and state-dependence; on real consequences, that inelastic demand misallocates far less capital than real investment frictions. Inside the estimated model, instrumenting observed market value with flow shocks recovers 1.3; instrumenting price pressure directly recovers 1.6. The difference is the fundamental improvement firms deliver, which the first estimate counts as pressure. And 1.6 is itself an average over equilibrium outcomes rather than a parameter, since what a price change does to investor demand depends on what moved it, how far, and the firm it happened to. On the real side, three counterfactuals (more volatile sentiment-driven flows, the shift from active to passive management, and tighter limits on arbitrageurs' risk-bearing capacity) move aggregate TFP by less than 0.03% in either direction, against 0.14% from a modest reduction in real capital adjustment costs. Two forces keep the effect small: source- and size-dependence cap how much distortion price pressure creates, and state-dependence undoes part of the rest, because price pressure also makes financing cheap where firms are already short of capital.

Empirically, a one-standard-deviation flow shock raises a firm's stock price by 1.9% on impact, with only partial reversal, estimated using impulse response functions (Jordà, 2005) following Dou et al. (2022) and Chaudhary et al. (2022). Firms respond by issuing equity (0.2% of assets), raising investment, and paying down debt. These responses are strongest among firms with high growth opportunities. The shocks hold up under a range of exogenous-shares and exogenous-shifts checks (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2018).

On the model side, firms, mutual funds, and residual investors each move the price that the others are also responding to. Firms maximize fundamental value through investment and financing, facing issuance costs that fall as the firm becomes overvalued and repurchase costs that fall as it becomes undervalued (Hayashi, 1982; Zhang, 2005; Belo et al., 2019). Mutual funds receive the exogenous flow shocks and adjust portfolios subject to a benchmarking constraint that keeps them from absorbing those flows immediately. What they do not absorb falls to a residual pool of other investors, who face a trading cost that rises with trade size, generating a downward-sloping demand curve for price pressure. Market clearing across the three agents then pins down market value and, with it, price pressure. A two-period version of the model makes this mechanism transparent. I then estimate a full dynamic version, disciplined to match the data.

An exogenous demand shock is not enough for standard estimation to isolate investors' response to price pressure, fundamentals held fixed: firms endogenously respond to that pressure too, violating the exclusion restriction and biasing the estimate down. Market value, the only observable, blends price pressure with the fundamental-value response that issuance triggers. That fundamental-value response accounts for 17% of the initial price response to a flow shock. An estimate that cannot separate the two counts all of it as price pressure, overstating how far price pressure itself moved for a given change in holdings and making investors look less responsive than they are. Run on data simulated from the estimated model, where both components are known, the standard approach of instrumenting observed market value with flow shocks recovers 1.3, against a true elasticity of 1.6.

That 1.6 is still only an average, and the elasticity it averages is source-, size-, and state-dependent. Start with source: a price rise headed for reversal and one that is not call for different responses from investors, so demand elasticity is not defined until the source of the price change is named. Price pressure is the component that will be undone, so residual investors take the other side; their elasticity to it, 1.6, sets how much of a flow shock survives their absorption. A rise in fundamental value carries no such correction, so residual investors, who trade only against deviations from it, absorb only what constrained funds cannot. Mutual funds do respond to higher fundamental value, though not because anything is reverting. A larger firm with fewer growth opportunities offers a lower expected return, so funds reduce their positions, at an elasticity of 0.7, less than their sensitivity to price pressure. The two numbers answer different questions, and it is the 1.6, not the 0.7, that governs how much price pressure a non-fundamental demand shock creates.

Size-dependence reflects the self-limiting responses a shock triggers: a larger shock does not produce a proportionally larger price impact, so the effective elasticity it meets rises with its size even though the underlying structural parameters are unchanged. Price impact measures the change in market value a shock delivers once firms and investors have both responded, so its inverse, the effective elasticity, is an equilibrium outcome rather than a ceteris paribus one. On the firm side, the additional supply a firm issues absorbs some of the very pressure that made issuing attractive, so each further unit of issuance is less rewarding than the last. On the investor side, residual investors and mutual funds anticipate a sharper reversal as price pressure grows, and push back harder against it. The pattern also appears in an untargeted check on the data: the per-unit price response to small shocks is about 60% larger than to shocks of average size.

The price impact a flow shock produces depends on the firm's state, because the responses behind it do. Two dimensions of that state do most of the work: growth opportunities, whether the firm has investment worth funding; and issuance capacity, how much equity it can sell before doing so erodes the pressure it exploits. A firm with growth opportunities issues more, and its price impact is larger: capital is worth more in its hands, and expanding eases the reversal investors would otherwise expect. Issuance capacity turns on leverage: levered firms are more exposed to price fluctuations, which shows up jointly as a larger price impact and a sharper pullback by funds. So a levered firm exhausts that capacity sooner, and its equity issuance stays flat even as the incentive to issue keeps rising. In the data, sorting on leverage alone shows both at once: a larger price impact for high-leverage firms, and weaker equity issuance. Neither this pairing nor the growth-opportunity pattern is targeted in estimation, and the model delivers both.

Correcting the elasticity upward closes only a little of the distance to perfectly elastic demand, so pervasive demand shocks should still swing price pressure widely, giving firms ample incentive to over- or under-invest, misallocating capital in aggregate. The chain is real, but it does not run far: the same responses that make elasticity source-, size-, and state-dependent place a low ceiling on how much capital inelastic demand misallocates. To quantify it, I take the estimated model and vary how much price pressure a demand shock generates and how much of it investors absorb, holding technology and real investment frictions fixed. I run three counterfactuals: raising the volatility of sentiment-driven flows, shrinking the actively managed sector, and tightening limits on arbitrageurs' risk-bearing capacity. Yet all three move aggregate TFP by less than 0.03% in either direction, against the 0.14% that a modest reduction in real capital adjustment costs delivers. Even eliminating price pressure entirely, so that no firm ever invests on a market value detached from its fundamental value, raises aggregate TFP by only 0.12%, still less than that same real-friction change.

Two forces keep misallocation that small. To a close approximation, misallocation is dispersion in firms' marginal product of capital (MPK). Price pressure adds a wedge to each firm's MPK, and that wedge changes dispersion in two ways. Source- and size-dependence cap the first, how widely the wedges vary across firms, which accounts for about a third of the rise in dispersion. State-dependence governs the second and larger channel, whether a wedge pushes its firm's MPK further from the middle or back toward it. There the wedges amplify on net, though state-dependence corrects more than half of what they would otherwise add.

Source- and size-dependence hold those wedges down in different ways: source sets how much price pressure a demand shock creates, and size keeps it from scaling up as shocks grow. Because an estimate that holds equity supply fixed understates how elastic demand is, investors absorb more of every flow than it implies, and the wedges left behind are smaller than it predicts. And because firms and investors both push back harder as pressure grows, a larger shock does not translate into a proportionally larger wedge.

The corrective force comes from state-dependence, because price pressure is not only a burden; it is also cheap financing in an economy where collateral constraints and issuance costs already leave some firms short of capital. Growth opportunities and issuance capacity jointly decide which firms can take it. Where firms can convert it, the wedge is corrective. They act on price pressure in either direction, issuing into it and repurchasing against it. And they act most where their capital sits furthest from what productivity warrants, so their MPK moves toward the middle and dispersion narrows. Where they cannot, it amplifies. Issuing erodes the pressure they would exploit, and the constraint that blocks them is the one that left them short of capital, so the wedge lands on firms whose MPK was already high, widening dispersion instead. On net, amplification wins, but the correction removes more than half of it. A framework that treats elasticity as a single number, and does not model which firms can convert price pressure into capital, misses both: it overstates the wedges and cannot see the correction at all.

Related Literature. First, this paper contributes to demand-based asset pricing (Koijen and Yogo, 2019; Davis et al., 2022). Haddad et al. (2021) show that when a group of investors changes its behavior, other investors counteract only half of the direct impact, so competition among investors does not restore elastic demand.[1] The strategic interaction here runs between firms and investors, and through the supply of shares rather than portfolio reallocation.

A growing body of work asks what an estimated elasticity recovers, questioning the tastes it assumes, the dynamics it imposes, and the instruments it relies on (Fuchs et al., 2023; van Binsbergen et al., 2025; He et al., 2025; Haddad et al., 2025). These concerns are all on the demand side, and take supply as fixed. Here both sides respond and clear against each other, with two consequences. Part of what the instrument moves is fundamental value rather than price pressure, so the estimate blends two elasticities that the model separates. And because the responses absorb the pressure that triggered them, multipliers fall as shocks grow. Chaudhry and Li (2025) document that pattern and match it to models with fixed adjustment

[1] Aggregate flow-to-price multipliers are large (Gabaix and Koijen, 2022), and supply interacts with demand in a comparable way in segmented bond markets (Vayanos and Vila, 2021; Siani, 2022; Coppola, 2025), though equity issuance changes the fundamental value of the claim being traded and not only its quantity.

costs or endogenous inattention; here it comes from how the two sides clear, with neither friction.

Second, this paper connects investor demand to the real effects of non-fundamental prices and to the measurement of capital misallocation. Whether the stock market is a sideshow for investment is an old question (Morck et al., 1990). It is not, for firms that depend on external equity: mispricing moves their issuance and investment, and flow-driven variation identifies financing rather than learning as the channel (Baker et al., 2003; Lou and Wang, 2018; Khan et al., 2012; Edmans et al., 2012; Polk and Sapienza, 2009).[2] Together with inelastic demand these findings suggest a large distortion: demand shifts are pervasive, inelastic demand turns them into sizable price movements, and firms invest on the prices they face. Turning that into a number requires a measure of misallocation.

Following Hsieh and Klenow (2009), dispersion in the marginal product of capital measures the distortion.[3] Choi et al. (2025) bring investor demand into this measurement, combining a dynamic investment model with a logit demand system, and find that demand fluctuations account for 23.4% of productivity losses, the natural quantitative prior for the exercise here. This paper replaces the fitted demand system with optimizing investors, a still parsimonious demand side that generates richer dynamics with untargeted empirical support, and identifies the shifts in demand from plausibly exogenous variation. Demand fluctuations account for less of the misallocation here.

Third, this paper provides a structural interpretation of the empirical literature on the price impact of fund flows. Earlier work established flow-driven price pressure, the tools to measure it, and how much larger elasticities look when estimated from trades (Coval and Stafford, 2007; Lou, 2010; van der Beck, 2022). The flow shocks here are constructed following Dou et al. (2022) and Chaudhary et al. (2022). Firms are a principal counterparty to these flows: Sammon and Shim (2024) find that they are the largest and most responsive providers of shares to passive investors, and aggregate and sentiment-based evidence points the same way (Baker and Wurgler, 2003; Frazzini and Lamont, 2008; Chinco and Sammon, 2024), though Chiu and Kini (2014) trace the pattern to retail rather than institutional flows. Index inclusion shows what turns on that response. It raises the inelastic demand a stock attracts (Pavlova and

[2] A parallel branch locates the real effect in the information a price carries (Chen et al., 2007; Dessaint et al., 2018; Bond et al., 2012). The mechanism here is the financing one: the firm responds to the price at which it can transact, not to what it infers from the price.

[3] With the caveat that the measured residual absorbs much besides distortion (Haltiwanger et al., 2018; David et al., 2022; Kılıç and Tüzel, 2026; Whited and Zhao, 2021).

Sikorskaya, 2023), and it lowers investment by raising the cost of equity managers believe they face (Kontz and Hanson, 2026). The shock differs, but the setting is the one this framework is built for: a demand shift, responses from firms and investors, and a price that clears against both. What a demand shift does cannot be read off the demand side alone, and which responses to model is a question about the counterfactual.

Finally, this paper contributes to the literature on time-varying equity financing costs, which studies firms' response while taking the variation as given. Warusawitharana and Whited (2016) estimate a dynamic investment model with misvaluation and find that it moves financial policy more than investment, and related work estimates the cost of external finance and documents firms timing their own securities (Baker and Wurgler, 2002; Eisfeldt and Muir, 2014; Begenau and Salomao, 2019; Belo et al., 2019; Ma, 2019). Here that variation originates in investor demand and responds to the firm's own issuance, which makes it measurable from observed flows and carries the firm's response through to misallocation.

The remainder of the paper is organized as follows. Section 1 introduces data, presents the empirical results, and conducts heterogeneity analyses to provide targets and initial guidance for the model. Section 2 presents a simple two-period model to illustrate the key mechanisms. Section 3 extends the model to a dynamic setting and defines the equilibrium. Section 4 discusses the calibration and estimation results. Section 5 analyzes the policy functions and key mechanisms. Section 6 conducts counterfactual experiments. Finally, Section 7 concludes the paper.

1 Data and Empirical Findings

This section presents empirical findings that both motivate and serve as estimation targets for the model. I document the impact of mutual fund flow shocks on stock prices and equity issuance. The impulse response analysis reveals demand-side frictions and highlights how firms endogenously respond to price movements, underscoring the importance of incorporating firm actions in demand elasticity estimation.

1.1 Data Sources

Mutual fund holdings data are from CRSP Survivor-Bias-Free US Mutual Fund Database. I focus on mutual funds with domestic equity holdings (CRSP style code starting with "ED" or

”M”). Since holdings data are only available at quarterly frequency in earlier years and are more consistent at quarterly frequency in later years, I aggregate returns and total net assets (TNA) to quarterly level. Following Elton et al. (2001), I require the lagged TNA to be larger than \$15 million. The final sample is from 2003Q1 to 2021Q4. On average, there are 3966 distinct mutual fund portfolios per quarter, a number that grows over time.

Firm characteristics are from CRSP/Compustat Merged Database. Section A.1 provides a detailed description of variable construction. Following Eisfeldt and Muir (2014), Net Equity Issuance (NEI) is defined as Sale of Common and Preferred Stock net of Repurchase of Common and Preferred Stock and Dividends[4].

1.2 Mutual Fund Flow Shocks

Following Dou et al. (2022), for each mutual fund m in period t, flows are calculated by

$$\text{Flow}_{m,t} = \frac{\text{TNA}_{m,t} - \text{TNA}_{m,t-1} \times (1 + R_{m,t})}{\text{TNA}_{m,t-1}}$$

To extract flow shocks $\hat{\eta}_{m,t}$, for each T-quarter rolling window (here $T = 16$), I regress flows on fund and portfolio characteristics $\mathbf{X}_{m,t}$:

$$\begin{aligned} \text{Flow}_{m,t} &= \mathbf{X}_{m,t}\boldsymbol{\beta}_\tau + \eta_{m,t} \quad \text{for } t = \tau+1, \tau+2, \ldots, \tau+T-1, \\ \hat{\eta}_{m,\tau+T} &= \text{Flow}_{m,\tau+T} - \mathbf{X}_{m,\tau+T}\hat{\boldsymbol{\beta}}_\tau \end{aligned}$$

The fund characteristics include lagged flow, fund excess return relative to the market return, and the portfolio characteristics are value-weighted characteristics in the Fama-French five-factor model (i.e. log market equity, book-to-market ratio, profitability, investment, and market beta) (Fama and French, 1993). The portfolio characteristics have significant predictive power and are time-varying as is shown in Figure A.1.

Then I calculate the idiosyncratic component of the flow shocks $f_{m,t}$ (with a slight abuse of notation) by regressing $\hat{\eta}_{m,t}$ on time and fund fixed effects to remove potential common factors

$$\hat{\eta}_{m,t} = \alpha_m + \delta_t + f_{m,t}. \tag{1}$$

Let $S_{i,m,t}$ denote the the number of shares of firm i held by fund m, firm-level shock $f_{i,t}$ is

[4] Compustat item `sstk - prstkc - dv`

calculated by

$$f_{i,t} = \frac{\sum_m S_{i,m,t-1} f_{m,t}}{\sum_m S_{i,m,t-1}}. \tag{2}$$

After aggregating flow shocks to firm level, I standardize flow shocks for easier comparison. Because shocks are constructed from one-step-ahead predictions using a $T = 16$-quarter rolling window starting in 2003Q1, the final firm-level sample spans 2007Q1 to 2021Q4.

1.3 Firm Responses

Table 1 presents the summary statistics of the firm panel. Mutual funds hold 24.6% of firm equity on average, with large variations across firms and time. Net equity issuance is positive on average (0.94% of total assets) and exhibits large heterogeneity, with firms engaging in substantial issuances and repurchases. As a validation check, I construct an alternative equity measure using book equity changes (total assets net of retained earnings and total liabilities), which exhibits patterns similar to the baseline cash flow-based NEI measure.

Table 1 Here

I estimate the following local projection in the spirit of Jordà (2005)

$$\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t} + X_{i,t-1} + \varepsilon_{m,t+h} \text{ for } h = 0, 1, \dots, 4, \tag{3}$$

where control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics: log market equity, log book-to-market ratio, cash scaled by asset, and investment. β_h can be interpreted as the $h-$quarter ahead impulse response from the mutual fund flow shocks. The coefficients are plotted in Figure 1. The shaded area indicates 90% pointwise confidence bands using standard errors clustered at firm level.

Figure 1 Here

Panel A shows a one-standard deviation flow shock leads to a 1.9% stock price increase. This price impact suggests constraints on both mutual funds and other investors. When an unconstrained mutual fund receives unexpected inflows, it does not need to scale up its holdings, avoiding positive price pressure. Similarly, if other investors were unconstrained, they should be able to absorb the excess demand from unexpected mutual fund flow shocks. Instead of an immediate price correction, the price reversal in subsequent quarters is much smaller than the initial impact. Section A.3 reports the corresponding increase in mutual fund ownership underlying this price pressure.

Panel B demonstrates that firms react to these price impacts by issuing equity. The size of NEI (a cash flow measure) is 0.2% of total assets, which is corroborated by the growth of total assets (a balance sheet measure, Table 2 Column (5)). This firm response highlights the identification challenge. Even if the initial flow shocks are plausibly exogenous, firms' optimal financing responses generate endogenous changes in fundamental value. Since equity issuance can fund investment or reduce leverage, thereby increasing future cash flows or reducing risks, part of the persistent price increase may reflect improved fundamentals rather than pure price pressure. This complicates the interpretation of demand elasticities.

Table 2 presents the response on impact ($h = 0$ in equation (3)) on additional firm-level variables. Columns (1) and (2) provide further validation for the baseline results on equity issuance. In Column (1), book equity increases by about 0.1%. In Column (2), when focusing on Sale of Common Stock and apply the 2% market equity filter following McKeon (2015), the estimate (0.16%) is slightly smaller than the baseline specification (0.2%) since the share repurchases are excluded. As expected from equity issuance, cash stock goes up in Column (3), while physical capital rises by 0.2% (Column (4)), indicating investment in productive assets. Column (5) shows that total assets, an alternative growth measure, rise by 0.2% as well, corroborating the increase in physical capital. Column (6) shows that leverage decreases by 0.03 percentage points, reducing risks. Moreover, since the leverage decline is smaller than the asset increase, firms are increasing debt financing, possibly with the help of increased collateral capacity.

Table 2 Here

1.4 Heterogeneity Analysis

1.4.1 Positive and Negative Flow Shocks

While NEI is positive on average, Table 1 shows that there is a significant portion of observations with negative NEI, reflecting net repurchases. Do firms respond to flow shocks differently when faced with positive and negative price pressures? To test for such asymmetry, I interact $f_{i,t}$ with two dummy variables, each indicating whether flow shocks are positive or negative.

Table 3 shows the response on impact ($h = 0$ in equation (3)) for positive and negative flow shocks. Column (2) shows that the price impacts are roughly symmetric. However, share repurchases following negative flow shocks are weaker than share issuances following positive shocks. This is consistent with the fact that share repurchases, as a form of payout, are generally smaller than share issuances. Despite the asymmetry, the results demonstrate that firm managers actively respond to price pressures in both directions, exploiting both temporary overvaluations through issuance and undervaluations through repurchases.

Table 3 Here

1.4.2 Active and Passive Funds

Based on fund classification information from WRDS, I separately calculate flow shocks for active and passive funds, with detailed construction and properties described in Section A.1.1. On average, passive and active funds own 9% and 15% of firm equity, respectively. After removing time fixed effects, the correlation between the two shocks is only 0.07. This low correlation provides supporting evidence for the exogeneity of flow shocks, as it suggests that fund flows are not common responses to firm fundamentals.

Figure 2 presents impulse response functions of firm returns to active and passive mutual fund flow shocks ($f_{i,t}^{T}$ for $T \in \{\text{Active}, \text{Passive}\}$). Despite similar standard deviations (0.362 versus 0.364), active and passive flow shocks generate different return responses. Passive fund flow shocks lead to a 2.0% contemporaneous return jump, nearly double the 1.1% response to active fund flow shocks. The subsequent reversals follow a similar pattern: passive fund flow shocks lead to a -0.5% reversal compared to a more moderate -0.25% reversal for active fund flow shocks.

Figure 2 Here

Passive funds typically operate under strict mandates, so unexpected AUM increases mechanically translate into heightened demand for constituent stocks. However, active funds should theoretically be able to avoid exerting positive price pressures on their holdings. Two factors likely explain why active funds still generate significant price pressures. First, active funds face portfolio constraints that limit their trading flexibility such as dividend reinvestment requirements (Schmickler and Tremacoldi-Rossi, 2022), benchmark tracking constraints (Kashyap et al., 2021), and client redemption pressures (Edmans et al., 2012). The weaker return responses relative to passive funds likely reflect these less stringent but still effective constraints. Second, as documented by Chinco and Sammon (2024), some nominally active funds are in fact index-tracking. Consequently, the impulse responses potentially reflect a mixture of truly active and passive investment strategies.

1.4.3 Firm Characteristics

To examine whether the baseline effects vary across firm types and provide suggestive evidence on mechanisms that are important for the model, I estimate the baseline specification (3) across subsamples based on key firm characteristics. Within each Fama-French 12 industry, I split the sample at the median of three lagged variables: mutual fund ownership, Tobin's Q, and leverage. Table 4 reports the return and NEI responses on impact ($h = 0$).

Table 4 Here

Panel A shows that firms with higher mutual fund ownership mechanically exhibit stronger responses in both returns and NEI. In Panel B, firms with higher growth potential (high Tobin's Q) demonstrate stronger responses across both dimensions. Return responses are about 50% stronger for the high-Q firms. Low-Q firms show essentially no reaction to price pressures through equity issuance, statistically indistinguishable from zero, consistent with their reduced need for external financing to fund expansion opportunities. While part of this pattern reflects the ownership channel as high-Q firms attract somewhat more mutual fund investment, this modest difference in ownership cannot fully account for the observed differences.

Panel C presents the results for subsamples partitioned by leverage. High-leverage firms have stronger return responses but weaker equity issuance responses than low-leverage firms, a pattern that at first appears puzzling, since the group with the larger price response reacts *less* through issuance. Mean mutual fund ownership is similar across the two groups, so this pattern is not primarily ownership-driven as in Panels A and B. Instead, as Section 5 shows, it reflects a composition of two offsetting forces that depend on firms' growth opportunities: low leverage provides financing flexibility that is valuable mainly for firms with investment opportunities to fund, while for firms with fewer such opportunities, high leverage instead creates room to benefit from paying down debt at favorable prices. Averaging across firms with different growth opportunities produces the pattern in this panel.

1.5 Additional Tests

The shift-share instrument is valid if it captures exogenous variation in investor demand, uncorrelated with changes in firm fundamentals. Validity can be established via exogenous shares (Goldsmith-Pinkham et al., 2020) or exogenous shifts (Borusyak et al., 2018, 2025). Section A.2 provides supporting evidence for both, and results are stable across all tests. As a further check, Section 5 shows that stock return and equity issuance responses estimated on small fund flow shocks alone are consistent with the model's self-limiting mechanism.

Under the share-based design, the concern is that lagged ownership shares reflect anticipatory allocation toward firms expected to perform well. Including characteristic-by-quarter fixed effects (industry, size, book-to-market, and profitability) absorbs systematic portfolio tilts along those dimensions. Reconstructing shocks using shares lagged four quarters addresses near-term firm-specific repositioning, with mild attenuation consistent with classical measurement error.

Under the shift-based design, the concern is that fund flows are correlated with anticipated firm fundamentals. Industry-by-quarter fixed effects address the concern that investors direct flows into industry-themed funds in anticipation of sectoral performance. Identification then comes from within-industry variation. Removing the largest firms within each industry addresses the possibility that flows are driven by the salient performance of superstar firms. There is also no pre-trend in firm fundamentals prior to flow shocks.

I also examine the sensitivity of the baseline results to alternative NEI definitions. First, excluding dividend payments from the baseline measure yields virtually identical findings, as expected given that dividends are less volatile than equity issuances and repurchases. Second, I construct gross equity issuance using the 2% market equity cutoff of McKeon (2015) and find similar results. I do not adopt this as the baseline, however, because the cutoff cannot be applied to equity repurchases, and other forms of stock issuance also represent costly equity financing[5].

2 A Two-Period Model

To illustrate the mechanisms in a simple setting, I present a two-period model with minimal ingredients and closed-form solutions for the trading stage. There are three types of agents: a

[5]For more discussions, see Huddart (1994); Fama and French (2005); Billett et al. (2019); Belo et al. (2019); Sammon and Shim (2024), among others.

firm making investment decisions, a mutual fund and a continuum of residual investors trading the firm's equity. Guided by the empirical evidence, the model centers on how exogenous fund flow shocks move asset prices when trading constraints limit investors' arbitrage capacity. These price movements influence the firm's equity issuance costs, leading to changes in fundamental values that feed back into investor demand.

The model delivers four properties of the demand elasticity. First, endogenous firm responses bias estimates built on observed prices. Second, the elasticity depends on the source of the price change. Third, responses by both the firm and the mutual fund attenuate price responses as shocks grow. Finally, the effects of price pressure depend on the firm's state. All four also govern how far the firm's investment falls short of the efficient level, and hence how price pressure distorts capital allocation.

2.1 Model Setup

The timeline is as follows. In the first period, the firm first chooses investment, then the mutual fund decides its portfolio allocation, and finally the residual investors trade to clear the market. Their trading capacity is limited, so a demand shock to the mutual fund drives the market value $\tilde{P}$ away from the fundamental value P. The ratio $\Lambda = \tilde{P}/P$ measures the price pressure, with $\lambda \equiv \log \Lambda$ denoting its log, and determines the issuance cost for the firm and returns for the mutual fund. In the second period, the firm produces output and distributes all profits.

The firm. The firm has initial capital K and therefore production K^{α}, where $\alpha \in (0, 1)$. With full depreciation, investment I becomes next-period capital K' as in (5). Net income E equals production minus investment, given by (6). When $E < 0$, the firm issues equity $H \equiv -E$ to cover the shortfall, incurring cost Ψ in (7). The cost decreases in price pressure Λ, as higher prices allow the firm to raise the same funds by issuing fewer shares. The firm maximizes its value V in (4), the sum of current and discounted future payouts. Payout in the first period equals net income minus any issuance cost. The present value of future payout is the expected value of next-period production $A'K'^{\alpha}$ discounted by the stochastic discount factor (SDF) M', where A' is a productivity shock with $\log A' \sim \mathcal{N}(0, \sigma_a^2)$.

$$\max_{I} \quad V = E - \mathbb{1}_{\{E<0\}}\Psi + \mathbb{E}[M'A'K'^{\alpha}] \tag{4}$$

$$\text{s.t.} \quad K' = I \tag{5}$$

$$E = K^{\alpha} - I \tag{6}$$

$$\Psi = \frac{c_h}{2}\left(\frac{E}{\Lambda K}\right)^2 K \tag{7}$$

In this setup, price pressure is the only channel through which the firm's investment affects its return. The ex-dividend price

$$P = \mathbb{E}[M'A'I^{\alpha}] = \hat{\beta}I^{\alpha} \tag{8}$$

depends only on $\hat{\beta} \equiv \mathbb{E}[M'A']$, not on the SDF or productivity shock separately. Combined with full profit distribution and no trading in the second period, this simplifies the return calculation:

$$R = \frac{V'}{\tilde{P}} = \frac{A'I^{\alpha}}{\Lambda\hat{\beta}I^{\alpha}} = \frac{A'}{\hat{\beta}\Lambda} \implies \log R = \log A' - \log \hat{\beta} - \lambda. \tag{9}$$

Consequently, expected log return $\mu = \mathbb{E}[\log R]$ is independent of investment I, and the log return variance $\sigma^2 = \mathrm{Var}(\log R) = \sigma_a^2$ is independent of both I and price pressure Λ. The restriction is deliberate. It shuts down any dependence of expected returns or risk on the firm's investment, which is what keeps the trading stage in closed form. In the full model of Section 3, investment moves both, so the firm's response reaches investor demand through risk as well as price pressure.

The mutual fund. The mutual fund chooses its portfolio weight ϕ to maximize its mean-variance preferences over log returns, as in (10), where γ_m is the risk aversion parameter and r_f is the risk-free rate. The fund also faces a quadratic adjustment cost with parameter c_ϕ, capturing benchmarking incentives tied to a target portfolio weight $\bar{\phi}$.

$$\max_{\phi} \quad \phi(\mu - r_f) - \frac{\gamma_m}{2}\phi^2\sigma^2 - \frac{c_\phi}{2}(\phi - \bar{\phi})^2 \tag{10}$$

Let Q denote the fund's size. The share of the firm held by the fund is

$$s^{M'} = \frac{\phi Q}{P}. \tag{11}$$

Residual investors. There is a continuum of one-period-lived residual investors representing all other shareholders. These investors price firms using the exogenously specified stochastic discount factor, the same one the firm applies to its future payouts, and trade accordingly. Absent trading costs, they would trade to ensure that each firm's market value equals its fundamental value. Trading constraints instead limit their arbitrage capacity: when the mutual fund increases (decreases) its holdings, possibly due to a positive (negative) flow shock, residual investors cannot adjust their positions sufficiently, creating temporary overvaluation (undervaluation).

Residual investors begin with shares s^R and choose new shares $s^{R'}$. They act as fundamental investors trading against log mispricing: by (9), a position bought at log price $\tilde{p} = p + \lambda$ and held to liquidation earns an expected log return of $-\lambda$ in excess of the fair compensation for risk. Their limited trading capacity is captured by a quadratic trading cost with parameter c_r (Gârleanu and Pedersen, 2013), which Section B.3 derives from a funding constraint. They solve:

$$\max_{s^{R'}} \quad s^{R'} (p - \tilde{p}) - \frac{c_r}{2} \left(s^{R'} - s^R\right)^2 \tag{12}$$

Market Clearing. Market clearing requires that the total demand for shares equals the total supply,

$$s^{M'} + s^{R'} = 1. \tag{13}$$

Equilibrium. An equilibrium consists of investment I^*, mutual fund portfolio weight ϕ^*, residual investor shares $s^{R'*}$, and market price pressure Λ^*, such that

1. Given optimal responses of the mutual fund and residual investors, the firm's investment I^* solves its optimization problem (4)–(7).

2. Given firm investment I^* and optimal responses of residual investors, the mutual fund's portfolio weight ϕ^* maximizes its objective (10).

3. Given firm investment I^* and fund portfolio weight ϕ^*, the residual investors' shares $s^{R'*}$ solve their optimization problem (12).

4. Market clearing (13) holds.

2.2 Optimality Conditions

I solve by backward induction, starting with the residual investors, then the mutual fund, and finally the firm. All derivations and proofs are in Section B.1.

Residual investors. The residual investor optimality condition is

$$s^{R'} = s^R - \frac{\lambda}{c_r}. \tag{14}$$

Substituting into the market clearing condition (13), and using the fact that initial shares sum to one ($s^M + s^R = 1$), yields log price pressure as a linear function of the mutual fund's portfolio adjustment:

$$\lambda = c_r \left(s^{M'} - s^M \right) = \xi\phi - c_r s^M, \qquad \xi \equiv \frac{c_r Q}{P}. \tag{15}$$

When $c_r \to 0$, residual investors have unlimited trading capacity. If the firm is overvalued ($\tilde{P} > P$), residual investors sell until price pressure is gone, driving $\tilde{P}$ back to P. The same mechanism applies when the firm is undervalued.

When fund ownership does not change, $s^{M'} = s^M$, there is no price pressure even with positive trading costs. A positive flow shock (larger Q) instead raises ownership to $s^{M'} > s^M$, since adjustment costs prevent the fund from lowering its weight enough to absorb the inflow, and with finite trading capacity ($c_r > 0$) residual investors cannot fully trade away the resulting pressure.

The coefficient $\xi = c_r Q/P$ in (15) is the fund's footprint, its marginal effect on price pressure, $d\lambda/d\phi = \xi$. Fund size Q and investment I matter only through this ratio, so price pressure is governed by a single object, the fund's holdings relative to fundamental value.[6]

The mutual fund. Internalizing how its actions affect price pressure, the mutual fund's optimality condition is

$$\phi = \frac{\mu - r_f + c_\phi \bar{\phi}}{\gamma_m \sigma^2 + c_\phi + \frac{d\lambda}{d\phi}}. \tag{16}$$

The first two terms in the numerator and the first term in the denominator represent the standard risk-return trade-off. The third term in the numerator and the second term in the denominator capture the effects of portfolio mandates. The mutual fund chooses a weighted

[6]Equations (11) and (15) are first-order approximations in λ. The full model in Section 3 uses the exact accounting, with trades clearing at market value, $s^{M'} = \phi Q/\tilde{P}$, and trading costs denominated in dollars, $\Lambda = 1/\left(1 + c_r(s^M - s^{M'})\right)$. Section B.2 shows that the properties below also hold in that setting.

average of the mean-variance optimal portfolio and the target portfolio $\bar{\phi}$. The final term in the denominator, $d\lambda/d\phi = \xi > 0$, reflects the strategic interaction with residual investors: increasing portfolio weight ϕ raises the firm's current valuation, which in turn reduces the marginal benefit of further increasing ϕ.

Because expected returns depend on the fund's choice only through λ in (9), combining (16) with (15) solves the trading stage in closed form given investment I:

$$\phi^* = \rho\,\phi^0, \qquad \lambda^* = \xi\phi^* - c_r s^M, \qquad \rho \;\equiv\; \frac{\gamma_m\sigma^2 + c_\phi}{\gamma_m\sigma^2 + c_\phi + 2\xi} \;\in\; (0,1), \tag{17}$$

where ϕ^0 is the weight the fund would choose if its trades moved no prices. The footprint ξ enters the denominator of ρ for two reasons: equilibrium price pressure lowers the return the fund expects, and the fund internalizes that its own buying creates that pressure.

The factor ρ is the fund's *flow pass-through*. Ownership is $s^{M'} = \phi Q/P$, so a one percent inflow raises the fund's stake by only ρ percent, the fund meeting the rest by lowering its weight and holding more of the risk-free asset. The pass-through is itself endogenous: it falls in fund size and rises in fundamental value, both through ξ.

Investment feeds back into price pressure. Along the fund's optimal response (17),

$$\frac{d\lambda}{dI} = -\frac{\alpha}{I}\,\lambda_Q < 0, \qquad \lambda_Q \;\equiv\; \left.\frac{\partial\lambda^*}{\partial\log Q}\right|_I \;=\; \rho\,\xi\phi^* \;=\; \rho\,c_r s^{M'}, \tag{18}$$

where higher investment requires supplying more shares, which reduces price pressure by absorbing some of the excess demand. The semi-elasticity λ_Q is the pressure a one-percent flow creates, and it factors transparently: the trading cost c_r, times the stake the fund holds, times the pass-through ρ.

The firm. The firm's optimality condition is

$$\hat{\beta}\alpha I^{\alpha-1} = 1 + \zeta, \qquad \zeta \;\equiv\; \mathbb{1}_{\{H>0\}}\,\frac{c_h H}{K}\,e^{-2\lambda}\left(1 + \alpha\,\frac{H}{I}\,\lambda_Q\right), \tag{19}$$

where the left-hand side is the standard marginal benefit of investment and the right-hand side is the marginal cost. When the firm does not issue, $\zeta = 0$ and the marginal cost is 1. When it issues, it incurs issuance costs. As with the mutual fund, the final factor reflects that the firm internalizes how investment affects price pressure and issuance costs, and it exceeds one because $d\lambda/dI < 0$. Efficient investment would equate the discounted marginal product of capital to its price, here unity, so ζ is the firm's investment wedge, the proportional gap between the two. Issuance costs make ζ positive for any firm that raises equity, and price pressure scales it.

2.3 Four Properties of the Elasticity

To make sure that equity issuance is relevant, I focus on parameter values such that in equilibrium the firm issues equity ($H^* > 0$) and the fund holds a positive position ($\phi^* > 0$).

Property 1 (Attenuation Bias)**.**

1. *Both equilibrium price pressure Λ^* and fundamental value P^* increase in mutual fund size Q.*

2. *Let $\eta \equiv 1/(c_r s^R)$ denote residual investors' elasticity with respect to price pressure. An estimator that instruments the observed change in market value with the fund flow shock recovers not η but*

$$\hat{\eta} = (1 - \pi)\,\eta \;<\; \eta, \qquad \pi \;\equiv\; \frac{dp^*/d\log Q}{d\tilde{p}^*/d\log Q} \;\in\; (0, 1), \tag{20}$$

the share of the observed price response attributable to fundamental value.

Property 1 states the paper's identification problem and locates the bias. When price pressure rises due to mutual fund inflows, equity issuance costs fall, encouraging higher investment and improving fundamental value. Nothing is wrong with the instrument: the flow shock is exogenous, and it moves price pressure as intended. The problem is the endogenous variable it instruments. Because the firm's issuance response raises fundamental value at the same time, observed market value moves by more than price pressure alone, so the estimator divides a correct numerator by an inflated denominator and reports investors as less responsive than they are. The attenuation factor π is the fundamental share of the price response, which the structural estimation in Section 4 recovers directly.

Property 2 (Source-Dependent Elasticities)**.** *For the residual investors, the elasticity with respect to price pressure Λ is negative, while the elasticity with respect to fundamental value P is zero. For the mutual fund, the elasticity with respect to price pressure Λ is negative, while the elasticity with respect to fundamental value P is positive.*

Property 2 says that a demand elasticity is defined only relative to the source of the price change it responds to, so naming a single elasticity is not enough. For residual investors the distinction is by construction: they trade only to correct deviations from fundamental value. For the mutual fund no such restriction applies, yet the two elasticities still differ in sign. When the fund expects a sharp correction in price pressure, and hence a reduction in returns, it lowers its portfolio weight. A rise in fundamental value need not reduce returns: it shrinks the fund's relative footprint $\xi = c_r Q/P$, so price pressure falls and expected returns rise. Where fundamental value changes for other reasons, with different implications for expected returns, the elasticity with respect to fundamental value can be positive or negative[7].

Property 3 (Size-Dependent Mitigation)**.**

1. *When fund size Q increases, the mutual fund optimally reduces its portfolio weight ϕ^*, and the firm optimally increases investment I^*, both partially offsetting the increase in price pressure Λ^*. Formally,*

$$\underbrace{\frac{d\Lambda^*}{dQ}}_{>0} = \underbrace{\frac{\partial\Lambda^*}{\partial Q}}_{>0} + \underbrace{\frac{\partial\Lambda^*}{\partial\phi^*}}_{>0} \cdot \underbrace{\frac{d\phi^*}{dQ}}_{<0} + \underbrace{\frac{\partial\Lambda^*}{\partial I^*}}_{<0} \cdot \underbrace{\frac{dI^*}{dQ}}_{>0} .$$

2. *The fund's mitigation strengthens with the shock. The fund's proportional retreat and the response of its ownership are*

$$\frac{d\log\phi^*}{d\log Q} = -\left(1-\rho\right), \qquad \frac{d\log s^{M'}}{d\log Q} = \rho, \tag{21}$$

and the pass-through ρ is strictly decreasing in the fund's footprint ξ, hence in fund size.

Property 3 shows the strategic interactions that attenuate price responses. Higher price pressure reduces expected returns for the mutual fund, leading it to reduce its portfolio weight, while lower issuance costs incentivize the firm to invest more. Both actions absorb part of the excess demand. These mitigating forces also determine how much of a demand shock reaches the real economy. For small shocks, issuance costs for the firm and benchmarking constraints for the fund still allow market prices to deviate from fundamental values. As shocks grow larger, the fund's mitigation strengthens: by (21), a larger fund has a larger footprint and

[7]In the full model in Section 3 below, larger firms have lower expected returns due to lower risk, implying a negative elasticity with respect to fundamental value.

therefore a smaller pass-through, so each additional percent of inflow raises ownership by less, and ownership changes are what price pressure is built from. The firm keeps acting as well, its investment easing price pressure. Whether that mitigation strictly strengthens with the shock, and so whether the price response ends up less than proportional, is a quantitative question the two-period model cannot settle, and Section 5 takes it up directly.

Property 4 (State-Dependent Price Pressure)**.** *Consider firms differing in initial capital K, each with an interior issuance equilibrium, facing a common fund size Q and holding initial ownership s^M fixed.*

1. *Capital-poor firms attract more price pressure. Because $P = \hat{\beta} I^{\alpha}$ is increasing in K, the fund's footprint $\xi = c_r Q/P$ is decreasing in K, and by (17) equilibrium λ^* is increasing in ξ.*

2. *Capital-poor firms exhaust that pressure faster. The rate at which a dollar of issuance erodes the pressure the firm is exploiting is*

$$\left|\frac{d\lambda}{dH}\right| = \frac{\alpha}{I}\,\lambda_Q, \qquad \frac{d\log|d\lambda/dH|}{d\log I} = -1 - \alpha\,(2\rho - 1) \;\in\; \big(-(1+\alpha),\, -(1-\alpha)\big), \quad (22)$$

   strictly negative for every $\xi > 0$, so the erosion rate is strictly decreasing in I and hence in K.

Property 4 says the price pressure a demand shock creates is state-dependent: the same flow lands differently across firms, so there is no single pass-through. Two forces shape the cross-section and pull against each other. A given fund position moves the price most in the smallest firms, because the fund's footprint is measured against firm size, so the fund holds a smaller weight there and retreats from it proportionally harder (part 1). But that same smallness lets the firm use less of the pressure it faces (part 2): each dollar issued raises fundamental value proportionally more when the firm is small, eroding the very pressure it is issuing into. This is the model's counterpart of issuance capacity. Here the state is capital alone, since the two-period firm has no debt; in Section 3 issuance capacity works through leverage, and expected returns carry a risk channel, so Section 5 traces this state-dependence over richer margins.

The four properties become statements about capital allocation through the wedge (19). Firms differ in log marginal product only through ζ, so the dispersion of ζ is the dispersion of marginal products, which lowers aggregate TFP (Section 6.2). Each property can then be read off a single expression. Property 1 and Property 2 both act on λ: the second settles which elasticity maps a demand shock into price pressure, and the first shows that a standard estimate understates it. Understating the elasticity overstates λ for any given flow, and so overstates every wedge. Property 3 acts on λ_Q, which enters (19) directly, so a larger shock contributes less than proportionally to the wedge. Property 4 acts on how price pressure is distributed across firms and on its multiplicative entry into (19): it governs the sign of the effect on dispersion, while the other three properties govern its magnitude.

The two-period model does not deliver magnitudes. Its state-dependence already pulls two ways: a given fund position moves the price most in the firms with the least capital and the largest wedge, a corrective pull, yet those same firms erode that pressure fastest. Which force dominates the cross-section the model cannot say. It has one firm, or at most a cross-section of otherwise identical firms indexed by K; it has no debt, so issuance capacity depends on size rather than leverage; it has two periods, so nothing that depends on the investor dynamics or on firms accumulating capital and debt over time can appear; and it has no risk channel, so the state-dependence of expected returns is absent by construction. Section 3 reinstates all four and defines the equilibrium, and Section 4 through Section 6 estimate it and quantify the dispersion of ζ.

3 Dynamic Model

This section extends the two-period model to a dynamic setting. The main limitation of the two-period model is that the firm side is highly simplified and cannot quantitatively capture the firm's investment, financing, and return dynamics. Therefore, it is difficult to assess the quantitative importance of fundamental value changes versus price pressure in driving empirical price responses, and the resulting implications for demand elasticity and capital misallocation. The dynamic equilibrium framework provides an environment to identify key parameters that would be difficult to estimate from reduced-form impulse responses alone and to evaluate the quantitative importance of the mechanisms.

3.1 Firms

Firms maximize their fundamental value, defined as the infinite sum of discounted payouts, through optimal investment and financing decisions, generalizing the two-period firm in Section 2. The costs of equity issuance and buyback depend on the firm's market valuation. When market value exceeds fundamental value, firms face lower equity issuance costs; conversely, undervaluation makes share buybacks less costly.

3.1.1 Technology

Firm i uses physical capital $K_{i,t}$ to produce $Y_{i,t}$:

$$Y_{i,t} = A_t X_{i,t} K_{i,t}^{\alpha}, \tag{23}$$

where A_t is the aggregate productivity and $X_{i,t}$ is the idiosyncratic productivity. The logarithm of aggregate productivity $a_t = \log A_t$ follows an AR(1) process,

$$a_{t+1} = (1 - \rho_a)\bar{a} + \rho_a a_t + \epsilon_{a,t+1},\ \epsilon_{a,t+1} \sim \mathcal{N}(0, \sigma_a^2), \tag{24}$$

where ρ_a, $\bar{a}$, and σ_a are the persistence, mean, and conditional volatility of the AR(1) process.

Define $x_{i,t} = \log X_{i,t}$ as the logarithm of idiosyncratic productivity, which follows an AR(1) process:

$$x_{i,t+1} = (1 - \rho_x)\bar{x} + \rho_x x_{i,t} + \epsilon_{x,i,t+1}, \tag{25}$$

where ρ_x and $\bar{x}$ are the persistence and mean of the AR(1) process. $\epsilon_{x,i,t+1}$ is an i.i.d. normal shock that follows $\epsilon_{x,i,t+1} \sim \mathcal{N}(0, \sigma_x^2)$, where σ_x is the conditional volatility.

Capital accumulation follows

$$K_{i,t+1} = (1 - \delta)K_{i,t} + I_{i,t}, \tag{26}$$

where δ is the depretiation rate, $I_{i,t}$ denotes investment, and $e^{\eta_{i,t+1}}$ is an i.i.d. capital quality shock.

Convex investment adjustment costs $G_{i,t}$ are given by

$$G_{i,t} = \frac{c_k}{2}\left(\frac{I_{i,t}}{K_{i,t}}\right)^2 K_{i,t}, \tag{27}$$

where cost parameter c_k determines the speed of adjustment.

3.1.2 Debt Financing

Each period, firms issue one-period debt L_t, which is to be repaid at $t+1$ at interest rate r_f. Following Hennessy and Whited (2005), collateral constraint is given by

$$L_{i,t+1} \leq \varphi K_{i,t+1}, \tag{28}$$

where $\varphi \in (0,1)$ denotes the borrowing capacity. When $L_t < 0$, the firm saves with interest rate $r_s = r_f - \kappa$, where $\kappa \in (0, r_f)$ denotes the wedge between borrowing and saving rates. Finally, $r_l = r_f \mathbb{1}_{\{L_t>0\}} + r_s \mathbb{1}_{\{L_t \leq 0\}}$ denotes the applicable interest rate.

3.1.3 Issuance, Repurchase, and Payout

With corporate taxes τ, the operating profit can be expressed as after-tax sales plus depreciation tax shield, net of investment, investment adjustment costs, and changes in debt financing:

$$E_{i,t} = (1-\tau)Y_{i,t} + \tau\delta K_{i,t} + \tau r_f L_t \mathbb{1}_{\{L_t>0\}} - I_{i,t} - G_{i,t} + L_{i,t+1} - (1+r_l)L_{i,t}. \tag{29}$$

If total operating costs exceed total profits, $E_{i,t} < 0$, firms can raise external equity,

$$H_{i,t} = \max\{0, -E_{i,t}\}, \tag{30}$$

which incurs costs

$$\Psi_{i,t} = c_{h0}K_{i,t} + \frac{c_{h1}}{2}\left(\frac{H_{i,t}}{\Lambda_{i,t}K_{i,t}}\right)^2 K_{i,t}, \tag{31}$$

where c_{h0} and c_{h1} are the fixed and variable issuance cost parameters and $\Lambda_{i,t}$ is the price pressure factor that fluctuates around 1 (an equilibrium outcome defined in Section 3.4). When a firm is overvalued ($\Lambda_{i,t} > 1$), the issuance size $H_{i,t}$ becomes relatively smaller compared to the firm's total market value, making issuance less costly. I adopt a reduced-form specification to capture the empirical relationship in Section 1.4.3, consistent with micro foundations in Baker and Wurgler (2002); Kim and Weisbach (2008); Bolton et al. (2013).

If the operating profit is positive, $E_{i,t} > 0$, following Warusawitharana and Whited (2016), the firm can pay out to shareholders by choosing any combination of dividend $D_{i,t}$ and share repurchase $B_{i,t}$ such that

$$D_{i,t} + B_{i,t} \leq E_{i,t}. \tag{32}$$

Dividend is subject to dividend tax τ_D, and buyback $B_{i,t}$ incurs costs

$$\Phi_{i,t} = c_{b0} K_{i,t} + \frac{c_{b1}}{2} \left(\frac{\Lambda_{i,t} B_{i,t}}{K_{i,t}} \right)^2 K_{i,t}, \tag{33}$$

where c_{b0} and c_{b1} are the fixed and variable repurchase cost parameters. Similar to the argument about issuance costs Ψ, when investor demand is weak and the firm is undervalued ($\Lambda_{i,t} < 1$), firms are more incentivized to repurchase stocks at a lower price.

Finally, the payout $O_{i,t}$ is operating profit minus issuance costs when operating profit is negative, or the combination of after-dividend tax dividend and repurchase net of repurchase costs when operating profit is possitive:

$$O_{i,t} = \begin{cases} E_{i,t} - \Psi_{i,t} & \text{if } E_{i,t} \leq 0, \\ (1 - \tau_D) D_{i,t} + B_{i,t} - \Phi_{i,t} & \text{if } E_{i,t} > 0. \end{cases} \tag{34}$$

Following Zhang (2005) I specify the stochastic discount factor (SDF) as

$$\log(M_{t+1}) = \log \beta - \gamma_t (a_{t+1} - a_t) - \frac{1}{2} \gamma_t^2 \sigma_a^2 \tag{35}$$

$$\gamma_t = \gamma_0 + \gamma_1 (a_t - \bar{a}), \tag{36}$$

where $\bar{a}$ is the long-run mean of aggregate productivity. Constant β determines the level of risk-free rate. Constants $\gamma_0 > 0, \gamma_1 < 0$ capture countercyclical price of risk.

Finally, the firm maximizes its continuation value

$$V_{i,t} = \max_{I_{i,t}, K_{i,t+1}} O_{i,t} + \mathbb{E}_t \left[M_{t+1} V_{i,t+1} \right] \equiv O_{i,t} + P_{i,t}, \tag{37}$$

subject to constraints (23) through (36).

3.2 Mutual Funds

There is one one-period living mutual fund for each firm, holding equity positions and earning a fixed fee based on their AUM. Fund AUM is influenced by returns from equity holdings and exogenous fund flow shocks. Idiosyncratic fund flow shocks introduce cross-sectional heterogeneity in fund sizes, providing the variation needed for identification. Position adjustment costs prevent funds from achieving the optimal portfolio allocations.

A mutual fund i manages its inherited AUM $Q_{i,t}$ by allocating its portfolio weights ϕ_{t+1} in firm i and a risk-free asset to gain portfolio returns $R^M_{i,t+1}(\phi_{t+1}) = R_f + \phi_{i,t+1}(R_{i,t+1} - R_f)$.

Let lower case letters denote the logarithm of their upper case counterparts, the size of the mutual fund evolves following

$$\tilde{q}_{i,t+1} = q_{i,t} + r^M_{i,t+1}.$$

The mutual fund manager will then calculate the fixed fee based on $\tilde{q}_{i,t+1}$.

Following the same setup as the SDF [8], up to a constant scaling factor (e.g. a fixed 2% management fee), the fund manager's discounted payoff is then

$$\frac{\beta}{1-\gamma_m}\mathbb{E}_t\left[\tilde{Q}^{1-\gamma_m}_{i,t+1}\right] = \frac{\beta}{1-\gamma_m}\mathbb{E}_t\left[e^{(1-\gamma_m)\tilde{q}_{i,t+1}}\right]. \tag{38}$$

For tractability, I assume that after the returns are realized and the fund managers collect their fees, mutual fund clients will withdraw the returns from the funds and rebalance their portfolios across mutual funds, so that the AUM for each mutual fund available for investment is the sum of an aggregate component and an idiosyncratic component:

$$q_{i,t+1} = \bar{q} + \omega_{i,t+1}, \tag{39}$$

where $\bar{q}$ is the average size of the mutual fund industry and $\omega_{i,t+1}$ is specific to mutual fund i, denoting its deviation from average size.

The idiosyncratic size deviation $\omega_{i,t+1}$ follows an AR(1) process that captures the persistence of each fund's size

$$\omega_{i,t+1} = (1-\rho_\omega)\bar{\omega} + \rho_\omega\omega_t + \epsilon_{\omega,i,t+1}, \tag{40}$$

where $\bar{\omega}$ and ρ_ω are the long run mean and persistence of idiosyncratic fund size. Idiosyncratic fund size shock $\epsilon_{\omega,i,t+1}$ is drawn from an i.i.d. normal distribution with conditional volatility σ_ω.

The amount of proceeds invested to firm i is the portfolio weight for next period $\phi_{i,t+1}$

[8] With a CRRA preference, the manager's disconted utility is $\beta\tilde{Q}^{1-\gamma_m}_{i,t+1}/(1-\gamma_m)$. Taking the first order derivative with respect to $\tilde{Q}_{i,t+1}$ gives us the marginal utility $\beta\tilde{Q}^{-\gamma_m}_{i,t+1}$. The log marginal utility is therefore $\log\beta - \gamma_m\tilde{q}_{i,t+1}$, consistent with (36).

multiplied by fund size today $Q_{i,t}$. Divide it by the total market value $\tilde{P}_{i,t}$, we get the mutual fund ownership for next period:

$$s^M_{i,t+1} = \frac{Q_{i,t}\phi_{i,t+1}}{\tilde{P}_{i,t}}. \tag{41}$$

A standard Taylor approximation yields $r^M_{i,t+1} \approx r_f + \phi_{i,t+1}(r_{i,t+1} - r_f) + \frac{1}{2}\phi_{i,t+1}(1-\phi_{i,t+1})\sigma^2_{i,t}$, where $\sigma^2_{i,t} = \mathrm{Var}_t(r_{i,t+1})$ is the conditional variance of returns. Given normally distributed returns and dropping the constant terms, the fund manager's discounted payoff can be rewritten as

$$u^M = \phi_{i,t+1}(\mu_{i,t} - r_f) + \frac{1}{2}\phi_{i,t+1}(1-\phi_{i,t+1})\sigma^2_{i,t} + \frac{1}{2}(1-\gamma_m)\phi^2_{i,t+1}\sigma^2_{i,t}, \tag{42}$$

where $\mu_{i,t} = \mathbb{E}_t[r_{i,t+1}]$ is the conditional expected return.

Fund managers face two portfolio adjustment constraints. First, managers avoid deviating too far away from targeted positions $\bar{\phi}$ due to investment mandates, risk management requirements, and index benchmarking. This constraint is captured by the portfolio adjustment cost $\frac{c_\phi}{2}(\phi_{i,t+1} - \bar{\phi})^2$, where c_ϕ controls the size of the cost. Second, executing trades is costly, leading managers to spread trades over time (Kyle, 1985). Since the model features one-period living funds, I capture this smoothing incentive by an ownership adjustment cost $\frac{c_s}{2}(s^M_{i,t+1} - s^M_{i,t})^2$, where c_s controls the cost of deviating from previous period's ownership.

Finally, the maximization problem can be written as

$$\max_{\phi_{i,t+1}} \; u^M(\phi_{i,t+1}) - \frac{c_\phi}{2}(\phi_{i,t+1} - \bar{\phi})^2 - \frac{c_s}{2}(s^M_{i,t+1}(\phi_{i,t+1}) - s^M_{i,t})^2 \tag{43}$$

subject to constraints (39) through (42). Importantly, as is analyzed in detail in Section 3.4, conditional returns and volatilities that affect payoff u^M are all functions of the choice variable $\phi_{i,t+1}$.

3.3 Residual Investors

Each firm is paired with a continuum of one-period-lived residual investors representing all other shareholders. As in Section 2, they trade against mispricing subject to a quadratic trading cost with parameter c_r. The level-price counterpart of (12) (first-order equivalent for small λ) is

$$\max_{s^R_{i,t+1}} \; s^R_{i,t+1}(P_{i,t} - \tilde{P}_{i,t}) - \frac{c_r}{2}\left(s^R_{i,t+1} - s^R_{i,t}\right)^2 \tilde{P}_{i,t}, \tag{44}$$

with first-order condition

$$s_{i,t+1}^R = s_{i,t}^R + \frac{P_{i,t} - \tilde{P}_{i,t}}{c_r \tilde{P}_{i,t}} \tag{45}$$

and demand elasticity $1/(c_r s_{i,t}^R)$ with respect to price pressure.

3.4 Market Clearing and Price Pressure

Substituting (45) into market clearing, $s_{i,t+1}^R + s_{i,t+1}^M = 1$, yields the exact (non-linearized) counterpart of (15):

$$\tilde{P}_{i,t} = \frac{1}{1 + c_r \left(s_{i,t}^M - s_{i,t+1}^M \right)} P_{i,t} \equiv \Lambda_{i,t} P_{i,t}, \tag{46}$$

where $\Lambda_{i,t}$, fluctuating around 1, denotes the degree of price pressure.

3.5 Equilibrium

Let S denote the vector of state variables $(A_t, X_{i,t}, Q_{i,t}, L_{i,t}, K_{i,t}, s_{i,t}^R)$. With some abuse of notation, the recursive equilibrium is defined as

1. a cum-dividend fundamental value function for firms $V(S)$ (and hence ex-dividend value $P(S)$);
2. a set of policy functions for firms $K'(S), L'(S)$;
3. portfolio holding decisions for mutual funds $\phi'(K', L', S)$;
4. price pressure function for firms $\Lambda(S)$;

such that in each period

1. Taking $\phi'(K', L', S)$ as given, the firm chooses $K'(S)$ and $L'(S)$ to maximize $V(S)$
2. Observing K', L', taking $\Lambda(S)$ and $V(S)$ as given, mutual funds calculate expected returns and variances and choose portfolio weights $\phi'(K', L', S)$.
3. Given K', L' and $s^{M'}$, residual investors choose $s^{R'}$.

4. Markets clear for each state. Realized price pressures are consistent with $\Lambda(S)$,

$$\frac{1}{1 + c_r \left[(1 - s^R) - \frac{Q\phi'\left(K'(S), L'(S), S\right)}{\Lambda(S) V(S)} \right]} = \Lambda(S) \tag{47}$$

The timing of events within each period is as follows:

1. All shocks are realized and assets are reshuffled among newly established mutual funds.
2. Firms make investment and financing decisions.
3. Mutual funds choose portfolio weights.
4. Residual investors determine their positions.
5. Transactions are executed, markets clear, and prices are realized.

Details of the computation algorithm are provided in Appendix C. Several key features help make this system numerically tractable. First, the sequential nature means early movers only need to form beliefs about late movers' decisions rather than solving the optimization for all potential strategy combinations. Second, firms' investment and borrowing decisions determine both next-period state variables and current-period issuance and buyback decisions. This approach prevents the state space for investors, whose decisions depend on the firm's actions, from becoming unmanageable. Third, given firm and mutual fund actions, residual investors' optimal decisions have closed-form solutions that, combined with market clearing, directly yield the price pressure rule.

Mutual funds calculate the return of firm i by

$$R_{i,t+1} = \frac{\tilde{P}_{i,t+1} + O_{i,t+1}}{\tilde{P}_{i,t}}. \tag{48}$$

While at the optimum, it is still true that

$$V_{i,t} = O_{i,t} + \mathbb{E}_t[M_{t,t+1} V_{i,t+1}] \implies 1 = \mathbb{E}\left[M_{t,t+1} \frac{V_{i,t+1}}{V_{i,t} - O_{i,t}} \right] \tag{49}$$

$\mathbb{E}_t[M_{t,t+1} R_{i,t+1}] \neq 1$ due to the price pressure terms.

The equilibrium return differs from traditional dynamic firm models (e.g., Zhang, 2005), due to stock market segmentation, where each mutual fund and residual investor can only invest in their assigned firm. This segmentation is similar to the preferred-habitat literature on the term structure of interest rates (e.g., Vayanos and Vila, 2021). While residual investors follow the market-wide SDF (36), their trading is constrained by trading costs. Consequently, firm pricing reflects the baseline market-wide marginal utility adjusted by a factor measuring residual investor's marginal trading costs.

4 Calibration and Estimation

This section presents the calibration and estimation of model parameters by targeting data moments. I also employ instrumental variable estimation within the model-simulated economy to illustrate the source and magnitude of the bias from ignoring endogenous firm actions.

To maintain consistency with the empirical analysis, the model is solved at a quarterly frequency. I calculate the model-implied moments by simulating 10 samples and reporting the cross-sample average. Each sample contains 3,000 firms simulated over 1,000 quarters. To mitigate the effects of initial conditions, I discard the first 400 quarters and treat the remaining simulated data as drawn from the economy's stationary distribution.

4.1 Calibration

Table 5 reports the calibrated parameters in the baseline model. To avoid complicating the estimation with an excess of parameters, I calibrate the parameters that do not directly affect the demand and supply dynamics by first estimating them outside of the dynamic model. If a direct estimation is not possible, the parameters are set by matching some selected moments or using values reported in previous studies.

Table 5 Here

Aggregate parameters. The long-run mean of mutual fund size $\bar{q}$ is calibrated such that the average mutual fund AUM is one-third of average firm market value. The inverse of risk-free rate β is set to be 0.995 at quarterly frequency. The risk aversion parameters γ_0 and γ_1 are set to match an average aggregate stock market return of 3% per quarter and an annual Sharpe ratio of 0.35.

Mutual Funds. The persistence and volatility of idiosyncratic flow shocks are directly estimated from data. On average, mutual funds own 20% of a firm. The firm's market value is three times the size of the fund. This implies that targeted position $\bar{\phi} = \bar{s} \times (\bar{\tilde{P}}/\bar{Q}) = 0.2 \times 3 = 0.6$. The risk aversion coefficient γ^M is set to 2.0, close to the median value of 2.5 estimated by Koijen (2014), and also roughly consistent with the targeted position $\bar{\phi}$[9].

Firms. Following Zhang (2005) and Belo et al. (2019), the persistence ρ_x and the volatility σ_x of productivity shocks are set to 0.97^3 and 0.19, respectively. The curvature of the production function α, quarterly depreciation rate δ, corporate tax rate τ are standard parameters, set at 0.65, 0.03, and 0.3, respectively. Following Warusawitharana and Whited (2016), dividend tax rate τ_D is 0.15. The collateral constraint $\phi = 0.8$ and the quarterly saving-borrowing wedge $\kappa = 0.005/4$ follow Livdan et al. (2009); Belo et al. (2019); Choi et al. (2025).

4.2 Estimation

I estimate the seven key parameters that directly affect supply and demand dynamics using indirect inference.

On the firm side, the issuance cost parameters c_{h0} and c_{h1} and the repurchase cost parameter c_{b0} and c_{b1} directly determine both the average level of net equity issuance and the reaction of equity issuance to cost changes caused by price pressure. The investment adjustment cost parameter c_k governs the intertemporal Euler equation and implicitly determines both issuance and repurchase decisions.

For mutual funds, the portfolio deviation cost c_ϕ captures benchmarking intensity. In the limit where c_ϕ approaches infinity, mutual funds are fully passive so their portfolio ϕ stays constant, and all changes in s^M are driven solely by the fluctuation in their size.

Finally, for given changes in mutual fund holdings, the residual investor trading cost c_r directly controls the size of price pressure in (46). As derived in Section 3.3, the inverse of c_r represents the semielasticity with respect to price pressure.

I use eleven target moments for the indirect inference estimation.

The first five moments capture the dynamic interaction between mutual funds, residual investors, and firms. First, I target the IRF of returns at $h = 0$ and $h = 1$[10]. The initial response

[9] Absent the position adjustment costs and price impacts, the optimal position of a mutual fund with CRRA utility is $\phi = (\mu - r_f)/(\gamma\sigma^2)$, where $\mu - r_f$ is the risk premium and σ^2 is the variance of returns. With an annual risk premium of 6% and a volatility of 20%, the optimal position is 0.75 when $\gamma = 2$.

[10] In the model the IRF of returns is defined as the change in log market value $\Delta \log \tilde{P}$ for easier comparison and decomposition in Section 4.4. Empirically the IRFs of $\Delta \log \tilde{P}$ and R are similar. This is because $\Delta \log \tilde{P} = \log\left(1 + \frac{\tilde{P}_{t+1} - \tilde{P}_t}{\tilde{P}_t}\right) \approx \frac{\tilde{P}_{t+1} - \tilde{P}_t}{\tilde{P}_t} = R_{t+1} - 1 - \frac{O_{t+1}}{\tilde{P}_t}$, and the payout term $O_{t+1}/\tilde{P}_t$ is an additive level not

($h = 0$) is directly related to the cost parameters. The return response at $h = 1$ captures dynamic adjustments. Stronger return reversals occur when initial reactions are primarily driven by price pressure or when mutual funds pay less effort to smooth their position changes. I focus on the first two periods because the model lacks mechanisms for significant long-term responses, and empirical evidence suggests that longer-term responses are not significantly different from zero. Second, I use the IRF of NEI at $h = 0$. High firm issuance and repurchase costs reduce firms' responsiveness to price pressure. Third, I target the variance of changes in mutual fund ownership share (Δs^M), which governs the extent of active portfolio rebalancing and determines the magnitude of return responses. Fourth, I target the variance of returns. Given the volatility of productivity shocks and fund flow shocks, it reveals market participants' capacity to absorb shocks.

The next six moments discipline parameters related to firm technology and financing. To match the dynamics of NEI, I calculate the mean and variance of NEI conditional on being non-negative or non-positive.[11]. I also target the regression coefficient of $SALE/K$ on I/K. A large coefficient indicates investment dynamics driven primarily by productivity shocks, while a small coefficient suggests a more important role for fund flow shocks, which affect issuance costs. Finally, I target the variance of investment rate, which directly enters financing and investment costs.

Since this model is overidentified, I construct the weighting matrix for the objective function using the influence function approach in Erickson and Whited (2002). Let x denote the data, b denote the vector of parameters, g denote the difference between data and simulated moments, and $\hat{W}$ denote the weighting matrix. The indirect inference estimator of b is then defined as the solution to the minimization of

$$\hat{b} = \arg\min_{b} g(x, b)' \hat{W} g(x, b).$$

Computational details are provided in Appendix C. Table 6 presents the estimation results.

Table 6 Here

Panel A of Table 6 reports the estimated parameters. The key parameter of interest is

directly affected by the shock realized at t, so it drops out of the impulse response.

[11]Zero-NEI observations contribute to both averages. Since many observations are small but non-zero, this approach captures NEI levels without imposing arbitrary thresholds for "active" versus "inactive" NEI, which could introduce biases to the extensive margin of NEI. Since the firm side is overidentified, other moments help indirectly identify relevant parameters.

the trading cost for residual investors, c_r. Since residual investors on average hold 80% of a firm's shares, their elasticity with respect to price pressure, $1/(c_r s^R)$, averages approximately 1.6 across the model's stationary cross-section of s^R. This is consistent with the empirical evidence provided by Schmickler and Tremacoldi-Rossi (2022).

Panel B of Table 6 compares the moments calculated from the model with those from the data. Despite overidentification, the model fits the data well overall. The return reversal at $h = 1$ is stronger than in the data, which is expected given that one-period-living investors generate more immediate and pronounced reactions. The discrepancy is weaker in the longer horizon. For example, the cumulative return from $h = 1$ to $h = 3$ is -0.81 in the data and -1.66 in the model.

While the standard errors provide initial evidence on local identification, I further investigate the sources of identification by computing the sensitivity matrix of $\hat{b}$ to $\hat{g}(b)$ using $-(G'WG)^{-1}G'W$, where G is the Jacobian, following Andrews et al. (2017). As the raw sensitivity measures are not scale invariant, I standardize the sensitivity matrix by scaling each element with the square root of the ratio of moment variance to parameter variance to facilitate interpretation. Table 7 presents the local elasticities of parameter estimates with respect to the targeted moments, calculated at estimated parameters.

Table 7 Here

The sensitivity matrix highlights why the model's parameters must be estimated jointly. The variance of returns responds to all seven parameters (magnitudes 0.124 to 0.492), including firm-side cost parameters such as c_{b0} (0.211) and c_{b1} (-0.211); the variance of investment likewise responds to the residual investor cost c_r (-0.139) alongside the firm-side cost parameters. Fund benchmarking intensity c_ϕ and residual investor cost c_r both shape price pressure but are identified through different moments. c_r's largest loadings are on the return response (Return IRF $h = 0$: 0.643; $h = 1$: -0.446; Var(R): -0.492), with a stronger return response associated with a higher estimated c_r and thus a lower demand elasticity, while c_ϕ's largest loading is on the variance of ownership changes (Var(Δs^M): 0.650), a moment c_r barely moves (-0.148). On the firm side, the capital adjustment cost c_k and the issuance/repurchase cost levels c_{h0}, c_{b0} are most strongly identified by NEI-related moments (NEI IRF $h = 0$: c_k 0.692, c_{b0} 0.716; c_{h0}'s largest loading is Var(I/K), 0.531). These NEI moments are themselves jointly determined by all four issuance and repurchase cost parameters rather than by one side alone: the mean and variance of NEI conditional on issuing respond about as strongly to the repurchase costs c_{b0}, c_{b1} (-0.199, 0.200 and 0.469, -0.479) as to the issuance costs c_{h0}, c_{h1} (-0.187, 0.232 and 0.492, -0.705), and it is this joint response that pins down the curvature parameters c_{h1}, c_{b1} specifically through the variance of NEI conditional on issuing or repurchasing (-0.705, -0.479).

4.3 Return Response Decomposition

The observable return response to a fund flow shock combines two distinct components: the transitory price pressure wedge and the change in fundamental value it triggers. Following (46),

$$d \log \tilde{P} = d \log \Lambda + d \log P, \tag{50}$$

and Table 8 reports these two components of the initial return response. The price pressure response (1.39%) is weaker than the return response (1.67%) at $h = 0$ due to improved fundamental value following NEI responses. The gap between the observable return response and the unobservable price pressure response is the source of bias in the demand elasticity estimate. To explore heterogeneity across firms, I independently sort firms into terciles by beginning-of-period Tobin's Q and by beginning-of-period leverage, and report the four groups formed by intersecting the top and bottom terciles of each sort.

Table 8 Here

In the full sample, 17% of the initial return response is driven by improvements in fundamental value, while the remaining 83% is attributed to price pressure. There is substantial heterogeneity across firms. Consistent with the empirical findings in Table 4, firms with high Tobin's Q exhibit both stronger return responses and a larger fundamental value share.

Table 4's unconditional leverage sort looks puzzling because return and NEI responses do not move together: high-leverage firms show a stronger return but a weaker NEI response than low-leverage firms (2.243%, 0.147% vs. 1.624%, 0.229%). The model resolves this as a composition of two regimes that run in opposite directions depending on firms' growth opportunities. Among low-Q firms, high leverage is associated with a larger fundamental value response, consistent with cheap issuance being used mainly to pay down debt when investment opportunities are limited. Among high-Q firms, the pattern reverses: low leverage is associated with a larger fundamental value response, consistent with financing flexibility being used to fund expansion when investment opportunities are abundant. This composition of offsetting forces across firms with different growth opportunities is analyzed further in Section 5.

4.4 Recovering the Corrected Elasticity: An Instrumental Variable Comparison

This section delivers the paper's two headline elasticity numbers: instrumenting observed price changes, the standard approach, recovers a biased estimate of 1.3, while instrumenting price pressure directly recovers the corrected elasticity of 1.6. I estimate both demand elasticities, for residual investors and for mutual funds, using an instrumental variable approach on the model-simulated economy.

To estimate the demand elasticity for residual investors, I need a residual supply shock that is orthogonal to the residual investor demand (45). In the model, fund flow shocks are demand shocks independent of residual investor demand by construction, seemingly satisfying the traditional exclusion restriction. However, as shown in Section 4.3, this approach still yields biased results in our context because we cannot disentangle changes in firm value driven by price pressure Λ from those driven by fundamental value P, introducing inevitable omitted variable bias.

To illustrate this potential bias, I conduct instrumental variable estimation using the model-simulated economy, as reported in Panel A of Table 9. I regress changes in log ownership by residual investors against different measures of price changes, using fund flows as instruments.

Table 9 Here

In columns 1 and 2, the independent variable is changes in log market value $d\log\tilde{P}$. Without instrumenting, the estimate is positive, illustrating the traditional omitted variable bias. When firm fundamentals change, mutual funds endogenously rebalance their portfolios to achieve the optimal return-variance trade-off, so the estimate represents a weighted average of optimization responses by firms, mutual funds, and residual investors. In Column 2, using fund flow shocks as instruments yields a demand elasticity estimate of approximately 1.3.

Columns 3 and 4 use changes in log price pressure $d\log\Lambda$ as the independent variable, which is unobservable in actual data but can correctly remove the endogenous fundamental value changes. Without instruments (column 3), the estimate is already -0.6, reflecting that residual investors reduce holdings in response to positive price pressure to mitigate value deviation. With proper instrumenting, Column 4 estimates the true demand elasticity of 1.6, about 20% higher than the estimate in Column 2.

The specification in columns 3-4 remain slightly misspecified. Following (45), the correct specification should use deviations from fundamental value $(P_{i,t} - \tilde{P}_{i,t})/\tilde{P}_{i,t}$ instead of changes in price pressure. Columns 5 and 6 present the estimates using this specification. With or without instruments, the estimates are of similar magnitude and close to the estimate in Column 4, indicating that the bias from using changes in price pressure instead of the deviation measure is small.

I estimate demand elasticity for mutual funds in Panel B of Table 9. Following Koijen and Yogo (2019), the demand curve is downward sloping with respect to firm market value (rather than price pressure) because larger firms on average have lower expected returns. The demand elasticity estimate thus shows how the size premium is reflected in mutual funds' portfolio choices, which is not directly related to the price pressure. In this case, productivity shocks help firms grow but do not directly affect mutual fund demand, satisfying the exclusion restriction.

Columns 1 to 3 provide the reduced form, first stage, and IV estimates using changes in log market value. The first stage in Column 1 is biased upward. While mutual funds do cut holdings when firms grow bigger, they also increase holdings due to positive fund flow shocks, which rasies the market value through price pressure. With proper instrumenting, the IV estimate in Column 3 is -0.7, close to elasticity estimates surveyed in Gabaix and Koijen (2022). This estimate is still biased because when mutual funds adjust their holdings following productivity shocks, they also create price pressure.

Columns 4 to 6 use changes in log fundamental value, which is unobservable in actual data but can correctly remove the endogenous price pressure changes. Comparing Columns 4 and 6, the bias in the first stage is now small because biases introduced by price pressure are already removed. The elasticity estimate in Column 6 is smaller in magnitude than the estimate in Column 3, suggesting that mutual funds are more sensitive to price pressure increases than to fundamental value increases. This makes sense because the reduction in expected returns from price pressure should be sharper than that from fundamental value increases. Without additional instruments in the model, we cannot separately identify their demand elasticity with respect to price pressure as they are the ones introducing price pressure in the first place.

5 Mechanism

Section 4 established a corrected demand elasticity of 1.6, versus a biased estimate of 1.3, and showed that 17% of the initial price response reflects firms' fundamental-value improvement rather than price pressure. Section 2 showed analytically why responses to a price change should depend on its source, its size, and the firm's state; this section shows how these same three forces operate in the full dynamic model's policy functions, generating both the numbers above and the diminishing marginal price effects documented below.

5.1 Firm Optimality Conditions

The firm's investment, capital, and leverage choices are pinned down by three optimality conditions that make the price-pressure channel explicit: alongside the standard marginal costs and benefits of investing and borrowing, each condition carries a term through which firms internalize their effect on mutual fund trading and, through it, their own cost of capital. Let q and μ be the Lagrange multipliers for the capital accumulation (26) and borrowing constraint (28). The first-order conditions with respect to I, K', and L' are

$$-\frac{\partial O}{\partial I} = q \tag{51}$$

$$q = \mathbb{E}\left\{M'\left[\frac{\partial V'}{\partial K'}\right]\right\} + \mu\varphi + \frac{\partial O}{\partial \Lambda}\frac{\partial \Lambda}{\partial \phi'}\frac{\partial \phi'}{\partial K'} \tag{52}$$

$$\mu - \mathbb{E}\left\{M'\left[\frac{\partial V'}{\partial L'}\right]\right\} = \frac{\partial O}{\partial E}\frac{\partial E}{\partial L'} + \frac{\partial O}{\partial \Lambda}\frac{\partial \Lambda}{\partial \phi'}\frac{\partial \phi'}{\partial L'}. \tag{53}$$

The left-hand side of (51) represents the marginal cost of investment: investing more reduces current payout. The first two terms on the right-hand side are standard marginal benefits of capital accumulation. More capital increases future production capacity and relaxes the borrowing constraint. The last term reflects the novel firm-investor interaction: firms recognize that their investment decisions affect mutual fund trading through $\frac{\partial \phi'}{\partial K'}$, which in turn influences their cost of capital and payout $\frac{\partial O}{\partial \Lambda}\frac{\partial \Lambda}{\partial \phi'}$.

Similarly, the left-hand side of (53) represents the marginal cost of borrowing: potentially binding borrowing constraints and future debt repayment. The marginal benefit on the right-hand side includes higher current cash flows, and the effect of borrowing on mutual fund trading and thus payout costs through $\frac{\partial \phi'}{\partial L'}$.

5.2 Equilibrium Price Formation

The full equilibrium price-formation process shows that price pressure alone does not pin down expected returns: how price pressure and expected returns co-move depends on the shape of both mutual funds' and firms' equilibrium responses, not on the magnitude of price pressure by itself. Figure 3 illustrates this equilibrium price formation process for a firm seeking to increase investment. Following backward induction[12], the top panel shows equilibrium expected returns and volatility as functions of mutual fund portfolio weights, given optimal firm decisions. When mutual funds increase weights, current price pressure rises, reducing the expected returns. For this firm, conditional volatility also decreases, creating a trade-off between lower returns and lower volatility for mutual funds. However, since the volatility variation is relatively muted, the trade-off between returns and adjustment costs in (43) likely dominates.

Figure 3 Here

Moving backward in the analysis, firm decisions directly affect conditional expected returns and volatility by changing both current and future fundamental values. These changes influence mutual funds trade-offs, which affect price pressure, and feed back into expected returns and volatility. The middle panel shows how returns and volatility depend on firm investment decisions, taking mutual fund portfolio responses as given. The dashed black line indicates current capital level, while the dotted red line shows equilibrium optimal capital for next period. Firm actions have a particularly large impact on volatility.

Considering both steps, the bottom panel plots equilibrium price pressure and mutual fund portfolio weights as functions of firm investment decisions. As shown in (46), price pressure closely tracks mutual fund holdings. However, higher price pressure does not automatically translate into lower expected returns. The range of movement in μ (the left panel in the second row) is significantly larger than the range of movement in Λ (the left panel in the bottom row), and the shape of the two curves differ significantly. This is exactly the equilibrium chain that the state- and size-dependence results below trace out across firms and shock sizes.

[12]The last step where residual investors form their portfolios follows (45).

5.3 Response to Fund Flow Shocks

The equilibrium price-formation chain traced above operates for every firm, but how strongly it operates depends on the firm's state. Figure 4 plots policy functions against idiosyncratic mutual fund size Ω, weighted by the model's stationary distribution over firm states. The Overall row shows the baseline mechanism: mutual funds cut their next-period portfolio weight ϕ' in anticipation of the price correction that a larger shock implies, but portfolio adjustment costs limit how far they can unwind, so mutual fund ownership and price pressure Λ still rise with shock size. Net equity issuance rises in tandem, since the same price pressure funds are trading against makes new equity temporarily cheap for firms to issue, and firms put the proceeds to both uses: investment I/K rises and next-period leverage L'/K' falls modestly.

Figure 4 Here

Leverage is one state variable that reshapes this response. High-leverage firms are more prone to price fluctuations and risk. This shows up jointly as a sharper cut in mutual funds' portfolio weight ϕ' (from 0.62 to 0.25, versus 0.55 to 0.41 for low-leverage firms) and a larger rise in price pressure Λ. Firms' ability to exploit that price pressure still diverges sharply by leverage, even though investment rates I/K track closely across the split. Low-leverage firms respond elastically: net equity issuance is steep and sign-switching, turning to repurchases when the fund is small and to heavy issuance once the fund is large. High-leverage firms would also like to issue as price pressure rises. But their greater exposure to price fluctuations makes issuance self-defeating: issuance costs cut further into an already-strained investment budget, and issuing itself drags Λ down more sharply for such a firm. Net equity issuance for these firms drifts only slightly above zero and stays there, even as the incentive to issue keeps rising. Leverage L'/K' correspondingly stays high, close to the model's collateral limit ($\varphi = 0.8$), rather than declining further.

The bottom row splits firms by whether they are net issuers or net repurchasers. Issuers' net equity issuance is bounded well away from zero even when the fund is at its smallest, starting near 3% of capital and rising to about 10% as fund size grows; repurchasers' net equity issuance stays flat and small throughout, never far from zero. This gap is a direct signature of the fixed issuance cost c_{h0} in the issuance cost function (31): since a firm pays this cost regardless of how much it issues, issuing only a marginal amount is never optimal, so firms either forgo issuance entirely or issue a materially sized lump. Investment tracks the same divide: issuers' I/K rises from about 7.5% to 10.5%, while repurchasers' stays flat near 2.5-3%, indicating that firms only pay the fixed cost to issue when they have real growth opportunities to fund. Leverage moves comparatively little for either group, drifting down only modestly for issuers (from about 0.28 to 0.24) and staying flat near 0.42 for repurchasers, suggesting that paring down leverage is a secondary motive relative to funding growth opportunities.

5.4 Ownership as an Endogenous State Variable

Traditional corporate finance models summarize a firm's state using variables that appear on its financial statements: capital, leverage, cash. Here, a firm's investor base, the share of its equity held by mutual funds s^M, is itself a state variable that shapes its financing environment, even though it appears on no balance sheet. Figure 5 traces out this effect, plotting policy functions against s^M, weighted by the model's stationary distribution over firm states.

Figure 5 Here

The figure shows the underlying mechanism. Mutual funds aim for a target portfolio weight, and when current ownership s^M is far below that target, closing even part of the gap requires a large purchase; this is what generates high price pressure Λ, which falls from about 1.05 when s^M is at its lowest to about 0.68 when s^M is at its highest, as the required adjustment shrinks. The remaining three policy functions follow directly from Λ, exactly as in the response to fund flow shocks above: with price pressure highest when s^M is lowest, equity is at its cheapest then, so net equity issuance and investment are at their highest and leverage is at its lowest; as s^M rises and price pressure fades, issuance and investment recede and firms lean relatively more on debt, so leverage rises.

5.5 Size-Dependence: Diminishing Marginal Effects

Source and state are not the only forces shaping a shock's effects; its size matters too, because the same equilibrium forces documented above limit how far a larger shock's effects can grow. When firms face positive price pressure, they issue equity to improve fundamental values, narrowing the gap between market value and fundamental value. Simultaneously, mutual funds endogenously reduce their holdings in anticipation of the resulting price correction, further dampening price pressure directly. Consequently, even with a fixed elasticity for residual investors with respect to price pressure, larger mutual fund flow shocks do not translate proportionally into larger price fluctuations due to strategic responses from both mutual funds and firms.

These effects diminish sharply at the margin as shock size grows, and Figure 6 illustrates why, plotting the initial impulse responses of market value, price pressure, and net equity issuance under different mutual fund flow shock sizes. Panel A shows the IRFs to a one-standard-deviation shock for varying shock sizes. As shock size increases, market value responses (blue solid line) rise from 1.06% at $\sigma_{\omega i} = 0.09$ to 1.71% at the baseline, but with sharply diminishing marginal effects: moving from $\sigma_{\omega i} = 0.09$ to 0.15, from 0.15 to 0.21, and from 0.21 to 0.27 yields incremental market value increases of 0.46, 0.19, and 0.09 percentage points, respectively. Price pressure (red dashed line) rises more gradually across the same range and essentially plateaus beyond $\sigma_{\omega i} = 0.24$ (1.224% at $\sigma_{\omega i} = 0.24$ and 1.220% at $\sigma_{\omega i} = 0.27$), reflecting that strategic responses from firms and mutual funds increasingly absorb larger shocks. Panel B shows the corresponding per-unit responses normalized to the baseline shock size ($\sigma_{\omega i} = 0.21$). All three per-unit responses decline rapidly as shock size increases, confirming the self-limiting nature of the equilibrium.

Figure 6 Here

As an untargeted check on this mechanism, Table 10 reports per-unit stock return and equity issuance responses estimated on the subsample of small empirical flow shocks: normalized coefficients are larger than one for both returns and equity issuance, consistent with the same diminishing marginal effects found in the model, though returns decay somewhat faster than equity issuance empirically.

Table 10 Here

Together, these results show that a demand shock's price impact depends on its source, its size, and the firm's state, and that firms' and investors' strategic responses are self-limiting along each dimension.

6 Counterfactuals

This section asks how much capital misallocation inelastic investor demand causes, and why. Section 6.1 measures the magnitude: I perturb four parameters that govern how demand shocks propagate into prices and investment, and read off the resulting change in aggregate TFP. I examine four scenarios: (1) higher mutual fund flow shock volatility ($\sigma_{\omega i}$ from 0.21 to 0.24), (2) a smaller mutual fund sector ($\exp(\bar{q})$ from 0.29 to 0.22), (3) a less elastic residual-investor sector (c_r from 0.878 to 1.00), and (4) lower capital adjustment costs (c_k from 0.492 to 0.45). The three financial-side scenarios move aggregate TFP by less than 0.03% in either direction, an order of magnitude below the real-friction comparison. Section 6.2 then decomposes why: the same source-, size-, and state-dependence that shape the corrected demand elasticity in Section 5 also bound how much misallocation price pressure can generate, capping the dispersion it adds to firms' marginal products and governing where that dispersion lands.

6.1 MPK and Capital Misallocation

Following Hsieh and Klenow (2009), Haltiwanger et al. (2018), and Baqaee and Farhi (2017) I measure capital misallocation using aggregate TFP, calculated as $\text{TFP}_{\text{agg}} = Y_{\text{agg}}/K^{\alpha}_{\text{agg}}$, where Y_{agg} and K_{agg} denote aggregate output and capital, respectively. Higher aggregate measured TFP indicates more efficient capital allocation across firms. Table 11 presents the percentage changes of aggregate TFP in counterfactual specifications from the baseline.

Table 11 Here

The channels through which price pressure is generated and absorbed have evolved in recent years. Assets have shifted from active to passive management. Sentiment- and narrative-based trading, often linked to retail investor enthusiasm, plays a strong role in price formation. And traditional arbitrageurs' risk-bearing capacity varies with capital regulation, leverage constraints, and institutional mandates. With investor demand inelastic, do these changes matter for capital misallocation? I examine this with three counterfactuals, each perturbing one parameter that governs how much price pressure a flow shock generates, holding the rest of the model at baseline. When the mutual fund sector shrinks 24% ($\exp(\bar{q})$ from 0.29 to 0.22), narrowing the pool of frictional intermediaries that stand between flow shocks and prices, aggregate TFP rises by 0.006%. A 14% increase in flow-shock volatility ($\sigma_{\omega i}$ from 0.21 to 0.24) instead lowers aggregate TFP by 0.014%, and a roughly 12% decline in residual investors' effective elasticity to price pressure (c_r from 0.878 to 1.00, since elasticity equals $1/(c_r s^R)$) lowers it by 0.028%. All three move aggregate TFP by less than 0.03% in either direction.

To see why these price-pressure perturbations move aggregate TFP so little, I next consider the polar case. Setting $\Lambda \equiv 1$ turns the price-pressure channel off altogether, so that price pressure never separates market value from fundamental value. This shuts down every channel the three counterfactuals above engage only partially, regardless of which direction each parameter is moved. Yet it raises aggregate TFP by just 0.120%, confirming that it is not the size of these particular parameter changes that limits their effect, but a modest ceiling on how much capital misallocation the price-pressure channel can generate at all.

As a point of comparison, I also perturb a real, non-price-pressure friction: an 8.5% reduction in the capital adjustment cost (c_k from 0.492 to 0.45) raises aggregate TFP by 0.14%, exceeding even the full $\Lambda \equiv 1$ effect of shutting down price pressure entirely. Estimated investor demand is inelastic, so price pressure following demand shocks can be large. Equity issuance, itself estimated to respond significantly to that pressure, is the very channel through which it distorts investment. Together, these features point toward substantial misallocation. Yet price pressure's misallocation cost stays modest, both at these marginal perturbations and at its full elimination. Appendix C extends the model to heterogeneous exposure, where the production function is specified as $Y_{i,t} = A_t^{b_i} X_{i,t} K_{i,t}^{\alpha}$, with $b_i \sim \mathcal{N}(1, 0.2)$, following David et al. (2022) and Choi et al. (2025), and confirms that the misallocation results above are robust to this extension.

6.2 MPK-Dispersion Decomposition

The financial counterfactuals in Table 11 move aggregate TFP by less than 0.03% in either direction. This section decomposes why, tracing that small headline number to the same source-, size-, and state-dependence documented in Section 5.

Aggregate TFP is $\text{TFP}_{\text{agg}} = Y_{\text{agg}}/K_{\text{agg}}^{\alpha}$. Let $m_i \equiv \log \text{MPK}_i$ denote the log marginal product of capital at firm i. Following the wedge-based approach to measuring misallocation in Hsieh and Klenow (2009), a second-order expansion of TFP_{agg} around the cross-sectional distribution of firm capital shares gives

$$\Delta \log \text{TFP}_{\text{agg}} \approx -\frac{1}{2}\frac{\alpha}{1-\alpha}\Delta \text{Var}(m_i), \tag{54}$$

where aggregate TFP moves in the opposite direction of the cross-sectional dispersion of m_i. Intuitively, because MPK differs across firms, moving a unit of capital from a low-MPK firm to a high-MPK firm raises output at the receiving firm by more than it lowers output at the losing firm. A given aggregate capital stock therefore produces less output the more dispersed MPK is across firms, since more of it sits away from its most productive use. See Section B.5 for the derivation of Equation (54).

Equation (54) is a second-order approximation, in the spirit of the Jensen's-gap aggregation results in Baqaee and Farhi (2017), that assumes a single static factor market, abstracting from the dynamic adjustment costs and financing frictions the full model carries. Even so, it performs well in practice. Applied to the baseline economy against its $\Lambda \equiv 1$ counterpart, Equation (54) implies a wedge-based TFP gain of 0.102%, close to the exact gain of 0.120% in Table 11.

The price-pressure channel is capped by $\Lambda \equiv 1$, so decomposing $\Delta^0 \mathrm{Var}(m_i)$ explains why every price-pressure scenario in Table 11 stays small, where Δ^0 denotes the change from baseline to $\Lambda \equiv 1$. Let m_i^0 denote firm i's log MPK in the $\Lambda \equiv 1$ economy, and define the price-pressure wedge $w_i \equiv m_i - m_i^0$. The wedge is a difference across two economies for the same firm, not a distance from a target: since $\mathrm{MPK} = \alpha Y/K$, a positive wedge means price pressure left the firm holding less capital relative to its productivity, and a higher marginal product as a result. The variance of a sum then gives $\mathrm{Var}(m_i) = \mathrm{Var}(m_i^0) + \mathrm{Var}(w_i) + 2\,\mathrm{Cov}(m_i^0, w_i)$, so

$$\Delta^0 \mathrm{Var}(m_i) \equiv \mathrm{Var}(m_i) - \mathrm{Var}(m_i^0) = \mathrm{Var}(w_i) + 2\,\mathrm{Cov}(m_i^0, w_i). \tag{55}$$

$\mathrm{Var}(w_i)$ measures how much new MPK dispersion price pressure creates on its own. $2\,\mathrm{Cov}(m_i^0, w_i)$ captures where that distortion lands relative to the dispersion already there: whether the wedges push firms further in the direction their MPK already deviated from the cross-sectional average, stretching the existing spread, or back toward it, compressing the spread instead.

The two terms in Equation (55) map onto the same source-, size-, and state-dependence that shaped the corrected elasticity in Section 5: source- and size-dependence bound how large $\mathrm{Var}(w_i)$ can grow, while state-dependence governs the alignment term $2\,\mathrm{Cov}(m_i^0, w_i)$. Together they keep $\Delta^0 \mathrm{Var}(m_i)$ small.

Source-dependence sets the level of the mapping from a flow shock to the wedge it produces, through the elasticity governing how much of the shock residual investors absorb. Instrumenting price pressure directly recovers a corrected elasticity of 1.6; instrumenting the observed price response instead, which conflates price pressure with the genuine fundamental improvement (Table 8), recovers a downward-biased 1.3 (Table 9), for the reasons detailed in

Section 4. A higher elasticity means residual investors absorb more of any given flow, so the same shock moves Λ, and hence w_i, by less to begin with. Using the naive 1.3 in place of the corrected 1.6 therefore overstates how much wedge a given shock generates, and with it, how large $\text{Var}(w_i)$ can grow.

Size-dependence complements source-dependence: where source sets the level of the shock-to-wedge mapping, size flattens its slope as the shock grows. Mutual funds unwind more of their position in anticipation of a sharper price correction as a shock grows, and firms find issuance increasingly self-defeating as price pressure rises, so per-unit price and net-equity-issuance responses shrink with shock size rather than staying constant (Section 5), a self-limiting mechanism with untargeted empirical support (Table 10). This is why a 14% increase in flow-shock volatility ($\sigma_{\omega i}$ from 0.21 to 0.24) moves aggregate TFP by only 0.014% in Table 11: a larger shock does not translate proportionally into a larger wedge, because the responses generating that wedge are themselves diminishing. Source- and size-dependence together keep the wedge itself small: $\text{Var}(w_i)$ is just 0.034 percentage points, about a third of the full 0.102 percentage-point change in $\text{Var}(m_i)$ (Table 12).

Table 12 Here

The remaining two-thirds of the change comes not from how large the wedges are but from where they land: alignment contributes $2\,\text{Cov}(m_i^0, w_i) = 0.068$ percentage points, twice $\text{Var}(w_i)$'s 0.034. Because alignment is a covariance across firms, the natural way to decompose it is by splitting the cross-section into groups. For firms partitioned into groups g, write $m_i^0 - \bar{m}^0 = (m_i^0 - \bar{m}_g^0) + (\bar{m}_g^0 - \bar{m}^0)$, and likewise for $w_i - \bar{w}$, where $\bar{m}^0, \bar{w}$ are the full-sample capital-share-weighted means and $\bar{m}_g^0, \bar{w}_g$ are group g's. Expanding the product $(m_i^0 - \bar{m}^0)(w_i - \bar{w})$ and capital-share-weighting within each group, the two cross terms vanish, since within-group deviations sum to zero against the group's own weights by construction, leaving

$$\text{Cov}(m_i^0, w_i) = \sum_g s_g \underbrace{\text{Cov}_g(m_i^0, w_i)}_{\text{within}} + \sum_g s_g \underbrace{(\bar{m}_g^0 - \bar{m}^0)(\bar{w}_g - \bar{w})}_{\text{between}}, \tag{56}$$

where g indexes a partition of firms into groups, $s_g \equiv \sum_{i \in g} s_i$ is group g's aggregate capital share, and $\text{Cov}_g(\cdot)$ denotes the covariance computed using only firms in g, weighted by their within-group capital shares s_i/s_g. The first sum is within-group alignment and the second is between-group alignment. Because misallocation is the spread of m_i across firms, each term has a direct reading: within-group alignment asks whether firms inside a group spread further apart or bunch closer together once price pressure is added, and between-group alignment asks the same of the group averages. A positive term is spreading, and therefore amplification; a negative term is bunching, and therefore correction.

Converting price pressure into capital asks two things of a firm: an opportunity worth funding, and the balance-sheet room to fund it. These are the two state variables the firm's own optimality conditions turn on, growth opportunities through q in (51), whose empirical counterpart is the market-to-book ratio, and financing capacity through the collateral-constraint multiplier μ in (53); the same cut already organizes the paper's other state-dependence evidence (Table 4, Table 8). Table 12 accordingly sorts firms into four leverage-by-Q groups and reports each group's within- and between-group contribution, additive by construction.

Alignment matters for a second reason beyond its magnitude: unlike $\mathrm{Var}(w_i)$, it carries a sign. A wedge registers as misallocation no matter which firm it lands on, so $\mathrm{Var}(w_i)$ can only accumulate. Alignment can cancel, and along this partition it cancels heavily: summed across the eight entries of Table 12, amplifying contributions come to +0.142 percentage points and corrective ones to −0.075, leaving the net +0.068; absent the corrective entries, $\Delta^0\mathrm{Var}(m_i)$ would have been 0.176 rather than 0.102 percentage points, some 70% larger. Amplification is the unsurprising half; what the decomposition has to explain is where the corrective force comes from. A framework that models how much price pressure a demand shock generates, but not which firms end up absorbing it, cannot answer that question properly, and can therefore be wrong about the magnitude of misallocation as well as its direction.

The two halves of Equation (56) divide these forces cleanly. Between-group alignment is net amplifying at +0.084 percentage points and holds the bulk of the amplification in the table; within-group alignment is net corrective at −0.016. The rest of this subsection takes them in turn, within-group first.

Within-group alignment is signed by financing capacity, negative for both low-leverage groups (−0.033 at low Q, −0.031 at high Q) and positive for both high-leverage groups (+0.024 and +0.023), with Q moving it by no more than 0.002 anywhere. Those two negative entries are 85% of all the correction in Table 12, and a single mechanism produces both signs. Inside any group, the firm shortest of capital has the most to gain from converting price pressure into investment; leverage decides whether it can. Low-leverage firms have the flexibility to act on price pressure in either direction. High-leverage firms do not: issuing is self-defeating in the sense Section 5 documents, since it drags Λ down more sharply for such a firm even as the incentive to issue keeps rising. The policy functions in Section 5 show both patterns: net equity issuance is steep and sign-switching over fund size for low-leverage firms, turning from repurchases to heavy issuance as the fund grows, and nearly flat for high-leverage firms. The consequence for allocation follows the same split. Where the routes stay open, they are used most where the marginal value of capital is highest, so the firms already shortest of capital absorb the smallest wedge and the group bunches together. Where the routes are closed, the inability to act binds hardest on those same firms, which absorb the largest wedge instead, and the group spreads apart.

Between groups, the two dimensions have to be read jointly. The group average wedges $\bar{w}_g$ make this direct. Among high-Q firms, financing capacity is nearly everything: the high-leverage group's average wedge is 0.0161 against the low-leverage group's 0.0023, a sevenfold gap. Among low-Q firms it is worth almost nothing, 0.0097 against 0.0093. Financing flexibility is valuable only to a firm with something to fund, and opportunities are worth having only to a firm that can act on them. Where that burden lands is what turns it into misallocation, and it lands regressively, like a tax taking the most from the firms that had the least. A burden borne equally would cancel out of the variance; what misallocates is that it falls unevenly, and unevenly in a particular direction. The firms that cannot convert price pressure into capital are the high-leverage firms, and the collateral constraint that closes their debt route is the same friction that left them short of capital to begin with, so their baseline MPK is the highest in the cross-section ($\bar{m}_g^0 - \bar{m}^0 > 0$ for both). Price pressure is filtered through the same friction that generated the dispersion it is added to, and amplification follows as the default. Both high-leverage groups accordingly contribute positively and unremarkably, +0.021 at low Q and +0.027 at high Q.

Two entries depart from that mechanism. The first is the high-Q, low-leverage group, which amplifies not because it is burdened but because it is spared. These firms are the most capital-abundant in the cross-section and carry the smallest average wedge of any group. Low leverage allows them to convert the price-pressure windfall into real capital rather than absorbing it as a distortion: in the group-level breakdown of Table 8 they have the highest fundamental-value share of the price response of any group, 22%. Both of their deviations are negative ($\bar{w}_g - \bar{w} < 0$ and $\bar{m}_g^0 - \bar{m}^0 < 0$), so the product, and consequently the group's between-group contribution, is positive. Being spared at the capital-abundant end widens the gap exactly as being burdened at the capital-starved end does.

The second departure is the only corrective entry, and it is capacity without opportunity. Low-Q, low-leverage firms can convert price pressure into capital but have little reason to:

with few opportunities to fund, they do not want to pay the issuance cost. They therefore carry a constrained firm's wedge while sitting on the capital-abundant side of the distribution. The breakdown in Table 8 bears this out: their fundamental-value share is the lowest of any group, 13%, a Q pattern Section 4 matches to the data. The mismatched deviations ($\bar{w}_g - \bar{w} > 0$ against $\bar{m}_g^0 - \bar{m}^0 < 0$) pull them toward the middle rather than away from it, at -0.011. Corrective forces, within groups and between them alike, arise only where the financing routes stay open: used, inside a group, to reallocate toward the firms shortest of capital; unused, between groups, to leave a capital-abundant group carrying a constrained firm's wedge.

Source-, size-, and state-dependence together explain Table 11's shape, and they act at different points. Source- and size-dependence cap the wedge itself, holding $\mathrm{Var}(w_i)$ to 0.034 percentage points and keeping it nearly uniform across firm states. State-dependence governs the larger remainder: two-thirds of $\Delta^0\mathrm{Var}(m_i)$ runs through where the wedge lands rather than how large it is. That alignment term is substantially self-limiting, since the corrective entries remove more than half of the gross amplification. The financial counterfactuals move aggregate TFP as little as they do because both forces operate at once: source and size bound the wedge, and state-dependent transmission, though net amplifying, is cut by more than half wherever firms retain the room to reallocate.

7 Conclusion

This paper asks what a standard demand-elasticity estimate recovers once firms respond to the price pressure inelastic demand creates, and how much capital that demand misallocates. Using mutual fund flow shocks in a shift-share design, I show that a flow shock raises a firm's stock price with only partial reversal and induces equity issuance, investment, and debt paydown. To interpret that response, I estimate by indirect inference a dynamic model in which firms, mutual funds, and residual investors each move the price the others are responding to, so that market clearing splits market value into fundamental value and price pressure.

Firm issuance breaks the exclusion restriction standard estimation relies on: market value blends price pressure with the fundamental improvement issuance delivers, 17% of the initial price response, so instrumenting it with flow shocks recovers 1.3 against a true elasticity of 1.6. That 1.6 is an average over equilibrium outcomes rather than a parameter, and the elasticity it averages is source-, size-, and state-dependent. Investors sell into a price rise they expect to unwind but not into one they do not, so demand elasticity is not defined until the source of the price change is named. Firms and investors both push back harder as pressure grows, so a larger shock does not produce a proportionally larger price impact. And the price impact a shock produces varies with the firm's growth opportunities and its issuance capacity.

Correcting the elasticity upward closes little of the distance to perfectly elastic demand, so pervasive demand shocks should still give firms ample incentive to over- or under-invest. That chain is real, but it does not run far. Three demand-side counterfactuals move aggregate TFP by less than 0.03% in either direction, and eliminating price pressure altogether raises it by only 0.12%, still less than the 0.14% a modest reduction in real capital adjustment costs delivers. Two forces keep it that small: source- and size-dependence hold down the wedges price pressure adds to firms' marginal products, and state-dependence corrects part of what remains, since price pressure is also cheap financing for firms short of capital.

The broader lesson is that demand elasticity in equity markets has more structure than a single number, and that the structure is estimable. A single elasticity number often blurs a subtle distinction: a ceteris paribus derivative holding everything but price fixed, versus a total derivative agnostic to which channel moved price. Different counterfactuals load on different channels. Measuring investment distortion requires an equilibrium system in which price pressure is separately identified, not an elasticity alone, since price pressure is what real decisions get made on and what firms' own responses erode. Building the model around the counterfactual of interest supplies both the elasticity it depends on and the equilibrium it runs through.

The framework points past the setting it was estimated on. It needs the elasticity of whoever absorbs a flow, so more comprehensive holdings data would show whether the residual pool hides distinct clienteles. It also models individual firms in isolated markets, so extending it to aggregate quantities and prices would let demand shocks compete for a common pool of capital. And the feedback it turns on is not specific to equity flows: debt markets carry a version of it, and so do large-scale central bank purchases, where the intervention moves prices and issuers respond by changing what is being bought. Whether the correction survives when the demand shift is aggregate rather than firm-specific is an open question. The same discipline extends to other counterfactuals in asset markets, wherever prices and the decisions made on them are jointly determined.

References

Andrews, Isaiah, Matthew Gentzkow, and Jesse M. Shapiro, "Measuring the Sensitivity of Parameter Estimates to Estimation Moments*," *The Quarterly Journal of Economics*, November 2017, *132* (4), 1553–1592.

Appel, Ian R., Todd A. Gormley, and Donald B. Keim, "Passive investors, not passive owners," *Journal of Financial Economics*, July 2016, *121* (1), 111–141.

Baker, Malcolm and Jeffrey Wurgler, "Market Timing and Capital Structure," *The Journal of Finance*, 2002, *57* (1), 1–32. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/1540-6261.00414.

_, Jeremy C. Stein, and Jeffrey Wurgler, "When Does the Market Matter? Stock Prices and the Investment of Equity-Dependent Firms*," *The Quarterly Journal of Economics*, August 2003, *118* (3), 969–1005.

Baker, Malcolm P. and Jeffrey Wurgler, "Investor Sentiment and the Cross-Section of Stock Returns," November 2003.

Bali, Turan, Robert Engle, and Scott Murray, "Empirical Asset Pricing: The Cross Section of Stock Returns: An Overview," in "Wiley StatsRef: Statistics Reference Online," John Wiley & Sons, Ltd, 2017, pp. 1–8.

Baqaee, David Rezza and Emmanuel Farhi, "Productivity and Misallocation in General Equilibrium," Technical Report w24007, National Bureau of Economic Research November 2017.

Begenau, Juliane and Juliana Salomao, "Firm Financing over the Business Cycle," *The Review of Financial Studies*, April 2019, *32* (4), 1235–1274.

Belo, Frederico, Xiaoji Lin, and Fan Yang, "External Equity Financing Shocks, Financial Flows, and Asset Prices," *The Review of Financial Studies*, September 2019, *32* (9), 3500–3543.

Billett, Matthew T., Ioannis V. Floros, and Jon A. Garfinkel, "At-the-Market Offerings," *Journal of Financial and Quantitative Analysis*, June 2019, *54* (3), 1263–1283.

Bolton, Patrick, Hui Chen, and Neng Wang, "Market timing, investment, and risk management," *Journal of Financial Economics*, July 2013, *109* (1), 40–62.

Bond, Philip, Alex Edmans, and Itay Goldstein, "The Real Effects of Financial Markets," *Annual Review of Financial Economics*, 2012, *4* (1), 339–360.

Borusyak, Kirill, Peter Hull, and Xavier Jaravel, "Quasi-Experimental Shift-Share Research Designs," September 2018.

_ , _ , and _ , “A Practical Guide to Shift-Share Instruments,” *Journal of Economic Perspectives*, February 2025, *39* (1), 181–204.

Chaudhary, Manav, Zhiyu Fu, and Jian Li, “Corporate Bond Multipliers: Substitutes Matter,” December 2022.

Chaudhry, Aditya and Jiacui Li, “Endogenous Elasticities: Price Multipliers Are Smaller for Larger Demand Shocks,” June 2025.

Chen, Qi, Itay Goldstein, and Wei Jiang, “Price Informativeness and Investment Sensitivity to Stock Price,” 2007, *20* (3), 619–650.

Chinco, Alex and Marco Sammon, “The passive ownership share is double what you think it is,” *Journal of Financial Economics*, July 2024, *157*, 103860.

Chiu, H.H. and O. Kini, “Equity Issuances, Equity Mutual Fund Flows, and Noise Trader Sentiment*,” *Review of Finance*, April 2014, *18* (2), 749–802.

Choi, Jaewon, Mahyar Kargar, Xu Tian, and Yufeng Wu, “Investor demand, firm investment, and capital misallocation,” 2025, *168*, 104039.

Coppola, Antonio, “In Safe Hands: The Financial and Real Impact of Investor Composition over the Credit Cycle,” *The Review of Financial Studies*, March 2025, p. hhaf017.

Coval, Joshua and Erik Stafford, “Asset fire sales (and purchases) in equity markets,” *Journal of Financial Economics*, November 2007, *86* (2), 479–512.

David, Joel M., Lukas Schmid, and David Zeke, “Risk-adjusted capital allocation and misallocation,” *Journal of Financial Economics*, September 2022, *145* (3), 684–705.

Davis, Carter, Mahyar Kargar, and Jiacui Li, “Why is Asset Demand Inelastic?,” August 2022.

Davis, Steven J., John C. Haltiwanger, and Scott Schuh, *Job Creation and Destruction*, MIT Press, 1996.

Dessaint, Olivier, Thierry Foucault, Laurent Frésard, and Adrien Matray, “Noisy Stock Prices and Corporate Investment,” February 2018.

Dou, Winston Wei, Leonid Kogan, and Wei Wu, “Common Fund Flows: Flow Hedging and Factor Pricing,” May 2022.

Edmans, Alex, Itay Goldstein, and Wei Jiang, “The Real Effects of Financial Markets: The Impact of Prices on Takeovers,” *The Journal of Finance*, 2012, *67* (3), 933–971.

Eisfeldt, Andrea L. and Tyler Muir, “Aggregate External Financing and Savings Waves,” September 2014.

Elton, Edwin J., Martin J. Gruber, and Christopher R. Blake, "A First Look at the Accuracy of the CRSP Mutual Fund Database and a Comparison of the CRSP and Morningstar Mutual Fund Databases," *The Journal of Finance*, 2001, *56* (6), 2415–2430.

Erickson, Timothy and Toni M. Whited, "Two-Step GMM Estimation of the Errors-in-Variables Model Using High-Order Moments," *Econometric Theory*, 2002, *18* (3), 776–799.

Fama, Eugene and Kenneth French, "Common risk factors in the returns on stocks and bonds," *Journal of Financial Economics*, 1993, *33* (1), 3–56.

Fama, Eugene F. and Kenneth R. French, "Financing decisions: who issues stock?," *Journal of Financial Economics*, June 2005, *76* (3), 549–582.

Frazzini, Andrea and Owen A. Lamont, "Dumb money: Mutual fund flows and the cross-section of stock returns," *Journal of Financial Economics*, May 2008, *88* (2), 299–322.

Fuchs, William, Satoshi Fukuda, and Daniel Neuhann, "Demand-System Asset Pricing: Theoretical Foundations," December 2023.

Gabaix, Xavier and Matteo Maggiori, "International Liquidity and Exchange Rate Dynamics," *The Quarterly Journal of Economics*, 2015, *130* (3), 1369–1420.

_ **and Ralph S. J. Koijen**, "In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis," May 2022.

Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift, "Bartik Instruments: What, When, Why, and How," *American Economic Review*, August 2020, *110* (8), 2586–2624.

Gârleanu, Nicolae and Lasse Heje Pedersen, "Dynamic Trading with Predictable Returns and Transaction Costs," *The Journal of Finance*, 2013, *68* (6), 2309–2340. Publisher: Wiley.

Haddad, Valentin, Paul Huebner, and Erik Loualiche, "How Competitive is the Stock Market? Theory, Evidence from Portfolios, and Implications for the Rise of Passive Investing," April 2021.

_ **, Zhiguo He, Paul Huebner, Peter Kondor, and Erik Loualiche**, "Causal Inference for Asset Pricing," 2025.

Haltiwanger, John, Robert Kulick, and Chad Syverson, "Misallocation Measures: The Distortion That Ate the Residual," Technical Report w24199, National Bureau of Economic Research January 2018.

Han, Chirok and Peter C. B. Phillips, "GMM Estimation for Dynamic Panels with Fixed Effects and Strong Instruments at Unity," *Econometric Theory*, 2010, *26* (1), 119–151.

Hayashi, Fumio, "Tobin's Marginal q and Average q: A Neoclassical Interpretation," *Econometrica*, 1982, *50* (1), 213–224. Publisher: [Wiley, Econometric Society].

He, Zhiguo and Arvind Krishnamurthy, "Intermediary Asset Pricing," *The American Economic Review*, 2013, *103* (2), 732–770.

_ , Peter Kondor, and Jessica S. Li, "Demand Elasticity in Dynamic Asset Pricing," June 2025.

Hennessy, Christopher A. and Toni M. Whited, "Debt Dynamics," *The Journal of Finance*, 2005, *60* (3), 1129–1165.

Hsieh, Chang-Tai and Peter J. Klenow, "Misallocation and Manufacturing TFP in China and India," February 2009.

Huddart, Steven, "Employee stock options," *Journal of Accounting and Economics*, September 1994, *18* (2), 207–231.

Kashyap, Anil K., Natalia Kovrijnykh, Jian Li, and Anna Pavlova, "Is There Too Much Benchmarking in Asset Management?," April 2021.

Khan, Mozaffar, Leonid Kogan, and George Serafeim, "Mutual Fund Trading Pressure: Firm-Level Stock Price Impact and Timing of SEOs," *The Journal of Finance*, 2012, *67* (4), 1371–1395.

Kim, Woojin and Michael S. Weisbach, "Motivations for public equity offers: An international perspective," *Journal of Financial Economics*, February 2008, *87* (2), 281–307.

Koijen, Ralph S. J. and Motohiro Yogo, "A Demand System Approach to Asset Pricing," *Journal of Political Economy*, 2019, *127* (4), 1475–1515.

Koijen, Ralph S.J., "The Cross-Section of Managerial Ability, Incentives, and Risk Preferences," *The Journal of Finance*, 2014, *69* (3), 1051–1098. Publisher: [American Finance Association, Wiley].

Kontz, Christian and Sebastian Hanson, "The Real Cost of Benchmarking," 2026.

Kyle, Albert S., "Continuous Auctions and Insider Trading," *Econometrica*, 1985, *53* (6), 1315–1335. Publisher: [Wiley, Econometric Society].

Kılıç, Mete and Şelale Tüzel, "Investing in misallocation," 2026, *176*, 104208.

Livdan, Dmitry, Horacio Sapriza, and Lu Zhang, "Financially Constrained Stock Returns," *The Journal of Finance*, 2009, *64* (4), 1827–1862. Publisher: [American Finance Association, Wiley].

Lou, Dong, "A Flow-Based Explanation for Return Predictability," June 2010.

Lou, Xiaoxia and Albert Y. Wang, "Flow-Induced Trading Pressure and Corporate Investment," *The Journal of Financial and Quantitative Analysis*, 2018, *53* (1), 171–201.

Ma, Yueran, "Nonfinancial Firms as Cross-Market Arbitrageurs," *The Journal of Finance*, 2019, *74* (6), 3041–3087.

McKeon, Stephen B., "Employee Option Exercise and Equity Issuance Motives," October 2015.

Morck, Randall, Andrei Shleifer, Robert W. Vishny, Matthew Shapiro, and James M. Poterba, "The Stock Market and Investment: Is the Market a Sideshow?," *Brookings Papers on Economic Activity*, 1990, *1990* (2), 157–215.

Pavlova, Anna and Taisiya Sikorskaya, "Benchmarking Intensity," *The Review of Financial Studies*, March 2023, *36* (3), 859–903.

Polk, Christopher and Paola Sapienza, "The Stock Market and Corporate Investment: A Test of Catering Theory," *The Review of Financial Studies*, January 2009, *22* (1), 187–217.

Sammon, Marco and John J. Shim, "Who Clears the Market When Passive Investors Trade?," September 2024.

Schmickler, Simon and Pedro Tremacoldi-Rossi, "Spillover Effects of Payouts on Asset Prices and Real Investment," October 2022.

Siani, Kerry, "Raising Bond Capital in Segmented Markets *," October 2022.

van Binsbergen, Jules H., Benjamin David, and Christian C. Opp, "How (Not) to Identify Demand Elasticities in Dynamic Asset Markets," July 2025.

van der Beck, Philippe, "On the Estimation of Demand-Based Asset Pricing Models," May 2022.

Vayanos, Dimitri and Jean-Luc Vila, "A Preferred-Habitat Model of the Term Structure of Interest Rates," *Econometrica*, 2021, *89* (1), 77–112.

Warusawitharana, Missaka and Toni M. Whited, "Equity Market Misvaluation, Financing, and Investment," *The Review of Financial Studies*, 2016, *29* (3), 603–654.

Whited, Toni M. and Jake Zhao, "The Misallocation of Finance," *The Journal of Finance*, 2021, *76* (5), 2359–2407.

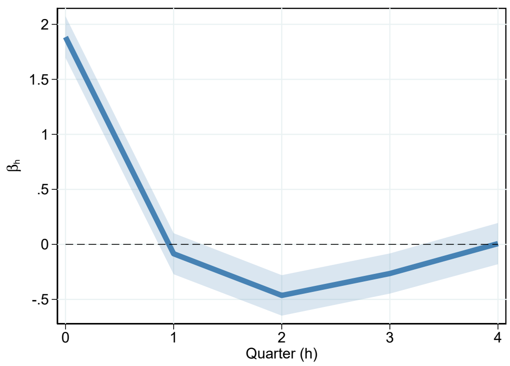
Zhang, Lu, "The Value Premium," *The Journal of Finance*, 2005, *60* (1), 67–103.

Òscar Jordà, "Estimation and Inference of Impulse Responses by Local Projections," *American Economic Review*, March 2005, *95* (1), 161–182.

Figures

Figure 1: Firm Responses

(a) $R_{i,t+h}$



(b) $NEI_{i,t+h}/K_{i,t+h-1}$

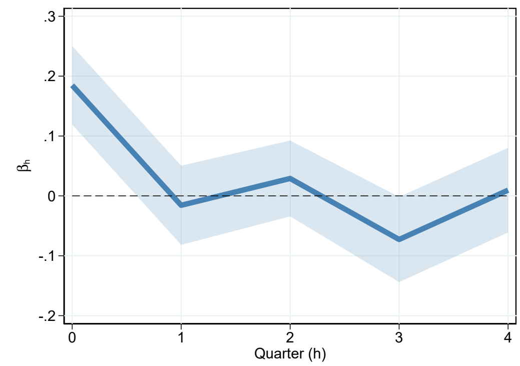


Figure 1 presents impulse response functions of returns and net equity issuance. The blue solid line plots coefficients β_h from estimating the local projection following Jordà (2005):
$\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t} + X_{i,t-1} + \varepsilon_{m,t+h}$ for $h = 0, 1, \ldots, 4$. The shaded area indicates 95% pointwise confidence bands using standard errors clustered at firm level. control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). The sample is from 2007Q1 to 2021Q4.

Figure 2: Firm Return Responses to Active and Passive Fund Flow Shocks

(a) Active (b) Passive

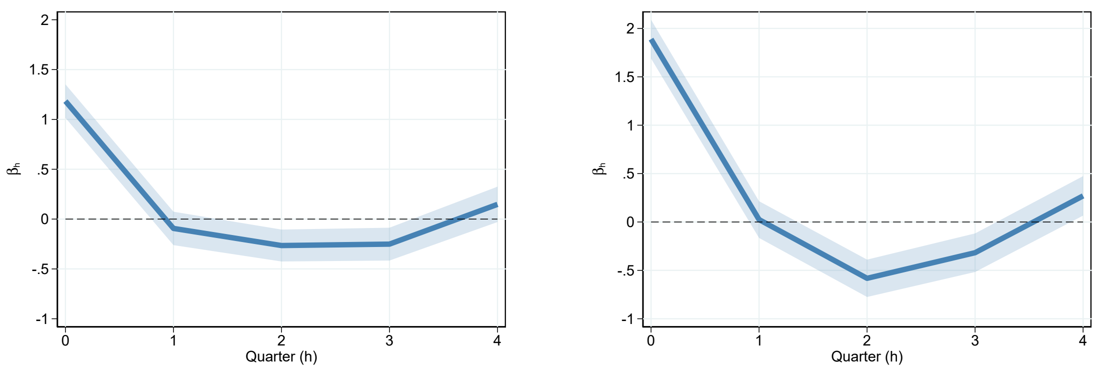


Figure 2 presents impulse response functions of firm returns to active and passive mutual fund flow shocks ($f_{i,t}^{T}$ for $T \in \{\text{Active}, \text{Passive}\}$). The blue solid line plots coefficients β_h from estimating the local projection following Jordà (2005): $\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t}^{T} + X_{i,t-1} + \varepsilon_{m,t+h}$ for $h = 0, 1, \ldots, 4$. The shaded area indicates 95% pointwise confidence bands using standard errors clustered at firm level. control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s_{i,t-1}^{M}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). The sample is from 2007Q1 to 2021Q4.

Figure 3: Equilibrium Price Formation

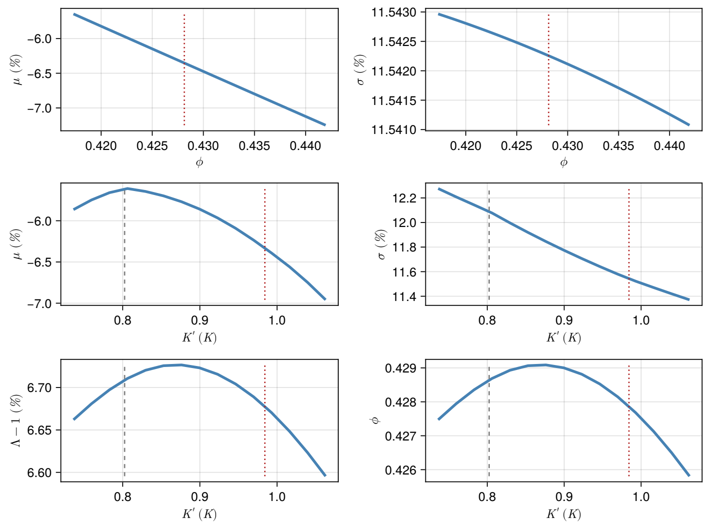


Figure 3 illustrates the equilibrium price formation process for a firm seeking to increase investment. The top panel plots equilibrium expected returns and volatility as functions of mutual fund portfolio weights, given optimal firm decisions. The middle panel plots equilibrium expected return and volatility as functions of the firm's investment decisions, taking optimal mutual fund responses as given. The bottom panel plots equilibrium price pressure and mutual fund holding rules as functions of the firm investment decisions. The dotted red line indicates the mutual fund's optimal portfolio weight in the top panels, and the firm's equilibrium optimal capital for the next period in the middle and bottom panels. The dashed black line indicates the firm's current level of capital.

Figure 4: Policy Function Over Fund Size

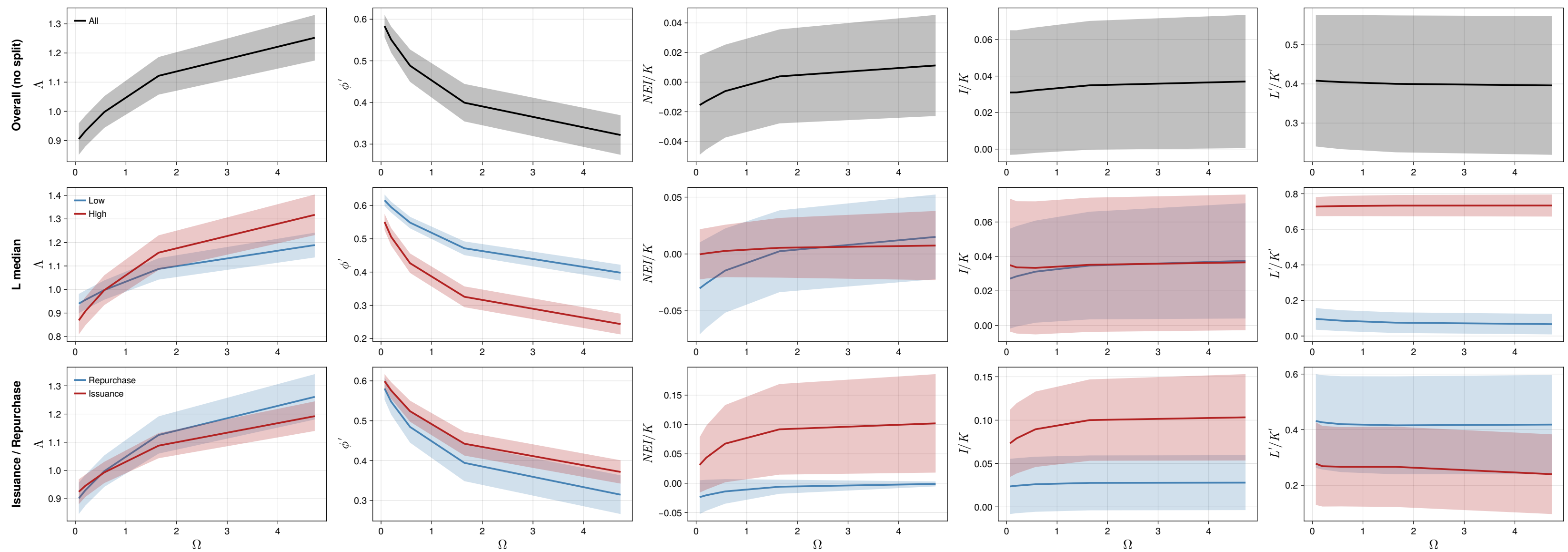


Figure 4 plots policy functions against idiosyncratic fund size Ω from the baseline model, weighted by the model's stationary distribution over states. The top row shows the full sample (no split). The middle row splits the sample by leverage L at its population median (Low/High in blue/red). The bottom row splits firms by equity issuance vs. repurchase behavior (Repurchase/Issuance in blue/red). The five columns are price pressure Λ, next-period mutual fund portfolio weight ϕ', net equity issuance scaled by capital NEI/K, investment rate I/K, and next-period leverage L'/K'. Lines show the weighted mean ±0.5 weighted standard deviation within each subsample.

Figure 5: Policy Function Over Mutual Fund Ownership

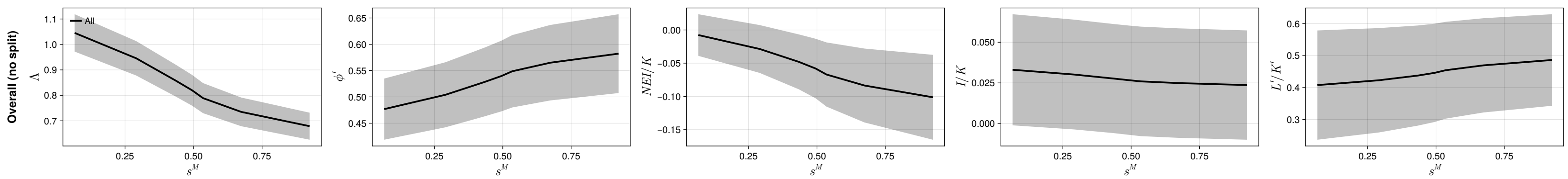
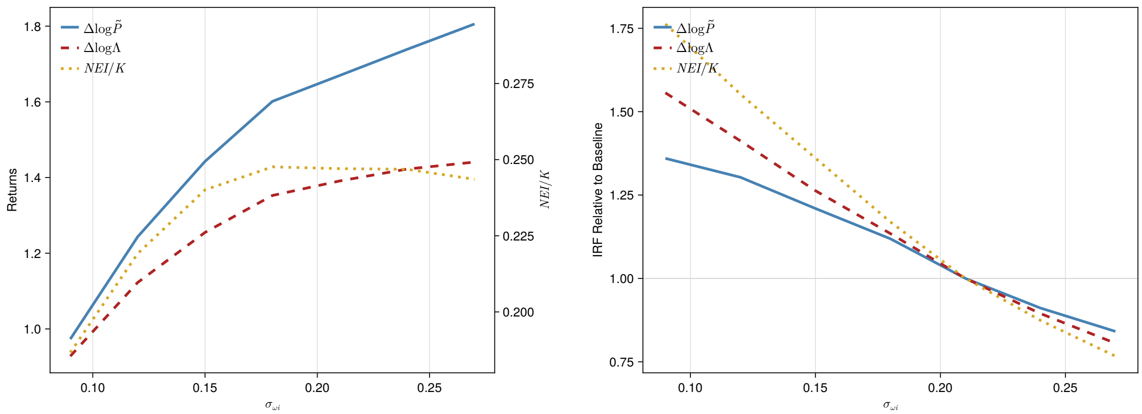


Figure 5 plots policy functions against mutual fund ownership s^M from the baseline model, weighted by the model's stationary distribution over states, for the full sample (no split). The five columns are price pressure Λ, next-period mutual fund portfolio weight ϕ', net equity issuance scaled by capital NEI/K, investment rate I/K, and next-period leverage L'/K'. Lines show the weighted mean ± 0.5 weighted standard deviation.

Figure 6: Responses Under Different Shock Sizes

(a) IRF to One-Standard-Deviation Shocks (b) Per-Unit IRF Normalized to Baseline



[Figure 6](#) illustrates the initial impulse responses of market value, price pressure, and NEI under different mutual fund flow shock sizes. Panel A presents the IRFs to one-standard-deviation shocks of different sizes. Panel B presents the per-unit IRFs normalized to those under the baseline shock size. For each $\sigma_{\omega i}$, the plotted value is $\frac{\mathrm{IRF}(\sigma_{\omega i})}{\mathrm{IRF}(0.21)} \frac{0.21}{\sigma_{\omega i}}$. The blue solid line shows the market value response, the red dashed line shows the price pressure response, and the yellow dotted line shows the net equity issuance response (on the right axis in Panel A). The responses are averages across 10 simulated panels, where each panel contains 3,000 firms over 1,000 quarters, with the first 400 quarters discarded.

Tables

Table 1: Summary Statistics

	count	mean	sd	min	max
$f_{i,t}$	172,584	-0.00	1.00	-2.97	3.04
$R_{i,t}$ (%)	172,584	2.51	25.28	-66.42	89.99
$NEI_{i,t}/K_{i,t-1}$ (%)	172,584	0.94	8.45	-9.35	62.89
$s^M_{i,t}$ (%)	172,568	24.57	16.71	0.07	60.32
$dK_{i,t}$ (%, AT)	172,584	1.57	11.22	-28.45	58.70
$dK_{i,t}$ (%, PPENT)	169,147	2.20	12.61	-29.64	72.31
$dBE_{i,t}$ (%)	171,054	1.92	13.58	-57.85	81.17

Table 1 presents the summary statistics of the merged panel. $f_{i,t}$ is holding-weighted and standardized idiosyncratic mutual fund flow shocks. $R_{i,t}$ is quarterly stock return from CRSP. $NEI_{i,t}$ is Sale of Common and Preferred Stock net of Repurchase of Common and Preferred Stock and Dividends. In robustness checks, I alternatively define this variable without subtracting dividends or apply a 2% market equity filter following McKeon (2015). In the baseline, to be consistent with the model, $K_{i,t-1}$ is lagged total assets. $s^M_{i,t}$ is mutual fund holding defined as total number of shares held by mutual funds divided by total number of shares outstanding. Investment $dK_{i,t}$ is the DHS growth rate (Davis et al., 1996) of total asset. Book equity is Total Assets net of Retained Earnings and Total Liabilities. $dBE_{i,t}$ is the DHS growth rate of book equity. The sample is from 2007Q1 to 2021Q4.

Table 2: Other Firm Fundamental Responses

	$dBE_{i,t}$	$GEI_{i,t}/K_{i,t-1}$	$dCash_{i,t}$	$dK_{i,t}$ (PPENT)	$dK_{i,t}$ (AT)	$\Delta\text{Leverage}_{i,t}$
	(1)	(2)	(3)	(4)	(5)	(6)
$f_{i,t}$	0.132***	0.157***	0.450***	0.209***	0.238***	-0.034***
	(3.15)	(4.70)	(2.88)	(4.26)	(5.27)	(-2.60)
Observations	157092	158593	158270	155387	158605	156590
R-Squared	0.064	0.263	0.120	0.208	0.115	0.078
Time	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Clustered By	Firm	Firm	Firm	Firm	Firm	Firm

Table 2 presents regression coefficients from $\Delta y_{i,t} = \alpha_i + \delta_t + \beta f_{i,t} + X_{i,t-1} + \varepsilon_{m,t}$. Dependent variables include the DHS growth rate of book equity, cash, physical capital (PPENT), total assets (AT), and leverage. Gross equity issuance is defined as Sale of Common and Preferred Stock when it is larger than 2% of lagged market equity following McKeon (2015), and zero otherwise. Control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). All dependent variables are in percentage points. Standard errors are clustered at firm level. The sample is from 2007Q1 to 2021Q4.

Table 3: Initial Return and Issuance Response, Positive and Negative Shocks

	$R_{i,t}$ (%)		$NEI_{i,t}/K_{i,t-1}$ (%)	
	(1)	(2)	(3)	(4)
$f_{i,t} \times 1\{f \geq 0\}$	2.111***	1.949***	0.219***	0.221***
	(13.03)	(11.82)	(3.67)	(3.68)
$f_{i,t} \times 1\{f < 0\}$	1.442***	1.821***	0.108**	0.149***
	(10.53)	(12.54)	(2.13)	(2.91)
Observations	216248	158605	216248	158605
R-Squared	0.249	0.307	0.249	0.298
Time	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Clustered By	Firm	Firm	Firm	Firm

Table 3 presents regression coefficients from $\Delta y_{i,t} = \alpha_i + \delta_t + \beta f_{i,t} + X_{i,t-1} + \varepsilon_{m,t}$. $\mathbb{1}\{f \geq 0\}$ ($\mathbb{1}\{f \geq 0\}$) is a dummy variable that equals one when the flow shock is greater than or eqaul to (smaller than) 0. Control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). Standard errors are clustered at firm level. The sample is from 2007Q1 to 2021Q4.

Table 4: Initial Return and Issuance Response, Subsamples

	(1) $R_{i,t}$ (%)	(2) $NEI_{i,t}/K_{i,t-1}$ (%)	(3) $R_{i,t}$ (%)	(4) $NEI_{i,t}/K_{i,t-1}$ (%)
Panel A: By Mutual Fund Ownership				
	High		Low	
$f_{i,t}$	3.392***	0.195***	1.303***	0.145***
	(20.75)	(4.03)	(10.73)	(3.35)
Observations	79567	79567	72177	72177
R-Squared	0.376	0.264	0.289	0.339
Panel B: By Q				
	High		Low	
$f_{i,t}$	2.181***	0.318***	1.440***	0.006
	(14.77)	(4.59)	(10.79)	(0.30)
Observations	71570	71570	73015	73015
R-Squared	0.320	0.368	0.373	0.297
Panel C: By Leverage				
	High		Low	
$f_{i,t}$	2.243***	0.147***	1.624***	0.229***
	(14.70)	(3.66)	(12.33)	(4.41)
Observations	72433	72433	76946	76946
R-Squared	0.361	0.344	0.301	0.329
Time	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Clustered By	Firm	Firm	Firm	Firm

Table 4 presents regression coefficients from $\Delta y_{i,t} = \alpha_i + \delta_t + \beta f_{i,t} + X_{i,t-1} + \varepsilon_{m,t}$. I sort firms by lagged mutual fund ownership, Tobin's Q, or leverage into two equal-sized groups within each Fama–French 12 industry. Control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). Standard errors are clustered at firm level. The sample is from 2007Q1 to 2021Q4.

Table 5: Calibrated Parameters

Panel A: Aggregate		
Long-run mean of fund size	$\exp(\bar{q})$	0.29
Persistence of aggregate productivity shocks	ρ_a	0.95
Volatility of aggregate productivity shocks	σ_a	0.006
Inverse of risk-free rate	β	0.995
Risk aversion, constant component	γ_0	30
Risk aversion, time-varying component	γ_1	-100
Panel B: Mutual Funds		
Persistence of idiosyncratic flow shocks	ρ_ω	0.98
Volatility of idiosyncratic flow shocks	σ_ω	0.21
Risk aversion	γ^M	2.0
Targeted portfolio position	$\bar{\phi}$	0.60
Panel C: Firm		
Persistence of idiosyncratic productivity shock	ρ_x	0.97^3
Volatility of idiosyncratic productivity shock	σ_x	0.19
Long-run mean of idiosyncratic productivity shock	$\bar{x}$	-3.08
Decreasing returns to scale	α	0.65
Depreciation rate	δ	0.03
Corporate tax rate	τ	0.3
Dividend tax rate	τ_D	0.15
Collateral constraint	φ	0.8
Saving-borrowing rate wedge	κ	0.005/4

Table 5 presents the calibrated parameters for the baseline model.

Table 6: Identified Parameters

Panel A: Parameter Estimates		
Parameter	Symbol	Value
Fixed issuance costs	c_{h0}	0.032
		(0.008)
Fixed buyback costs	c_{b0}	0.010
		(0.001)
Variable issuance costs	c_{h1}	0.552
		(0.072)
Variable buyback costs	c_{b1}	1.139
		(0.074)
Fund target deviation costs	c_{ϕ}	2.391
		(0.085)
Residual investor trading costs	c_r	0.878
		(0.031)
Capital adjustment costs	c_k	0.492
		(0.001)

Panel B: Targeted Moments		
Moment	Data	Model
Return IRF, $h = 0$ (%)	1.81	1.67
Return IRF, $h = 1$ (%)	0.11	-1.32
NEI IRF, $h = 0$ (%)	0.28	0.25
SD(Δs^M) (%)	3.40	3.26
$\mathbb{E}[NEI_t/K_{t-1} \mid NEI_t \geq 0]$ (%)	3.57	1.62
$\mathbb{E}[NEI_t/K_{t-1} \mid NEI_t \leq 0]$ (%)	-0.83	-1.02
SD($NEI_t/K_{t-1} \mid NEI_t \geq 0$) (%)	14.56	6.36
SD($NEI_t/K_{t-1} \mid NEI_t \leq 0$) (%)	1.65	1.68
β(SALE/K, I/K)	0.15	0.15
SD(R) (%)	23.87	16.84
SD(I/K) (%)	7.38	6.56

Table 6 presents estimated parameters and compares the moments from the simulated economy with data. Standard errors are in parentheses. The moments are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded. Data moments are from 2007Q1 to 2021Q4.

Table 7: Sensitivity of Parameters With Respect to Moments

	c_{h0}	c_{b0}	c_{h1}	c_{b1}	c_{ϕ}	c_r	c_k
Return IRF, $h = 0$ (%)	-0.018	-0.088	0.052	0.111	0.486	0.643	0.138
Return IRF, $h = 1$ (%)	-0.023	0.160	-0.094	-0.173	-0.351	-0.446	-0.017
NEI IRF, $h = 0$ (%)	-0.358	0.716	-0.364	-0.691	0.058	0.053	0.692
Var(Δs^M) ($\times 10^4$)	-0.109	-0.015	0.017	0.022	0.650	-0.148	0.183
$\mathbb{E}[NEI_t/K_{t-1} \mid NEI_t \geq 0]$ (%)	-0.187	-0.199	0.232	0.200	0.020	0.056	-0.305
$\mathbb{E}[NEI_t/K_{t-1} \mid NEI_t \leq 0]$ (%)	0.029	0.082	0.000	-0.088	-0.009	-0.013	-0.112
Var($NEI_t/K_{t-1} \mid NEI_t \geq 0$) ($\times 10^4$)	0.492	0.469	-0.705	-0.479	-0.042	-0.073	-0.183
Var($NEI_t/K_{t-1} \mid NEI_t \leq 0$) ($\times 10^4$)	0.044	0.187	0.011	-0.211	-0.026	-0.022	-0.117
β(SALE/K, I/K)	0.033	-0.052	0.023	0.051	0.018	0.031	-0.043
Var(R) ($\times 10^4$)	-0.453	0.211	0.124	-0.211	-0.346	-0.492	0.300
Var(I/K) ($\times 10^4$)	0.531	0.123	-0.444	-0.142	-0.037	-0.139	-0.364

Table 7 presents local elasticity of parameters (columns) with respect to moments (rows) at the estimated parameter values. The matrix is calculated following Andrews et al. (2017). Each element is scaled by the square root of the ratio of the moment variance to the parameter variance. Elasticities are calculated using central finite difference with a step size equal to 0.01 of the estimated parameter value. The moments are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded.

Table 8: Decomposition of Return Impulse Responses

	Full Sample	Low Q		High Q	
		Low Lev.	High Lev.	Low Lev.	High Lev.
Return	1.67	1.37	1.63	1.97	1.82
Price Pressure	1.39	1.19	1.37	1.53	1.50
Share (%)	*83*	*87*	*84*	*78*	*83*
Fundamental Value	0.28	0.18	0.26	0.44	0.32
Share (%)	*17*	*13*	*16*	*22*	*17*

Table 8 presents the decomposition of return impulse responses estimated from the baseline model. The moments are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded. Firms are sorted into terciles by beginning-of-period Tobin's Q and by leverage; the table reports the four groups formed by intersecting the top and bottom terciles of each sort.

Table 9: Instrumental Variable Estimation

Panel A: Residual Investor Elasticity						
	$d\log s^R$					
	(1)	(2)	(3)	(4)	(5)	(6)
$d\log\tilde{P}$	0.038	-1.305				
	(0.001)	(0.004)				
$d\log\Lambda$			-0.644	-1.567		
			(0.002)	(0.002)		
$(P-\tilde{P})/\tilde{P}$					-1.893	-1.609
					(0.004)	(0.002)
Estimator	OLS	IV	OLS	IV	OLS	IV
Panel B: Mutual Fund Elasticity						
	$d\log s^M$	$d\log\tilde{P}$	$d\log s^M$	$d\log s^M$	$d\log P$	$d\log s^M$
	(1)	(2)	(3)	(4)	(5)	(6)
$d\log\tilde{P}$	-0.174		-0.736			
	(0.004)		(0.003)			
$d\log X$		0.264			0.289	
		(0.000)			(0.000)	
$d\log P$				-0.562		-0.672
				(0.003)		(0.002)
Estimator	OLS	OLS	IV	OLS	OLS	IV

Table 9 presents instrumental variable estimations for the demand elasticity in simulated samples. Standard errors are clustered at firm level. The coefficients from the simulated sample are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded. Standard errors are in parentheses.

Table 10: Stock Return and Equity Issuance Responses to Small Fund Flow Shocks

	$R_{i,t}$ (%)		$NEI_{i,t}/K_{i,t-1}$ (%)	
	(1)	(2)	(3)	(4)
$f_{i,t}$	2.659***	3.150***	0.234***	0.205***
	(15.63)	(16.46)	(3.89)	(3.43)
Observations	127162	94005	127162	94005
R-Squared	0.232	0.299	0.284	0.334
Time	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes
Clustered By	Firm	Firm	Firm	Firm
Baseline coeff.	1.806	1.884	0.168	0.185
Normalized coeff.	1.472	1.672	1.389	1.112

Table 10 reports per-unit stock return and equity issuance responses to fund flow shocks, estimated on the subsample of small shocks. The sample is restricted to firm-quarters where $|f_{i,t}| \leq 1$ baseline standard deviation; within this restricted sample, the standard deviation of $f_{i,t}$ is approximately half that in the full sample. The coefficient on $f_{i,t}$ is the raw OLS estimate from the restricted subsample, in the original units (percent). The "Baseline coeff." row reports the corresponding full-sample coefficient for reference. The "Normalized coeff." row reports the ratio of the restricted-sample coefficient to the baseline coefficient, so that a value of 1.0 indicates the per-unit response in the restricted sample equals the full-sample estimate. All regressions include firm and time fixed effects with standard errors clustered by firm. Columns (2) and (4) additionally control for $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). Standard errors are clustered at firm level. The sample is from 2007Q1 to 2021Q4.

Table 11: Capital Misallocation: Percentage Change from Baseline

Param	Base	CF	Description	Agg TFP
		Panel A: Price-Pressure Channel		
$\exp(\bar{q})$	0.290	0.220	Small fund sector	0.006
$\sigma_{\omega i}$	0.210	0.240	High flow vol.	-0.014
c_r	0.878	1.000	High capacity constr.	-0.028
Λ	Varied	1.000	No price pressure	0.120
		Panel B: Real Channel		
c_k	0.492	0.450	Low capital adj. cost	0.142

Table 11 reports the percentage change in aggregate TFP relative to the baseline model under each counterfactual. Panel A's first three rows each shift one underlying parameter that drives price-pressure dynamics, holding the rest of the model at baseline: a smaller mutual fund sector ($\exp(\bar{q})$), higher flow-shock volatility ($\sigma_{\omega i}$), and less elastic residual investors (c_r), each moved from its baseline value to the counterfactual value shown in the table. The row below sets $\Lambda \equiv 1$, shutting off price pressure fluctuations economy-wide altogether. Panel B perturbs the capital adjustment cost (c_k), a real friction unrelated to price pressure. The moments are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded.

Table 12: MPK-Dispersion Decomposition, by Leverage and Q

	Full Sample	Low Q		High Q	
		Low Lev.	High Lev.	Low Lev.	High Lev.
ΔVar(log MPK)	0.102	-0.037	0.054	0.026	0.058
Var(w)	0.034	0.008	0.008	0.010	0.008
2Cov(m_0, w)	0.068	-0.044	0.046	0.016	0.050
Within	-0.016	-0.033	0.024	-0.031	0.023
Between	0.084	-0.011	0.021	0.047	0.027
$\propto\ s_g(\bar{m}_g^0 - \bar{m}^0)(\bar{w}_g - \bar{w})$:					
$\bar{m}_g^0$	-3.39	-3.48	-3.15	-3.53	-3.23
$\bar{w}_g$	0.0078	0.0093	0.0097	0.0023	0.0161
Capital share (%)	100	31	22	35	12

Table 12 decomposes the capital-weighted contribution of price pressure to the cross-sectional variance of $m_i \equiv \log \mathrm{MPK}_i$, $\Delta^0\mathrm{Var}(m_i) = \mathrm{Var}(w_i) + 2\mathrm{Cov}(m_i^0, w_i)$, where Δ^0 denotes baseline minus $\Lambda \equiv 1$ and $w_i \equiv m_i - m_i^0$ is firm i's price-pressure wedge, with m_i^0 its m_i in the $\Lambda \equiv 1$ economy (Table 11's $\Lambda \equiv 1$ row). Columns split the sample by leverage and Q medians. The $2\mathrm{Cov}(m_i^0, w_i)$ row is further split into its within-group and between-group components (Equation (56)). The two rows beneath Between report each group's mean baseline log MPK $\bar{m}_g^0$ and capital-share-weighted mean wedge $\bar{w}_g$, in the raw units of m_i^0 and w_i rather than percentage points, feeding Equation (56)'s between-group term; Full Sample entries are the capital-share-weighted grand means $\bar{m}^0$ and $\bar{w}$ the group deviations are taken against. All other values are in percentage points of implied TFP loss; capital shares are reported at the bottom of the table. The Full Sample $\Delta^0\mathrm{Var}(m_i)$ of 0.102 percentage points is an approximation to the exact TFP gain of 0.120 percentage points from eliminating price pressure reported in Table 11's $\Lambda \equiv 1$ row (Section 6.2).

A Empirical Appendix

A.1 Variable Definition

Variables from CRSP/Compustat Merged Database

I retrieve monthly returns `ret` from CRSP, adjust for delisting returns following Bali et al. (2017), and subtract the risk-free rate. I then aggregate the excess returns to quarterly level. Number of shares outstanding is the last observation of `shrout` for each quarter. Market capitalization is calculated as the absolute value of the product of `shrout` and `altprc`.

In the baseline results, following the convention in literature, NEI is defined as `sstk` - `prstkcy` - `dvy`. For robustness, I alternatively define NEI as `sstk` - `prstkcy` to exclude effects of dividends.

I include some other firm outcome variables. dBE is the DHS growth rate of `at` - `re` - `lt`. GEI/K is `sstk` divided by lagged `at` when `sstk` is larger than 2% of the market capitalization. $dCash$ is the DHS growth rate of `che`. $dK(PPENT)$ is the DHS growth rate of `ppent`. ΔLeverage is the first difference of leverage, which is defined as `dltt` + `dlc` scaled by `at`. Tobin's Q is defined as (market capitalization + `at` - texttceq) / `at`.

I include portfolio-level characteristics to predict mutual fund flows. Profitability is defined as (`sale` - `cogs` - `xsga` - `xint`)/Book Equity, where Book Equity is `seq` + `ceq` + `pstk` + `at` - `lt` + `txditc` - `pstk`. Log book-to-market ratio is the logarithm of Book Equity divided by market capitalization. Log market size is the logarithm of market capitalization. Log asset growth is the log difference of `at`. Market beta is calculated from rolling window regressions estimated using monthly return data. For each firm, I regress excess returns on excess market returns. The window size is 60 months and the firm must have at least 48 quarters of observations. Risk-free rate and market return is from Kenneth French's Data Library.

CRSP Survivor-Bias-Free US Mutual Fund Database

Mutual fund excess return is calculated at monthly level by subtracting market return from `mret` and then aggregated to quarterly level. Mutual fund size is `mtna`. I use `percent_tna` as weights to aggregate firm characteristics to portfolio level. I use lagged `nbr_shares` as weights to aggregate mutual fund flow shocks to firm level. In rare cases where the last observation of `nbr_shares` is missing, I use information up to two quarters to fill the missing values. Mutual fund ownership s^M is the sum of `nbr_shares` across mutual funds divided by `shrout` from CRSP.

Figure A.1 plots the coefficients estimated from the rolling window regressions. Consistent with the literature, lagged flows and excess returns have significant predictive power. However, the coefficients are time-varying. The coefficients on portfolio characteristics are also significant and time varying.

Figure A.1: Coefficients of Rolling Window Regressions

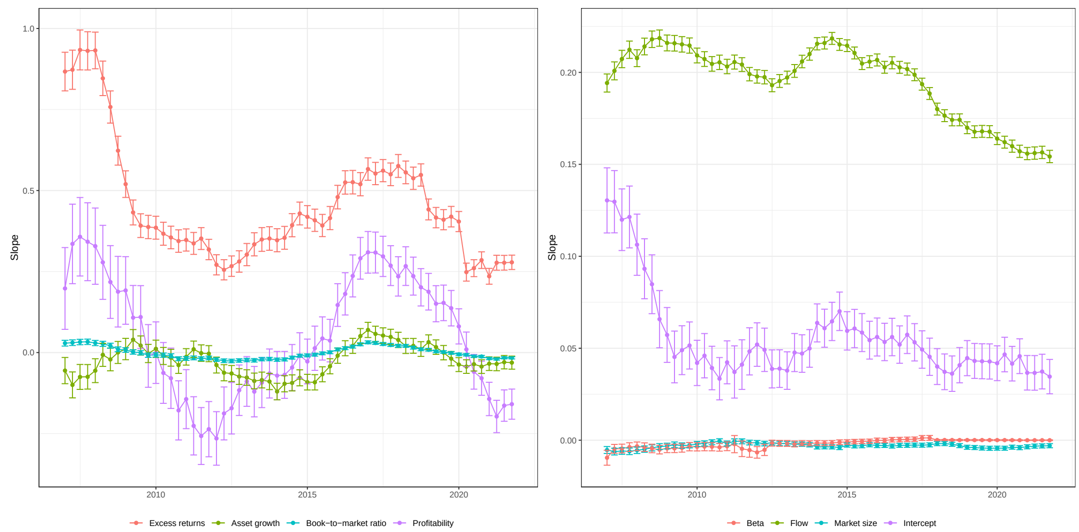


Figure A.1 presents coefficients estimated from regressing fund flows on lagged flow, fund excess returns, and lagged portfolio characteristics, including log market equity, book-to-market ratio, profitability, investment, and market beta. Each regression is performed on a panel with 16 quarters. All regressors are winsorized at 1% level.

A.1.1 Active and Passive Funds

In CRSP mutual fund database, the variable `index_fund_flag` identifies if a fund is an index fund. If a fund is a pure index fund (`index_fund_flag==D`), an index-based fund (`index_fund_flag==B`), or an index fund enhanced (`index_fund_flag==E`), I classify it as passive. Further, following Appel et al. (2016), I classify a fund as passively managed if its name includes a string that identifies it as an index fund. The strings include *"Index", "Idx", "Indx", "Ind ", "Russell", "S & P", "S and P", "S&P", "SandP", "SP", "DOW", "Dow", "DJ", "MSCI", "Bloomberg", "KBW", "NASDAQ", "NYSE", "STOXX", "FTSE", "Wilshire", "Morningstar", "100", "400", "500", "600", "900", "1000", "1500", "2000", "5000".*

A.2 Robustness

A.2.1 Endogeneity Concerns of Mutual Fund Flow Shocks

In shift-share instrument settings, validity can be established through either exogenous shares (Goldsmith-Pinkham et al., 2020) or exogenous shifts (Borusyak et al., 2018, 2025). I provide supporting evidence for both.

Share-based validity. Under the share-based design (Goldsmith-Pinkham et al., 2020; Chaudhary et al., 2022), identification requires that the lagged ownership shares $S_{i,m,t-1}$ in equation (2) are not themselves allocated in anticipation of firm investment and financing decisions. The concern is that fund managers with superior information or market-timing skills may identify promising sectors or characteristic-based trends—for instance, an anticipated industry-wide expansion such as the rise of electric vehicles, or an upcoming revival of value stocks—and overweight firms in those buckets ahead of the subsequent surge in firm investment and equity issuance. Table A.1 addresses this. Columns (1)–(2) use Fama–French 12 industry-by-quarter fixed effects, absorbing any time-varying sector-level tilt in fund portfolios. Columns (3)–(8) replace time fixed effects with size-decile-, book-to-market-decile-, and profitability-decile-by-quarter fixed effects, addressing analogous tilts based on firm characteristics. The coefficients on $f_{i,t}$ remain stable across all columns for both returns and net equity issuance.

Table A.1: Robustness: Characteristic-by-Time Fixed Effects

	Industry		Size		BM		Profitability	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$
$f_{i,t}$	1.556***	0.167***	1.753***	0.152***	1.764***	0.154***	1.806***	0.167***
	(16.20)	(4.97)	(17.51)	(4.50)	(18.18)	(4.50)	(19.26)	(5.19)
Obs.	158605	158605	158605	158605	158605	158605	158605	158605
R^2	0.342	0.304	0.321	0.335	0.321	0.304	0.333	0.325
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Clustered	Firm	Firm	Firm	Firm	Firm	Firm	Firm	Firm

Table A.1 presents regression results from $\Delta y_{i,t} = \alpha_i + \delta_{j,t} + \beta f_{i,t} + X_{i,t-1} + \varepsilon_{i,t}$, where $\delta_{j,t}$ denotes characteristic-by-time fixed effects. Columns (1)–(2) use Fama–French 12 industry-by-quarter fixed effects; columns (3)–(4) use size-decile-by-quarter fixed effects; columns (5)–(6) use book-to-market-decile-by-quarter fixed effects; columns (7)–(8) use profitability-decile-by-quarter fixed effects. Both dependent variables are in percentage points. Control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$, and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by assets, and investment). Standard errors are clustered at the firm level. The sample is from 2007Q1 to 2021Q4.

Following the reasoning in Chaudhary et al. (2022), Figure A.2 addresses the possibility of firm-specific anticipatory allocation. I reconstruct firm-level shocks using shares lagged four quarters, $f^{(4)}_{i,t} = \sum_m S_{i,m,t-4} f_{m,t} / \sum_m S_{i,m,t-4}$, ensuring that portfolio weights predate any near-term repositioning toward specific firms. The impulse responses for returns and net equity issuance are qualitatively unchanged, though the point estimates are slightly attenuated. This is expected: four-quarter-old weights are a noisier measure of contemporaneous fund exposure, as funds rebalance their portfolios over intervening quarters, introducing classical measurement error that biases the estimates toward zero.

Shift-based validity. The concern is that fund flow shocks $f_{m,t}$ reflect anticipated firm fundamentals

Figure A.2: Firm Responses: Four-Quarter Lagged Share Weights

(a) $R_{i,t+h}$ (b) $NEI_{i,t+h}/K_{i,t-1}$

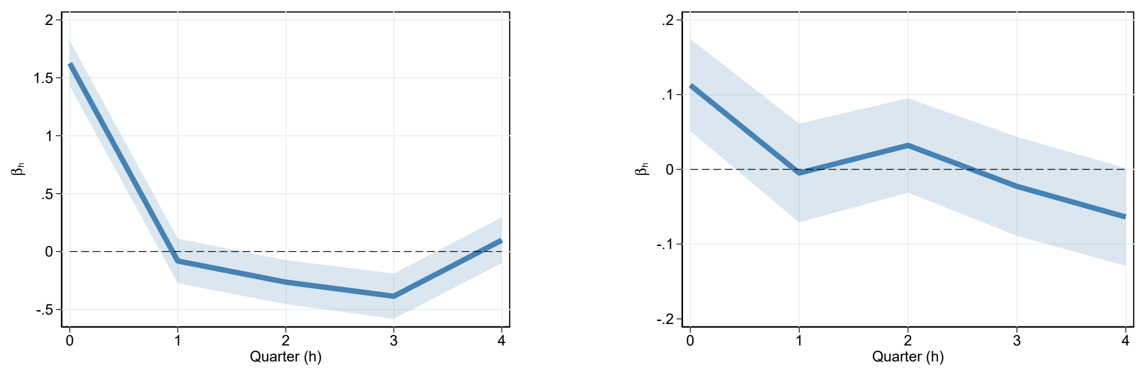


Figure A.2 presents impulse response functions of returns and net equity issuance. The blue solid line plots coefficients β_h from estimating the local projection following Jordà (2005): $\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t}^{(4)} + X_{i,t-1} + \varepsilon_{m,t+h}$ for $h = 0, 1, \ldots, 4$. The firm-level shock is constructed using shares lagged four quarters as weights, $f_{i,t}^{(4)} = \sum_m S_{i,m,t-4} f_{m,t} / \sum_m S_{i,m,t-4}$, where $S_{i,m,t}$ denotes the number of shares of firm i held by fund m at time t. The shaded area indicates 95% pointwise confidence bands using standard errors clustered at firm level. Control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}^{(4)}$, $s_{i,t-4}^{M}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). The sample is from 2007Q1 to 2021Q4.

rather than exogenous demand shocks, even after controlling for observable fund and portfolio characteristics. I address three specific concerns.

First, retail investors may anticipate factor-level performance and direct flows into funds with corresponding exposures. For instance, they might flow into industry funds ahead of a sectoral expansion, or into value funds ahead of a value-factor recovery, making fund flows endogenous to future fundamentals along those dimensions. Table A.1 addresses this by replacing time fixed effects with factor-group-by-quarter fixed effects: Fama–French 12 industry, size decile, book-to-market decile, and profitability decile. Identification then comes entirely from within-group variation. The equity issuance coefficients are stable across all columns. The return coefficient attenuates mildly in the industry specification (1.556 versus 1.9% in the baseline), consistent with cross-asset substitution within industries documented by Haddad et al. (2021) and Chaudhary et al. (2022); the characteristic-based columns show minimal attenuation. Since this paper targets the total price response to fund flow shocks, not the within-group net-of-substitution effect, the baseline specification remains appropriate.

Second, fund flows may be driven by the salient performance of superstar firms within an industry rather than fund-idiosyncratic exogenous demand shocks. Table A.2 re-estimates the baseline after removing the top $N \in \{2, 10, 20\}$ largest firms by market capitalization within each Fama–French 12 industry-quarter. Both the return and equity issuance coefficients remain stable and highly significant across all columns.

Table A.2: Robustness: Excluding the Largest Firms Within Each Industry

	Top 2		Top 10		Top 20	
	(1)	(2)	(3)	(4)	(5)	(6)
	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$	$R_{i,t}$	$NEI_{i,t}/K_{i,t-1}$
$f_{i,t}$	1.863***	0.185***	1.805***	0.187***	1.735***	0.189***
	(19.29)	(5.50)	(18.48)	(5.49)	(17.51)	(5.44)
Observations	157471	157471	152425	152425	146547	146547
R^2	0.307	0.298	0.306	0.297	0.306	0.297
Time	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Clustered	Firm	Firm	Firm	Firm	Firm	Firm

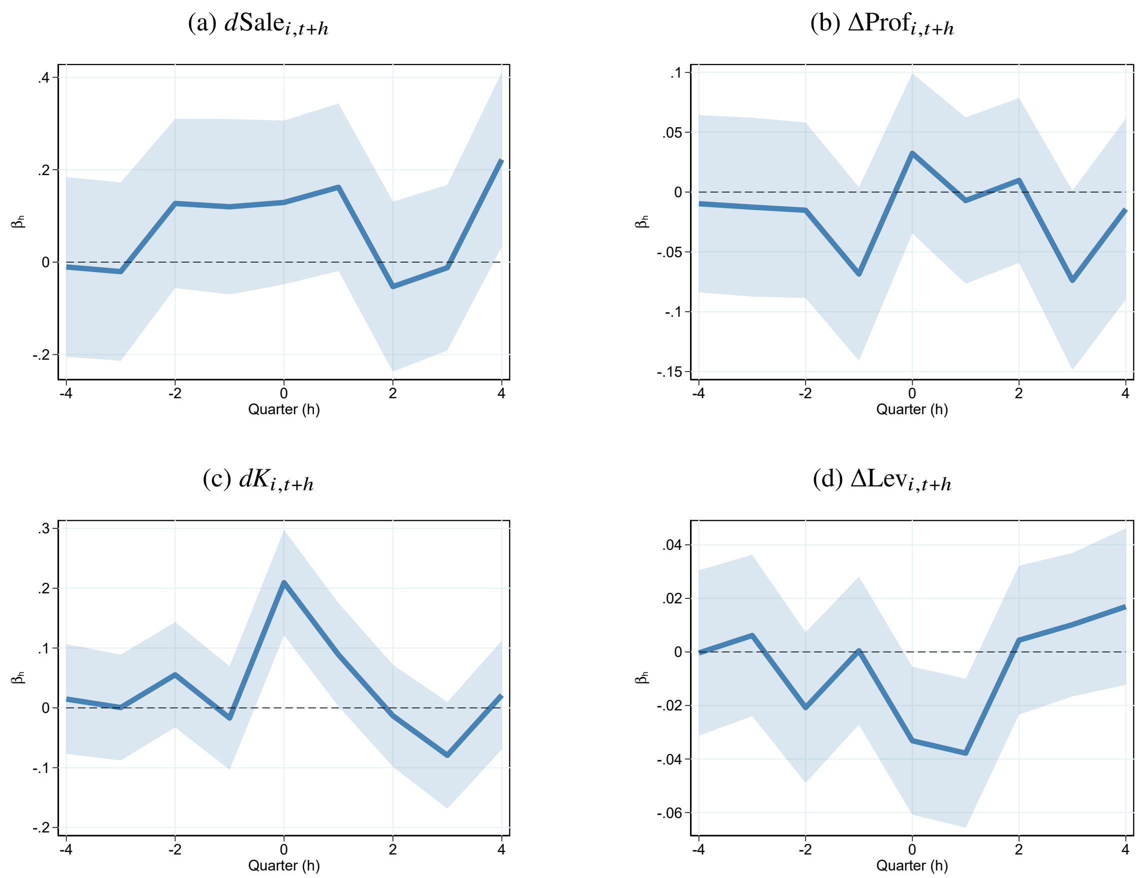
Table A.2 replicates the baseline regression on subsamples that exclude the top N largest firms by market capitalization within each Fama–French 12 industry-quarter, for $N \in \{2, 10, 20\}$, addressing the concern that fund flows are driven by the salient performance of the largest firms within industries. Both dependent variables are in percentage points. Control variables include lagged dependent variable, lagged flow shock, mutual fund ownership, and lagged firm characteristics (log market equity, log book-to-market ratio, cash-to-assets ratio, profitability, and log asset growth). All regressions include firm and quarter fixed effects. Standard errors are clustered at the firm level. The sample is from 2007Q1 to 2021Q4.

Third, investors might observe firm actions before flow shocks and allocate capital accordingly. To address this concern, I estimate the baseline specification for sales, profitability, investment, and leverage over quarters $h = -4, \ldots, -1, 0, 1, \ldots, 4$ relative to the flow shocks. Sales and profitability capture top-line and bottom-line performance, investment captures forward-looking decisions, and leverage captures capital structure decisions. Figure A.3 shows no significant pre-trend in any variable, ruling out systematic anticipatory capital allocation ahead of flow shocks.

A.2.2 Alternative Definition of NEI

Figure A.4 presents the impulse response from estimating (3). The dependent variable is defined as `sstk - prstkcy`. The result is very similar to the baseline results in Figure 1. It is to be expected since the variation of dividend in the short term is small.

Figure A.3: Pre-event Firm Actions



[Figure A.3](#) presents impulse response functions of sales growth, profitability, asset growth, and leverage. The blue solid line plots coefficients β_h from estimating the local projection following Jordà (2005):
$\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t} + X_{i,t-(h+1)} + \varepsilon_{m,t+h}$ for $h = -4, \ldots, -1, 0, 1, \ldots, 4$. The shaded area indicates 95% pointwise confidence bands using standard errors clustered at firm level. control variables $X_{i,t-(h+1)}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ if $h \geq 0$, $\Delta y_{i,t-(h+1)}$, $f_{i,t-(h+1)}$, $s^M_{i,t-(h+1)}$ if $h < 0$. and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). The sample is from 2007Q1 to 2021Q4.

A.3 Fund Flow Shocks

A.3.1 Fund Portfolio Responses

As an additional check to the results presented in Figure 1, I check whether mutual funds scale up their positions on firms they are currently holding. To that end, let $\phi_{i,m,t}$ denote the portfolio weight of mutual fund m in firm i, and $I_{m,t}$ denote the set of firms held by mutual fund m, I calculate the value-weighted increase in their current portfolio $\Delta \log S_{m,t}$ by

$$\Delta \log S_{m,t} = \sum_{i \in I_{m,t} \cap I_{m,t-1}} \phi_{i,m,t-1} (\log S_{i,m,t} - \log S_{i,m,t-1}) \tag{A.1}$$

Figure A.4: Firm NEI (Excl. Dividends) Response

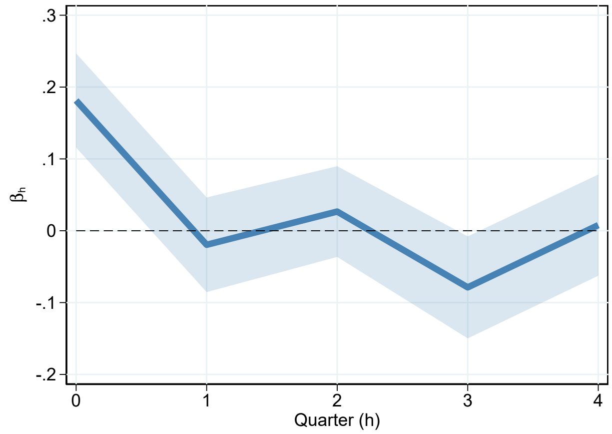


Figure A.4 presents impulse response functions of net equity issuance (excluding dividends), The blue solid line plots coefficients β_h from estimating the local projection following Jordà (2005): $\Delta y_{i,t+h} = \alpha_i + \delta_t + \beta_h f_{i,t} + X_{i,t-1} + \varepsilon_{m,t+h}$ for $h = 0, 1, \ldots, 4$. The shaded area indicates 95% pointwise confidence bands using standard errors clustered at firm level. control variables $X_{i,t-1}$ include $\Delta y_{i,t-1}$, $f_{i,t-1}$, $s^M_{i,t-1}$ and lagged firm characteristics (log market equity, log book-to-market ratio, cash scaled by asset, and investment). The sample is from 2007Q1 to 2021Q4.

I estimate the following local projection regression:

$$\Delta \log S_{m,t+h} = \alpha_m + \delta_t + \beta_h f_{m,t} + X_{i,t-1} + \varepsilon_{m,t+h} \text{ for } h = 0, 1, \ldots, 4, \tag{A.2}$$

where α_m and δ_t are mutual fund and time fixed effects, and control variables $X_{i,t-1}$ include lagged portfolio weight change and flow shocks. β_h can be interpreted as the $h-$quarter ahead impulse response from the mutual fund flow shocks. The coefficients are plotted in Figure A.5. A one percent unexpected increase in fund size leads funds to increase holdings in their current portfolio by 0.56 percent on impact. Consistent with the results on Δs^M at firm level, where mutual fund ownership rises 1.4% on impact, mutual funds continue gradually raising positions in existing holdings over several quarters. Four quarters after the shock, the cumulative increase reaches 0.95 percent. In other words, while pass-through is incomplete initially, it accumulates to nearly complete over time, rather than adjusting in one step. This gradual buildup, at both the fund and firm level, suggests that mutual funds aim to avoid price pressure and prefer a gradual adjustment.[13]

This pattern has important implications. First, assuming full pass-through on impact would overestimate price impact and underestimate demand elasticity. Second, the near-complete pass-through after several quarters supports the validity of the share-based identification design, as it

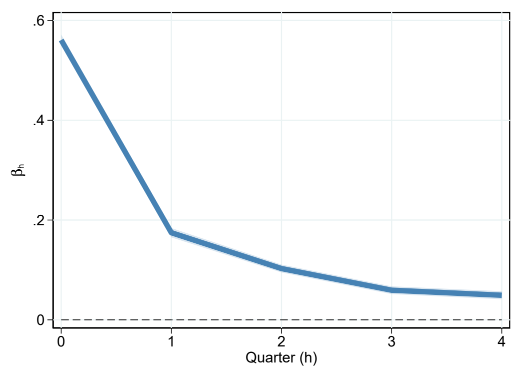
[13]Note that this is not due to persistent $f_{i,t}$ since the AR(1) coefficient of $f_{i,t}$ estimated following Han and Phillips (2010) is only 0.017. Also, $f_{i,t-1}$ is always controlled for in the regressions.

suggests funds do not systematically treat firms in their portfolios differently. Third, these dynamics underscore the importance of incorporating dynamic trading behavior into theoretical models.

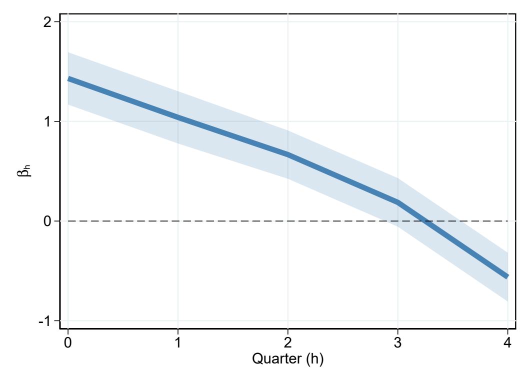
Combining this pass-through evidence with the price response documented in Figure 1, I compute a simple back-of-the-envelope demand elasticity. Following Gabaix and Koijen (2022), the price impact, defined as the percentage price change divided by the percentage change in mutual fund holdings (Δs^M, 1.4% on impact, Panel (b) below), is $1.9/1.4 = 1.36$, implying a pooled elasticity of $1/1.36 = 0.73$. This number is broadly in line with the elasticity calculated by Gabaix and Koijen (2022) using estimates from Lou (2010). However, as Section 3 argues, this calculation is a weighted average of several structural parameters and lacks clear economic interpretation for counterfactual analysis, motivating the structural estimation in Section 4.

Figure A.5: Portfolio Increases

(a) $\Delta \log S_{m,t+h}$



(b) $\Delta s^M_{i,t+h}$



Panel (a) of Figure A.5 presents impulse responses β_h estimated from

$$\Delta \log S_{m,t+h} = \alpha_m + \delta_t + \beta_h f_{m,t} + X_{i,t-1} + \varepsilon_{m,t+h} \text{ for } h = 0, 1, \ldots, 4,$$

where the dependent variable is value-weighted increase in mutual fund's current portfolio. α_m and δ_t are mutual fund and time fixed effects, and control variables $X_{i,t-1}$ include lagged portfolio weight change and flow shocks. β_h can be interpreted as the $h-$quarter ahead impulse response from the mutual fund flow shocks. Panel (b) presents the corresponding firm-level response, $\Delta s^M_{i,t+h}$, estimated from the baseline specification in equation (3). The sample is from 2007Q1 to 2021Q4.

B Model Appendix

B.1 Two-Period Model

Throughout, $H \equiv -E = I - K^{\alpha}$ denotes equity issuance and $\xi(I, Q) \equiv c_r Q/(\hat{\beta} I^{\alpha})$ the fund's price impact. Two constants recur: $D \equiv \gamma_m \sigma^2 + c_{\phi}$, the curvature of the fund's objective in ϕ, and $C \equiv \mathbb{E}[\log A'] - \log \hat{\beta} - r_f + c_{\phi}\bar{\phi} + c_r s^M$. In terms of these, the flow pass-through and the frictionless portfolio weight used in the main text are

$$\rho \;\equiv\; \frac{D}{D + 2\xi} \;\in\; (0,1), \qquad \phi^0 \;\equiv\; \frac{C - \theta}{D}, \tag{A.3}$$

so that $\phi^* = \rho\,\phi^0$, which is (17). The demand shifter θ, an exogenous additive term in the pressure equation, is carried throughout; it is zero in the baseline model and is varied only where Property 2 requires it.

B.1.1 The Trading Stage

Fix investment I, hence $P = \hat{\beta} I^{\alpha}$. The residual investors' optimality condition (14) and market clearing give the pressure equation (15), $\lambda(\phi) = \theta + \xi\phi - c_r s^M$, so the fund's price impact is the constant

$$\frac{d\lambda}{d\phi} = \xi > 0.$$

Substituting $\mu - r_f = \mathbb{E}[\log A'] - \log \hat{\beta} - r_f - \lambda(\phi)$ into the fund's objective (10), the first-order condition is

$$\underbrace{(\mu - r_f)}_{\text{return}} \;\underbrace{-\,\phi\xi}_{\text{price impact}} \;-\gamma_m \sigma^2 \phi - c_{\phi}(\phi - \bar{\phi}) = 0,$$

which is (16). The objective is globally strictly concave: its second derivative is $-D - 2\xi < 0$. Substituting the pressure equation and collecting constants yields the closed form (17):

$$\phi^*(\xi, \theta) = \frac{C - \theta}{D + 2\xi}, \qquad \lambda^*(\xi, \theta) = \theta + \xi\phi^*(\xi, \theta) - c_r s^M.$$

B.1.2 Price-Pressure Derivatives

Since $\xi = c_r Q/(\hat{\beta} I^{\alpha})$ depends on (Q, I) only through $Q I^{-\alpha}$, every trading-stage outcome satisfies the scale-invariance identity

$$\frac{\partial}{\partial \log I} = -\alpha \frac{\partial}{\partial \log Q}. \tag{A.4}$$

Define $g(\xi) \equiv \xi\phi^*(\xi) = (C-\theta)\xi/(D+2\xi)$, so that $\lambda^* = \theta + g(\xi) - c_r s^M$. Direct differentiation gives

$$g'(\xi) = \frac{(C-\theta)D}{(D+2\xi)^2} = \frac{D\,\phi^*}{D+2\xi} > 0, \qquad \frac{d\phi^*}{d\xi} = -\frac{2\phi^*}{D+2\xi} < 0.$$

Define the marginal impact of fund size on price pressure,

$$\lambda_Q(\xi) \equiv \left.\frac{\partial \lambda^*}{\partial \log Q}\right|_I = g'(\xi)\,\xi = \frac{D\,\xi\,\phi^*}{D+2\xi} > 0. \tag{A.5}$$

By (A.4), the pressure response to investment along the fund's optimal response is

$$\frac{d\lambda}{dI} = -\frac{\alpha}{I}\,\lambda_Q(\xi) < 0,$$

which is (18) in the main text. Finally, log-differentiating (A.5),

$$\xi\,\lambda_Q'(\xi) = \lambda_Q(\xi)\,\frac{D-2\xi}{D+2\xi} \;\le\; \lambda_Q(\xi), \tag{A.6}$$

with the inequality strict and the left side negative whenever $\xi > D/2$.

Consequently $\lambda_Q(\xi)$ is hump-shaped in ξ, maximized at $\xi = D/2$ with maximum $(C-\theta)/8$, so the marginal impact of fund size on price pressure is uniformly bounded: as the fund's footprint grows, its own retreat (ϕ^* falling in ξ) caps the pressure any further increase in size can create.

B.1.3 Proof of Property 1 and Property 3

By (19), the equilibrium investment I^* solves the single equation

$$F(I;Q,\theta) \equiv \hat{\beta}\alpha I^{\alpha-1} - 1 - \frac{c_h}{K}\,e^{-2\lambda}\,H\left(1 + \frac{\alpha H}{I}\,\lambda_Q\right) = 0, \tag{A.7}$$

where $H = I - K^\alpha$ and λ, λ_Q are evaluated at $\xi(I,Q)$ and θ. The interiority conditions are $H^* > 0$ and $\phi^* > 0$ (i.e., $C > \theta$). By (A.10) these imply $F_I(I^*) < 0$, so I^* is the unique interior solution.

Step 1: $F_Q > 0$. Using $\partial\lambda/\partial Q = \lambda_Q/Q$ and $\partial\lambda_Q/\partial Q = \lambda_Q'(\xi)\,\xi/Q$,

$$F_Q = \frac{c_h H e^{-2\lambda}}{KQ}\left[2\lambda_Q\left(1 + \frac{\alpha H}{I}\lambda_Q\right) - \frac{\alpha H}{I}\,\xi\lambda_Q'(\xi)\right].$$

By (A.6), $\xi\lambda_Q' \le \lambda_Q$, so the bracket satisfies

$$2\lambda_Q\left(1 + \frac{\alpha H}{I}\lambda_Q\right) - \frac{\alpha H}{I}\,\xi\lambda_Q' \;\ge\; \lambda_Q\left(2 - \frac{\alpha H}{I} + \frac{2\alpha H}{I}\lambda_Q\right) \;>\; \lambda_Q\,(2-\alpha) \;>\; 0,$$

using $H < I$ so that $\alpha H/I < \alpha < 1$. Hence $F_Q > 0$, and by the implicit function theorem

$$\frac{dI^*}{dQ} = -\frac{F_Q}{F_I} > 0, \qquad \frac{dP^*}{dQ} = \alpha\hat{\beta}I^{\alpha-1}\frac{dI^*}{dQ} > 0.$$

This proves the fundamental-value part of Property 1 and the investment part of Property 3.

Step 2: fundamental value rises less than one-for-one. Consider the joint variation $(d\log I, d\log Q)$ with $d\log Q = \alpha\, d\log I$, along which ξ, and therefore λ, λ_Q, and ϕ^*, is constant. Differentiating (A.7) along this path,

$$\begin{aligned} F_I + \frac{\alpha Q}{I}F_Q &= \hat{\beta}\alpha(\alpha-1)I^{\alpha-2} - \frac{c_h e^{-2\lambda}}{K}\frac{d}{dI}\left[H + \frac{\alpha\lambda_Q}{I}H^2\right] \\ &= \hat{\beta}\alpha(\alpha-1)I^{\alpha-2} - \frac{c_h e^{-2\lambda}}{K}\left[1 + \frac{\alpha\lambda_Q H}{I}\left(2 - \frac{H}{I}\right)\right] < 0, \end{aligned}$$

using $\alpha < 1$, $\lambda_Q > 0$, and $H/I \in (0,1)$. Together with $F_I < 0$ and $F_Q > 0$, this yields

$$\alpha\,\frac{d\log I^*}{d\log Q} = \frac{(\alpha Q/I)\,F_Q}{-F_I} < 1 \quad\Longleftrightarrow\quad \frac{d\log P^*}{d\log Q} < 1, \tag{A.8}$$

so that

$$\frac{d\log\xi^*}{d\log Q} = 1 - \alpha\,\frac{d\log I^*}{d\log Q} \in (0,1):$$

fund size relative to fundamental value rises with fund size.

Step 3: pressure and portfolio responses. Since $\lambda^* = \theta + g(\xi) - c_r s^M$ with $g' > 0$, and ϕ^* is decreasing in ξ,

$$\frac{d\lambda^*}{dQ} = g'(\xi)\,\frac{d\xi^*}{dQ} > 0, \qquad \frac{d\phi^*}{dQ} = \frac{d\phi^*}{d\xi}\,\frac{d\xi^*}{dQ} < 0.$$

This proves the price-pressure part of Property 1 and the portfolio part of Property 3.

Step 4: decomposition. From $\lambda = \theta + \xi(I,Q)\,\phi - c_r s^M$, the partial derivatives holding the other equilibrium objects fixed are

$$\frac{\partial\lambda}{\partial Q} = \frac{\xi\phi}{Q} > 0, \qquad \frac{\partial\lambda}{\partial\phi} = \xi > 0, \qquad \frac{\partial\lambda}{\partial I} = -\frac{\alpha\xi\phi}{I} < 0,$$

and the chain rule gives the decomposition displayed in Property 3 (stated there in terms of $\Lambda = e^{\lambda}$, which preserves all signs):

$$\frac{d\lambda^*}{dQ} = \frac{\partial\lambda}{\partial Q} + \frac{\partial\lambda}{\partial\phi}\,\frac{d\phi^*}{dQ} + \frac{\partial\lambda}{\partial I}\,\frac{dI^*}{dQ}.$$

The second and third terms are negative: the fund's retreat and the firm's issuance both absorb part of the excess demand.

Step 5: Property 1, part 2. Write $\nu \equiv d \log P^*/d \log Q = \alpha\, d \log I^*/d \log Q$, which lies in $(0, 1)$ by (A.8). Since $\lambda^* = \theta + g(\xi) - c_r s^M$ and $d \log \xi^*/d \log Q = 1 - \nu$, the two components of the observed price response to the flow shock are

$$\frac{dp^*}{d \log Q} = \nu, \qquad \frac{d\lambda^*}{d \log Q} = \lambda_Q\,(1-\nu), \qquad \frac{d\tilde{p}^*}{d \log Q} = \nu + \lambda_Q(1-\nu),$$

all strictly positive by Property 1. Residual investors' holdings respond only to price pressure, so by (14) the reduced form of the instrument on log holdings is $-\eta\,\lambda_Q(1-\nu)$ with $\eta = 1/(c_r s^R)$. The instrumental-variable estimand that divides this by the full price response is therefore

$$\hat{\eta} = \eta \cdot \frac{\lambda_Q(1-\nu)}{\nu + \lambda_Q(1-\nu)} = (1-\pi)\,\eta, \qquad \pi = \frac{\nu}{\nu + \lambda_Q(1-\nu)} \in (0, 1),$$

which is (20). The attenuation is strict whenever the firm issues, since $\nu > 0$ requires $dI^*/dQ > 0$; it vanishes as $\nu \to 0$, that is, when equity supply is fixed. $\square$

B.1.4 Proof of Property 2

Residual investors. From (14), $(s^{R'} - s^R)/s^R = (\theta - \lambda)/(c_r s^R)$: the elasticity with respect to price pressure is $-1/(c_r s^R) < 0$, and fundamental value P does not enter, so the elasticity with respect to fundamental value is zero.

The mutual fund. The fund's policy function is $\phi^*(\xi, \theta) = (C - \theta)/(D + 2\xi)$ with $\xi = c_r Q/P$.

For the elasticity with respect to price pressure, vary the demand shifter θ, which moves λ without any change in fund flows or fundamentals:

$$\frac{\partial \phi^*}{\partial \theta} = -\frac{1}{D + 2\xi} < 0, \qquad \left.\frac{d\lambda^*}{d\theta}\right|_I = 1 + \xi\,\frac{\partial \phi^*}{\partial \theta} = \frac{D + \xi}{D + 2\xi} > 0,$$

so

$$\left.\frac{d\phi^*}{d\lambda^*}\right|_{\theta\text{-induced}} = -\frac{1}{D + \xi} < 0.$$

For the elasticity with respect to fundamental value, vary P through the payoff scale (as issuance-driven fundamental improvements do in this model), which leaves expected returns and hence C unchanged and enters only through ξ:

$$\frac{\partial \phi^*}{\partial \log P} = -\frac{2(C-\theta)}{(D+2\xi)^2}\,\frac{\partial \xi}{\partial \log P} = \frac{2\xi\,\phi^*}{D + 2\xi} > 0.$$

Higher fundamental value shrinks the fund's relative footprint ξ, lowering equilibrium price pressure and raising expected returns, so the fund increases its weight. $\square$

B.1.5 Proof of Property 4

Write the firm's condition (A.7) in wedge form (19), $\hat{\beta}\alpha I^{\alpha-1} = 1 + \zeta$, and define the issuance intensity, the price-impact factor, and the elasticity of the wedge to investment

$$h \equiv \frac{H}{I} \in (0,1), \qquad a \equiv \alpha h \lambda_Q, \qquad W \equiv \left.\frac{\partial \log \zeta}{\partial \log I}\right|_{\lambda,\lambda_Q} = \frac{1}{h}\left(1 + \frac{a(1-h)}{1+a}\right) > 1, \tag{A.9}$$

using $\partial \log H/\partial \log I = 1/h$ and $\partial \log h/\partial \log I = (1-h)/h$, so that $\zeta = (c_h H/K)e^{-2\lambda}(1+a)$.

The second-order condition and uniqueness. Totalling the dependence of λ and λ_Q on investment through $\xi(I,Q)$, using $d\log\xi/d\log I = -\alpha$ and (A.6), and evaluating at a critical point of (A.7),

$$-IF_I = J + \zeta\alpha\left[2\lambda_Q - (2\rho - 1)\,\frac{a}{1+a}\right], \qquad J \equiv (1-\alpha)(1+\zeta) + \zeta W. \tag{A.10}$$

Both pieces are strictly positive. $J > 0$ because $\alpha < 1$, because $\zeta > 0$ whenever the firm issues, and because $W > 1$ by (A.9). The bracket is positive because $a/(1+a) < a = \alpha h \lambda_Q < \lambda_Q$ while $2\rho - 1 < 1$. The two terms it collects are the firm's own feedback onto price pressure: investment erodes the pressure the firm is issuing into, which steepens the marginal issuance cost, and it moves λ_Q along the hump of (A.5), which softens that cost when $\xi < D/2$ but never by enough to overturn the erosion. Since $-IF_I > 0$ at every critical point, F crosses zero at most once and the interior equilibrium is unique. The second-order condition is therefore implied by $H^* > 0$ and $\phi^* > 0$ rather than imposed alongside them. None of these signs requires the price-impact factor $1 + a$ to be small; each holds for any $a \geq 0$.

Capital-poor firms carry the largest wedge. Since λ and λ_Q depend on (I,Q) but not on K, initial capital enters (A.7) only through $H = I - K^\alpha$ and the scale factor $1/K$. Both H/K and h are decreasing in K, hence so is ζ at fixed I, and

$$F_K = -\frac{\partial}{\partial K}\left[\frac{c_h e^{-2\lambda}}{K} H(1+a)\right] > 0 \quad\Longrightarrow\quad \frac{dI^*}{dK} = -\frac{F_K}{F_I} > 0.$$

Since $\zeta = \hat{\beta}\alpha I^{\alpha-1} - 1$ with $\alpha < 1$, I^* increasing in K is equivalent to ζ^* decreasing in K. Capital-poor firms therefore carry both the larger issuance burden and the higher marginal product. The two parts below are demand-side and hold across firms facing a common fund size Q and a common initial ownership s^M, so that C and D in (17) do not vary across firms.

Part 1: capital-poor firms attract more pressure. Since I^* is increasing in K, $P = \hat{\beta} I^{*\alpha}$ is increasing in K and the footprint $\xi = c_r Q/P$ is decreasing in K. From (17), $\lambda^* = \theta - c_r s^M + (C-\theta) g(\xi)$ with $g(\xi) = \xi/(D+2\xi)$ and $g' = D/(D+2\xi)^2 > 0$, so λ^* is decreasing in K. The accompanying fund behavior follows from the same closed form: $\phi^* = (C-\theta)/(D+2\xi)$ is increasing in K, and

$$\frac{d\log\phi^*}{d\log Q} = -\frac{2\xi}{D+2\xi}, \qquad \frac{d\log s^{M'}}{d\log Q} = \frac{D}{D+2\xi},$$

the first strictly increasing in magnitude and the second strictly decreasing in ξ, hence respectively larger and smaller for capital-poor firms. These two expressions are (21), which Property 3 part 2 reads in the Q direction and Part 1 here reads in the K direction: both are statements about $\xi = c_r Q/P$, the first varying its numerator and the second its denominator.

Part 2: capital-poor firms exhaust that pressure faster. Since $dH = dI$ at fixed K, (18) gives $|d\lambda/dH| = (\alpha/I)\,\lambda_Q$ with $\lambda_Q = (C-\theta)D\xi/(D+2\xi)^2$. Taking logs and using $d\log\xi/d\log I = -\alpha$,

$$\frac{d\log|d\lambda/dH|}{d\log I} = -1 + \frac{\partial\log\lambda_Q}{\partial\log\xi}\cdot\frac{d\log\xi}{d\log I} = -1 - \alpha\,\frac{D-2\xi}{D+2\xi} = -1 - \alpha(2\rho-1),$$

using $\partial\log\lambda_Q/\partial\log\xi = (D-2\xi)/(D+2\xi) = 2\rho - 1$ from (A.6). Because that ratio lies in $(-1,1)$ for every $\xi > 0$ and $\alpha < 1$, the expression lies in $\big(-(1+\alpha), -(1-\alpha)\big)$ and is strictly negative, which is (22). The erosion rate is therefore strictly decreasing in I, hence in K, for all parameter values: the non-monotonicity of λ_Q in ξ (Section B.1.2) is dominated by the α/I factor and never overturns the sign. To a first approximation $|d\lambda/dH| \propto 1/I$, with the variation in λ_Q tilting the log-log slope by at most α around -1. $\square$

B.2 General Demand Structures

This section shows that Property 1, Property 3, and Property 2 are not artifacts of the linear pressure specification (15). They hold for a class of demand structures that includes the market-value ownership convention and hyperbolic pressure function used in the full model of Section 3. The firm side is kept parametric throughout; generality is on the demand side, where the specifications differ.

The class. Given (I, Q, θ), the trading stage determines (ϕ, λ) by the fund's problem (10), internalizing $d\lambda/d\phi$, and the pressure map

$$\lambda = \theta + L\left(s^{M'} - s^M\right), \qquad s^{M'} = \frac{\phi\, Q\, e^{-\omega\lambda}}{P}, \tag{A.11}$$

where $L(0) = 0$, $L' > 0$, and $\omega \in [0,1]$ indexes the ownership convention: $\omega = 0$ measures ownership at fundamental value, $\omega = 1$ at market value. Two members are of particular interest:

1. $L(z) = c_r z$ and $\omega = 0$: the specification in Section 2;

2. $L(z) = -\log(1 - c_r z)$ and $\omega = 1$: equivalently $\Lambda = 1/\left(1 + c_r(s^M - s^{M'})\right)$ with $s^{M'} = \phi Q/\tilde{P}$, the specification of the full model in Section 3.

Every member of the class inherits the scale invariance of the linear specification. Substituting $s^{M'} = \phi x e^{-\omega\lambda}$, both the pressure map (A.11) and the fund's problem (whose return depends on the choice only through λ) involve (I, Q) only through $x \equiv Q/P(I)$. Trading-stage outcomes can therefore be written as $\phi = f(x, \theta)$ and $\lambda = \ell(x, \theta)$, and, since $\log x = \log Q - \log \hat{\beta} - \alpha \log I$, every one of them satisfies

$$\frac{\partial}{\partial \log I} = -\alpha \frac{\partial}{\partial \log Q}. \tag{A.12}$$

The two assumptions below are stated in terms of this reduction.

Assumption 1 (Regular trading stage)**.** The trading stage has a unique interior solution, continuously differentiable in (x, θ), with $\lambda_Q(x, \theta) \equiv \partial\ell/\partial \log x > 0$, $f_x < 0$, $f_\theta < 0$, and $\ell_\theta > 0$.

Assumption 2 (No accelerating price impact)**.** $\dfrac{\partial \log \lambda_Q}{\partial \log x} \leq 2.$

Assumption 1 states that the trading stage is well-behaved: more fund demand relative to fundamental value raises pressure, and the fund retreats from a larger footprint or from an exogenous demand shift. Assumption 2 requires that the marginal price impact of fund size not grow too fast with the fund's relative footprint; it is the transparent counterpart of a regularity condition on second-order feedback effects.

Proposition 5. *In any member of the class satisfying Assumption 1 and Assumption 2, with an interior issuance equilibrium ($H^* > 0$, $\phi^* > 0$, $F_I(I^*) < 0$), Properties 1, 3, and 2 hold, including the attenuation result of Property 1, part 2. Property 4 holds in every member of the class without further conditions.*

Proof. The firm's problem is unchanged, so equilibrium investment solves (A.7) with $\lambda = \ell(x, \theta)$ and $\lambda_Q = \lambda_Q(x, \theta)$ evaluated at $x(I, Q)$. The proof retraces the four steps of Section B.1, noting where each assumption enters.

Step 1. Since $\partial\ell/\partial Q = \lambda_Q/Q$ and $\partial\lambda_Q/\partial Q = (x\lambda_{Qx})/Q$,

$$F_Q = \frac{c_h H e^{-2\ell}}{KQ}\left[2\lambda_Q\left(1 + \frac{\alpha H}{I}\lambda_Q\right) - \frac{\alpha H}{I} x\lambda_{Qx}\right].$$

By Assumption 2, $x\lambda_{Qx} \leq 2\lambda_Q$, so the bracket is at least $2\lambda_Q\left[1 - \frac{\alpha H}{I}(1-\lambda_Q)\right] > 0$: when $\lambda_Q \leq 1$ this uses $\alpha H/I < \alpha < 1$, and when $\lambda_Q > 1$ the subtracted term is itself negative. Hence $F_Q > 0$, giving $dI^*/dQ > 0$ and $dP^*/dQ > 0$.

Step 2. Along the variation $d\log Q = \alpha\, d\log I$, x, and therefore ℓ and λ_Q, is constant (by (A.12)), and the computation in Step 2 of Section B.1 goes through verbatim, using only $\lambda_Q > 0$: $F_I + (\alpha Q/I)F_Q < 0$, hence $d\log x^*/d\log Q \in (0,1)$.

Step 3. $d\lambda^*/dQ = \ell_x\, dx^*/dQ > 0$ and $d\phi^*/dQ = f_x\, dx^*/dQ < 0$ by Assumption 1.

Step 4. Implicit differentiation of (A.11) at fixed (ϕ, I) or (ϕ, Q) gives

$$\frac{\partial\lambda}{\partial\phi} = \frac{L' x e^{-\omega\lambda}}{1+\omega L'\phi x e^{-\omega\lambda}} > 0, \qquad \frac{\partial\lambda}{\partial Q} = \frac{\phi}{Q}\frac{\partial\lambda}{\partial\phi} > 0, \qquad \frac{\partial\lambda}{\partial I} = -\frac{\alpha Q}{I}\frac{\partial\lambda}{\partial Q} < 0,$$

where the denominator $1 + \omega L' s^{M'} > 0$ is part of interiority, and the last identity is (A.12) at fixed ϕ. The decomposition and its signs follow as before.

Property 2. The residual sector's elasticity with respect to price pressure is negative by $L' > 0$, and zero with respect to fundamental value by construction. For the fund, $d\phi/d\theta = f_\theta < 0$ with $d\lambda/d\theta = \ell_\theta > 0$ gives a negative elasticity with respect to pressure, and $d\phi/d\log P = -x f_x > 0$ (varying P through the payoff scale, which enters only through x) gives a positive elasticity with respect to fundamental value.

Property 1, part 2. Step 5 uses only $\nu \in (0,1)$ from Step 2 and the sign of λ_Q, so it carries over unchanged; the expression for π holds with $\lambda_Q = \lambda_Q(x,\theta)$.

Property 4, firm-side preliminaries. The second-order condition, uniqueness, and the comparative static $dI^*/dK > 0$ in Section B.1.5 use only the firm's problem: the wedge form (19) and the fact that price pressure enters the issuance cost through Λ^{-2}. They never use the pressure map L, the ownership convention ω, or the closed forms of the trading stage, and λ_Q enters only through the price-impact factor $1 + \alpha(H/I)\lambda_Q$, which no part of the proposition requires to be small (Section B.1.5). These results are thus firm-side, and generalizing the demand side leaves them untouched.

Property 4, parts 1–2. These are demand-side and do use the pressure map, but only through two of its properties. Part 1 needs ℓ increasing in $x = Q/P$, which holds by construction for any increasing L, so a firm with less fundamental value to absorb a given fund position faces more pressure. Part 2 needs the erosion elasticity $\partial\log|d\lambda/dH|/\partial\log I = -1 + \alpha\,\partial\log\lambda_Q/\partial\log x$ to be negative, which follows from $\alpha < 1$ together with $|\partial\log\lambda_Q/\partial\log x| \leq 1$, that is, Assumption 2. The linear member satisfies the latter strictly, as verified below. □

Verification for the two specifications. For the linear specification ($L(z) = c_r z$, $\omega = 0$), the closed forms of Section B.1.1 give $f = (C - \theta)/(D + 2c_r x)$ and $\lambda_Q = D\xi\phi^*/(D + 2\xi)$ with $\xi = c_r x$, so Assumption 1 holds directly and, by (A.6), $\partial \log \lambda_Q / \partial \log x = (D - 2\xi)/(D + 2\xi) \in (-1, 1)$, so Assumption 2 holds strictly for all parameter values. The same computation gives (22): $\partial \log |d\lambda/dH| / \partial \log I = -1 - \alpha(D - 2\xi)/(D + 2\xi) \in \big(-(1 + \alpha), -(1 - \alpha)\big)$, strictly negative for every $\xi > 0$ since $\alpha < 1$.

For the market-value specification ($L(z) = -\log(1 - c_r z)$, $\omega = 1$), direct computation gives

$$\lambda_Q(x) = \frac{D\,\xi\phi\,(1 + \xi\phi)}{D(1 + \xi\phi)^2 + \xi(2 + \xi\phi)}, \qquad \xi = c_r x,$$

which reduces to the linear case as $\xi\phi \to 0$. Assumption 1 requires strict concavity of the fund's problem, which holds for moderate price impact, and Assumption 2 holds in a neighborhood of $\Lambda = 1$ by continuity with the linear case; both can be verified numerically over any relevant parameter region. Since price pressure fluctuates near one in the estimated full model, the properties established here apply to its pressure specification as well. The linear specification is adopted in the main text because it alone delivers closed forms.

B.3 A Microfoundation for the Residual Sector

This section derives the residual sector's trading friction from a funding constraint in the spirit of Gabaix and Maggiori (2015), stated in the market-value units used in the full model of Section 3, and shows that the two-period objective (12) is the same problem expressed in return units.

Fix a period and a firm with market value $\tilde{P}$ and fundamental value P. The residual sector consists of passive legacy holders with stake s^R and a fixed unit mass of price-taking arbitrage cohorts that live one period. Each cohort intermediates the incremental stake $\Delta s \equiv s^{R'} - s^R$: it raises $\Delta s\,\tilde{P}$ from households, buys Δs shares at the market value $\tilde{P}$, and holds them until the firm's payoff V' is realized, at which point the proceeds $\Delta s\,V'$ revert to the households that funded it. Households value that payoff with their stochastic discount factor M', and fundamental value is by definition $P = \mathbb{E}[M'V']$. The value the cohort creates is therefore

$$\Pi(\Delta s) = \mathbb{E}\,[M'\,\Delta s\,V'] - \Delta s\,\tilde{P} = \Delta s\,(P - \tilde{P}).$$

Shares change hands only at $\tilde{P}$; P is a valuation, not a transaction price, and Π is the present value of the position to the households who ultimately own it, net of what it costs to acquire. The mass of cohorts is fixed, so entry does not compete the wedge away: this is the sense in which arbitrage capacity is limited.

The friction is limited commitment: after funds are committed, the cohort can abscond with part of its position, and households anticipate this. Following Gabaix and Maggiori (2015), the *fraction* of the position that can be diverted is proportional to its size, $c_r|\Delta s|$, so that the divertible amount is quadratic. Because diverted shares are disposed of at market prices, that amount is valued at $\tilde{P}$:

$$\text{divertible value} = \underbrace{(c_r\,|\Delta s|)}_{\text{fraction}} \cdot \underbrace{|\Delta s|\,\tilde{P}}_{\text{position value}} = c_r(\Delta s)^2\,\tilde{P}.$$

The size-dependent fraction is what makes capacity finite rather than merely costly: a cohort that doubles its block in a single name becomes twice as hard to monitor per dollar, and the block it would have to unwind on seizure recovers proportionally less. Prime brokers' concentration charges (margin rates that rise with the concentration of a position in one name) are its institutional counterpart. Households finance the position only if honesty dominates diversion:

$$\Delta s\,(P - \tilde{P}) \;\geq\; c_r(\Delta s)^2\,\tilde{P}. \tag{A.13}$$

Since the left side is linear in Δs and the right side quadratic, (A.13) holds strictly at small positions and fails at large ones. The cohort expands until the constraint binds: profit is increasing in Δs, so the funding constraint, not a first-order condition, is what stops it, giving

$$\Delta s = \frac{1}{c_r}\,\frac{P - \tilde{P}}{\tilde{P}},$$

which is the residual investors' portfolio rule in the full model and, to first order, the two-period rule (14) at $\theta = 0$. Equivalently, the sector behaves *as if* maximizing

$$s^{R'}(P - \tilde{P}) - \frac{c_r}{2}\left(s^{R'} - s^{R}\right)^2 \tilde{P},$$

the objective specified in the full model: the profit term is linear, so the first-order condition is identical whether profit is written on the total or the incremental stake.

Four remarks. First, the friction pins down the market-value scaling that distinguishes the full model's specification from the two-period model's: what can be diverted, and what collateral and margin are levied on, is valued at market prices, so the constraint scales with $\tilde{P}$, not with P. This scaling also keeps c_r a pure number: the implied elasticity $1/(c_r s^R)$ is invariant to firm size and the price level, which is what makes a single c_r comparable across the heterogeneous firms of the full model.

Second, the friction attaches to the *incremental* stake rather than the level of holdings because only new positions require new outside funding; legacy stakes sit with passive households. This is the funding-side counterpart of the transaction-cost technology of Gârleanu and Pedersen (2013), and it is what makes s^R a state variable and price pressure persistent in the full model.

Third, the proportional diversion technology is what delivers a constant-slope demand curve, and relaxing it maps into the general class of Section B.2. For a divertible fraction $f(|\Delta s|)$ with f increasing and $f(0) = 0$, the binding constraint gives $\Delta s = f^{-1}\big((P - \tilde{P})/\tilde{P}\big)$, a downward-sloping demand curve of general shape, and market clearing then yields a pressure map L satisfying $L(0) = 0$ and $L' > 0$. The properties therefore do not depend on the linear case; linearity is the tractability assumption, in the same role as the quadratic adjustment costs elsewhere in the model, while the market-value units are the derived part.

Fourth, dividing the as-if objective by $\tilde{P}$ expresses the same problem per dollar of market value,

$$s^{R'} \frac{P - \tilde{P}}{\tilde{P}} - \frac{c_r}{2}\left(s^{R'} - s^R\right)^2,$$

and since $(P - \tilde{P})/\tilde{P} = e^{-\lambda} - 1 = -\lambda + O(\lambda^2)$, its first-order form is the log objective (12) of the two-period model, with θ shifting the value the residual sector trades toward. The two specifications are therefore one problem in two units of account: dollars, where accounting across heterogeneous firms matters, and returns, where closed forms matter. Section B.2 verifies that the model's properties hold under both. Under this interpretation, c_r measures the tightness of the arbitrage sector's funding constraint (He and Krishnamurthy, 2013), the object varied in the residual-investor counterfactual in Section 6.

B.4 First order conditions

The firm problem is

$$V_{i,t} = \max_{I_{i,t}, K_{i,t+1}, L_{i,t+1}} O_{i,t} + \mathbb{E}_t\left[M_{t,t+1} V_{i,t+1}\right],$$

where

$$O_{i,t} = \begin{cases} E_{i,t} - \Psi_{i,t} \equiv O^{+}(E_{i,t}, K_{i,t}, \Lambda_{i,t}(\phi_{i,t+1}(K_{i,t+1}, L_{i,t+1}))) & \text{if } E_{i,t} \leq 0, \\ (1-\tau_D)(E_{i,t} - B_{i,t}) + B_{i,t} - \Phi_{i,t} \equiv O^{-}(E_{i,t}, \Lambda_{i,t}(\phi_{i,t+1}(K_{i,t+1}, L_{i,t+1}))) & \text{if } E_{i,t} > 0. \end{cases}$$

$$E_{i,t} = (1-\tau)A_t^{b_i} X_{i,t} K_{i,t}^{\alpha} + \tau\delta K_{i,t} + \tau r_f L_{i,t} \mathbb{1}_{\{L_{i,t}>0\}} - I_{i,t} - G_{i,t} + L_{i,t+1} - (1+r_l)L_{i,t}$$

$$K_{i,t+1} = (1-\delta)K_{i,t} + I_{i,t}$$

$$G_{i,t} = \frac{c_k}{2}\left(\frac{I_{i,t}}{K_{i,t}}\right)^2 K_{i,t}$$

$$\Psi_{i,t} = c_{h0}\frac{K_{i,t}}{\Lambda_{i,t}} + \frac{c_{h1}}{2}\left(\frac{H_{i,t}}{\Lambda_{i,t}K_{i,t}}\right)^2 K_{i,t}$$

$$\Phi_{i,t} = \frac{c_b}{2}\left(\frac{\Lambda_{i,t}B_{i,t}}{E_{i,t}}\right)^2 E_{i,t}$$

$$L_{i,t+1} \leq \varphi K_{i,t+1}.$$

First, when $E_{i,t} > 0$, $B_{i,t} = \frac{\tau_D}{c_b\Lambda^2}E_{i,t}$ and $O_{i,t} \equiv O^{+}_{i,t} = \left(1 - \tau_D + \frac{1}{2}\frac{\tau_D^2}{c_b\Lambda^2}\right)E_{i,t}$. Let q_t and λ_t denote the Lagrangian multipliers associated with (26) and (28). The first-order conditions with respect to I_t, K_{t+1}, and L_{t+1} are

$$-\frac{\partial O}{\partial I} = q \tag{A.14}$$

$$q = \mathbb{E}\left\{M'\left[\frac{\partial V'}{\partial K'}\right]\right\} + \lambda\varphi + \frac{\partial O}{\partial \Lambda}\frac{\partial \Lambda}{\partial \phi'}\frac{\partial \phi'}{\partial K'} \tag{A.15}$$

$$\lambda - \mathbb{E}\left\{M'\left[\frac{\partial V'}{\partial L'}\right]\right\} = \frac{\partial O}{\partial E}\frac{\partial E}{\partial L'} + \frac{\partial O}{\partial \Lambda}\frac{\partial \Lambda}{\partial \phi'}\frac{\partial \phi'}{\partial L'} \tag{A.16}$$

B.5 MPK-Dispersion Decomposition

This section derives Equation (54) in Section 6.2, the second-order relation between aggregate TFP and the cross-sectional dispersion of log MPK.

B.5.1 Setup

Consider a continuum of firms at a fixed point in time t, each producing

$$Y_i = A_t X_i K_i^{\alpha}, \tag{A.17}$$

where A_t is the aggregate productivity shock and X_i is firm i's idiosyncratic productivity, as in Equation (23). Because A_t is common to every firm at a fixed t, it factors out of every capital-share and dispersion object below and can be dropped: any comparison between two allocations evaluated at the same aggregate state holds A_t fixed, so it never enters a Δ. Let $K_{\text{agg}} = \int K_i \, di$ denote the aggregate capital stock and $s_i \equiv K_i / K_{\text{agg}}$ firm i's capital share. Aggregate TFP is

$$\text{TFP}_{\text{agg}} = \frac{Y_{\text{agg}}}{K_{\text{agg}}^{\alpha}} = \int X_i s_i^{\alpha} \, di. \tag{A.18}$$

B.5.2 The Efficient Allocation

The allocation s_i^* that maximizes TFP_{agg} subject to $\int s_i \, di = 1$ equalizes marginal products across firms, $\alpha X_i (s_i^*)^{\alpha-1} = \lambda$ for a constant λ, so that

$$s_i^* \propto X_i^{1/(1-\alpha)}. \tag{A.19}$$

A convenient property of the Cobb-Douglas case is that the efficient capital share then coincides with the efficient output share: since $X_i (s_i^*)^{\alpha} \propto X_i \cdot X_i^{\alpha/(1-\alpha)} = X_i^{1/(1-\alpha)} \propto s_i^*$,

$$s_i^* = \frac{Y_i^*}{Y_{\text{agg}}^*}. \tag{A.20}$$

B.5.3 Second-Order Expansion

Let $\theta_i \equiv \log s_i - \log s_i^*$ denote firm i's log deviation from its efficient capital share. Writing $s_i = s_i^* e^{\theta_i}$ and expanding s_i^{α} to second order in θ_i,

$$s_i^{\alpha} \approx (s_i^*)^{\alpha} \left(1 + \alpha \theta_i + \frac{\alpha^2}{2} \theta_i^2 \right), \tag{A.21}$$

so that, using $X_i (s_i^*)^{\alpha} = Y_i^* = s_i^* Y_{\text{agg}}^*$,

$$\frac{\text{TFP}_{\text{agg}}}{\text{TFP}_{\text{agg}}^*} \approx 1 + \alpha \int s_i^* \theta_i \, di + \frac{\alpha^2}{2} \int s_i^* \theta_i^2 \, di. \tag{A.22}$$

B.5.4 The Resource Constraint Kills the Linear Term

The linear term in (A.22) is not free: it is disciplined by the resource constraint $\int s_i \, di = 1$. Expanding $s_i = s_i^* e^{\theta_i} \approx s_i^*(1 + \theta_i + \theta_i^2/2)$ and using $\int s_i^* \, di = 1$,

$$\int s_i^* \theta_i \, di \approx -\frac{1}{2} \int s_i^* \theta_i^2 \, di, \tag{A.23}$$

i.e., the linear term is itself second-order in θ. Substituting into (A.22),

$$\frac{\text{TFP}_{\text{agg}}}{\text{TFP}^*_{\text{agg}}} \approx 1 - \frac{\alpha}{2} \int s_i^* \theta_i^2 \, di + \frac{\alpha^2}{2} \int s_i^* \theta_i^2 \, di \tag{A.24}$$

$$= 1 - \frac{\alpha(1-\alpha)}{2} \int s_i^* \theta_i^2 \, di, \tag{A.25}$$

so that, taking logs and writing $\alpha(1-\alpha)\theta_i^2 = \dfrac{\alpha}{1-\alpha} \left[(1-\alpha)\theta_i\right]^2$,

$$\log \left(\frac{\text{TFP}_{\text{agg}}}{\text{TFP}^*_{\text{agg}}} \right) \approx -\frac{1}{2} \frac{\alpha}{1-\alpha} \int s_i^* \left[(1-\alpha)\theta_i\right]^2 \, di. \tag{A.26}$$

B.5.5 From θ_i to Log MPK

Firm i's log MPK is $m_i \equiv \log \text{MPK}_i = \log(\alpha A_t X_i) + (\alpha - 1)\log K_i$, and at the efficient allocation $m_i^* = \log \lambda$ is the same constant for every firm. Since $K_i = s_i K_{\text{agg}}$,

$$m_i - m_i^* = (\alpha - 1)(\log s_i - \log s_i^*) = -(1-\alpha)\theta_i. \tag{A.27}$$

Substituting $(1-\alpha)\theta_i = -(m_i - m_i^*)$ into (A.26) gives

$$\log \left(\frac{\text{TFP}_{\text{agg}}}{\text{TFP}^*_{\text{agg}}} \right) \approx -\frac{1}{2} \frac{\alpha}{1-\alpha} \int s_i^* (m_i - m_i^*)^2 \, di. \tag{A.28}$$

Because $\int s_i^*(m_i - m_i^*) \, di$ is itself second-order (the same resource-constraint argument above, applied to $m_i - m_i^*$ in place of θ_i), the integral on the right is, to leading order, exactly the capital-share-weighted cross-sectional variance of m_i, giving Equation (54):

$$\Delta \log \text{TFP}_{\text{agg}} \approx -\frac{1}{2} \frac{\alpha}{1-\alpha} \Delta \text{Var}(m_i). \tag{A.29}$$

B.5.6 Applying the Result to a Non-Efficient Reference Point

The derivation above is stated relative to the fully efficient allocation s_i^*, for a clean interpretation of θ_i and m_i^*. Nothing in the argument requires s_i^* to be globally efficient, however: the second-order expansion, the resource-constraint cancellation of the linear term, and the resulting variance formula go through unchanged for *any* fixed reference allocation shared by the two economies being compared. In Section 6.2, the reference point used throughout is the $\Lambda \equiv 1$ economy rather than the fully efficient one: s_i^* becomes firm i's capital share when price pressure is shut off, and m_i^* becomes m_i^0, firm i's log MPK in that same $\Lambda \equiv 1$ economy. The $\Lambda \equiv 1$ economy still carries every other friction of the baseline model (adjustment costs, financing frictions, leverage constraints), so it is a fixed benchmark, not a claim that it is unconstrained-efficient.

C Computation Appendix

C.1 Algorithm to Solve the Model

1. Guess portfolio holding rule $\phi'^0(K', L', S)$, value function $V^0(S)$, and investment decision $K'^0(S)$.

2. Market clearing implies $s^{R'}(\phi'^0(K', L', S), K', L', S) = s^{R'0}(K', L', S)$ and therefore we have $\Xi(s^{R'0}(K', L', S), K', L', S) = \Xi^0(K', L', S)$. We also have the $\Xi^0(K'^0(S), L'^0(S), S) = \Xi^0(S)$.

3. Use $\Xi^0(K', L', S)$, run the Value Function Iteration routine. Get $V^1(S)$ and policy functions $K'^1(S)$.

4. Given $\Xi^0(S)$ (for tomorrow's value) and $V^0(S)$, calculate $\phi'^1(K', S)$.

5. Compare distances $|\phi^1(K', S) - \phi^0(K', S)|$ and $|V^1(S) - V^0(S)|$. Update $\phi^0(K', S)$ and $V^0(S)$. If larger than tolerance, go back to step 2.

C.2 Algorithm for the Indirect Inference Estimation

C.2.1 Constructing Model Moments and the Weight Matrix

I first regress firm returns, NEI, mutual fund portfolio holding, mutual fund positions, and changes in mutual fund positions on time and firm fixed effects and extract the residual to demean all relevant variables by firm and time. I calculate two dummies indicating non-negative and non-positive NEI (excluding dividends), and use the non-demeaned NEI data to calculate the conditional means. After calculating one-period forward of returns, I drop observations with any missing variable. This results in a panel of $N = 156,080$ observations. To keep the consistency between data and the model, I calculate the impulse response functions without added firm controls. As shown in Table 6, the regression coefficients are qualitatively similar to the baseline specification reported in Figure 1.

I follow Erickson and Whited (2002) to calculate the influence functions and covary the influence functions to calculate the weight matrix.

C.2.2 Estimation

Let x_n be a vector of data of dimension J, where J is the number of relevant variables. Let b be the vector of structural parameters to be estimated. Let $y_{n,k}(b)$ be the J-dimensional simulated data vector from simulation $k = 1, \ldots, K$. Let $x = \{x_1, \ldots, x_N\}$ and $y_k(b) = \{y_{1,k}, \ldots, y_{N,k}\}$ denote the data panel and the k-th simulated panel. I estimate b by matchign a set of simulated moments, denoted as $h(y_k(b))$, with the corresponding set of actual data moments, denoted as $h(x)$. Define

$$g(x,b) = h(x) - \frac{1}{K}\sum_{k=1}^{K} h(y_k(b)). \tag{A.30}$$

The indirect inference estimator of b is then defined as the solution to the minimization of

$$\hat{b} = \arg\min_{b} g(x,b)'\hat{W}g(x,b), \tag{A.31}$$

where $\hat{W}$ is the weight matrix calculated from the influence functions.

C.3 Robustness: Heterogeneous Productivity Exposure

The counterfactual results in Table 11 set firms' aggregate-shock exposure b_i in the production function $Y_{i,t} = A_t^{b_i} X_{i,t} K_{i,t}^{\alpha}$ to $b_i = 1$ for all firms. As a robustness check, I re-solve and re-simulate the model letting $b_i \sim \mathcal{N}(1, 0.2)$ vary across firms, following David et al. (2022) and Choi et al. (2025), and repeat the same four counterfactual scenarios. Table A.3 reports the results.

Table A.3: Capital Misallocation with Heterogeneous Productivity Exposure: Percentage Change from Baseline

Param	Base	CF	Description	Agg TFP
$\exp(\bar{q})$	0.290	0.220	Small fund sector	0.018
$\sigma_{\omega i}$	0.210	0.240	High flow vol.	-0.007
c_r	0.878	1.000	High capacity constr.	-0.011
c_k	0.492	0.450	Low capital adj. cost	0.150

Table A.3 repeats Table 11's four counterfactual scenarios with heterogeneous firm exposure to aggregate productivity shocks, $b_i \sim \mathcal{N}(1, 0.2)$, in place of the baseline's $b_i = 1$. The moments are calculated as the average across 10 simulated panels. Each simulated panel is composed of 3,000 firms over 1,000 quarters. The first 400 quarters are discarded.

The results are robust to this alternative specification. Reducing the capital adjustment cost still yields by far the largest aggregate TFP effect (0.15%), while the three financial-side counterfactuals remain an order of magnitude smaller, with the same signs as in the baseline: higher flow shock volatility lowers TFP (-0.007%), a smaller mutual fund sector raises TFP (0.018%), and less elastic residual investors lower TFP (-0.011%). This confirms that the paper's central finding, that financial-side frictions generate only modest capital misallocation relative to real investment frictions, does not depend on assuming firms are ex-ante homogeneous in their exposure to aggregate productivity shocks.